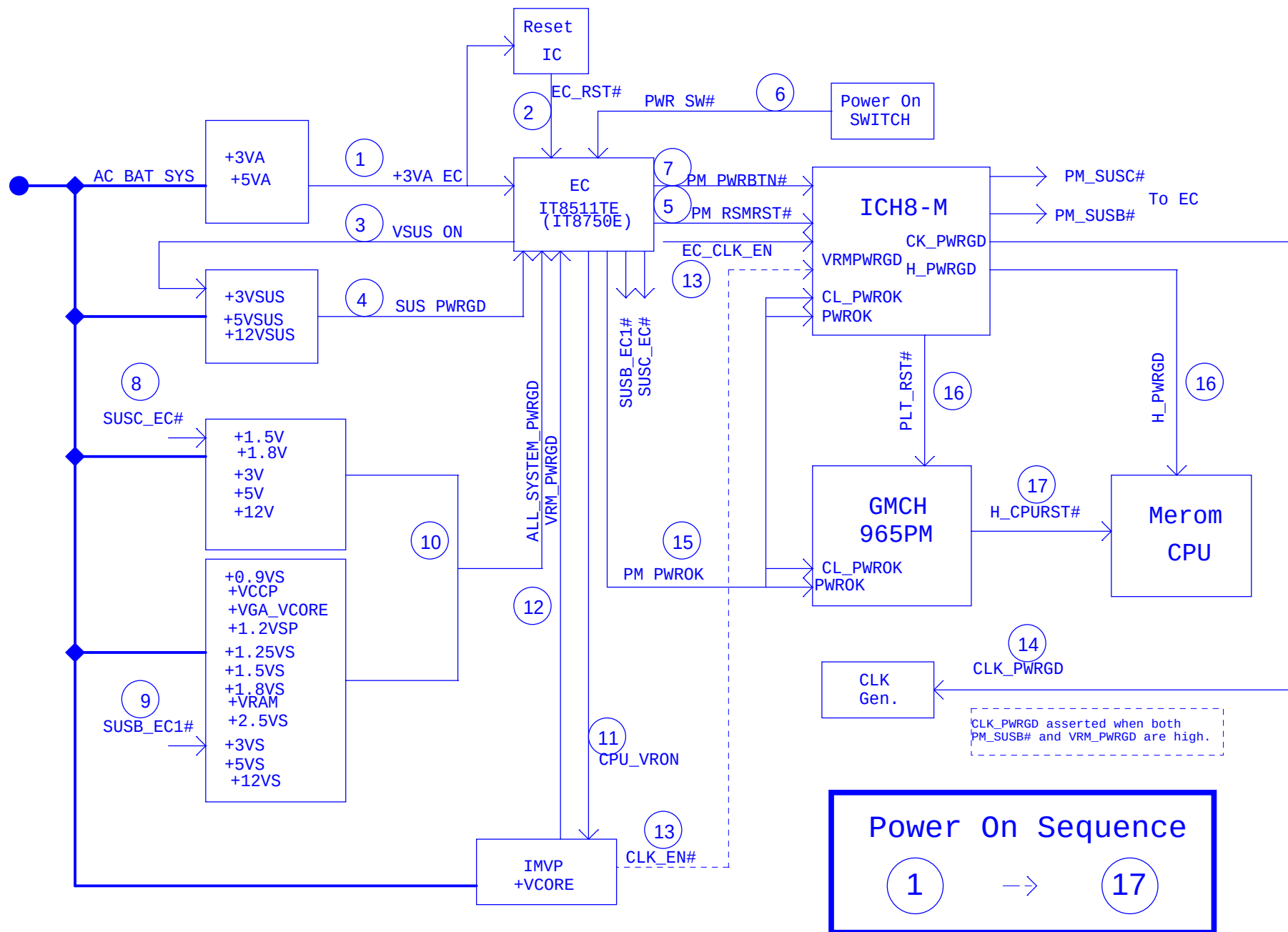
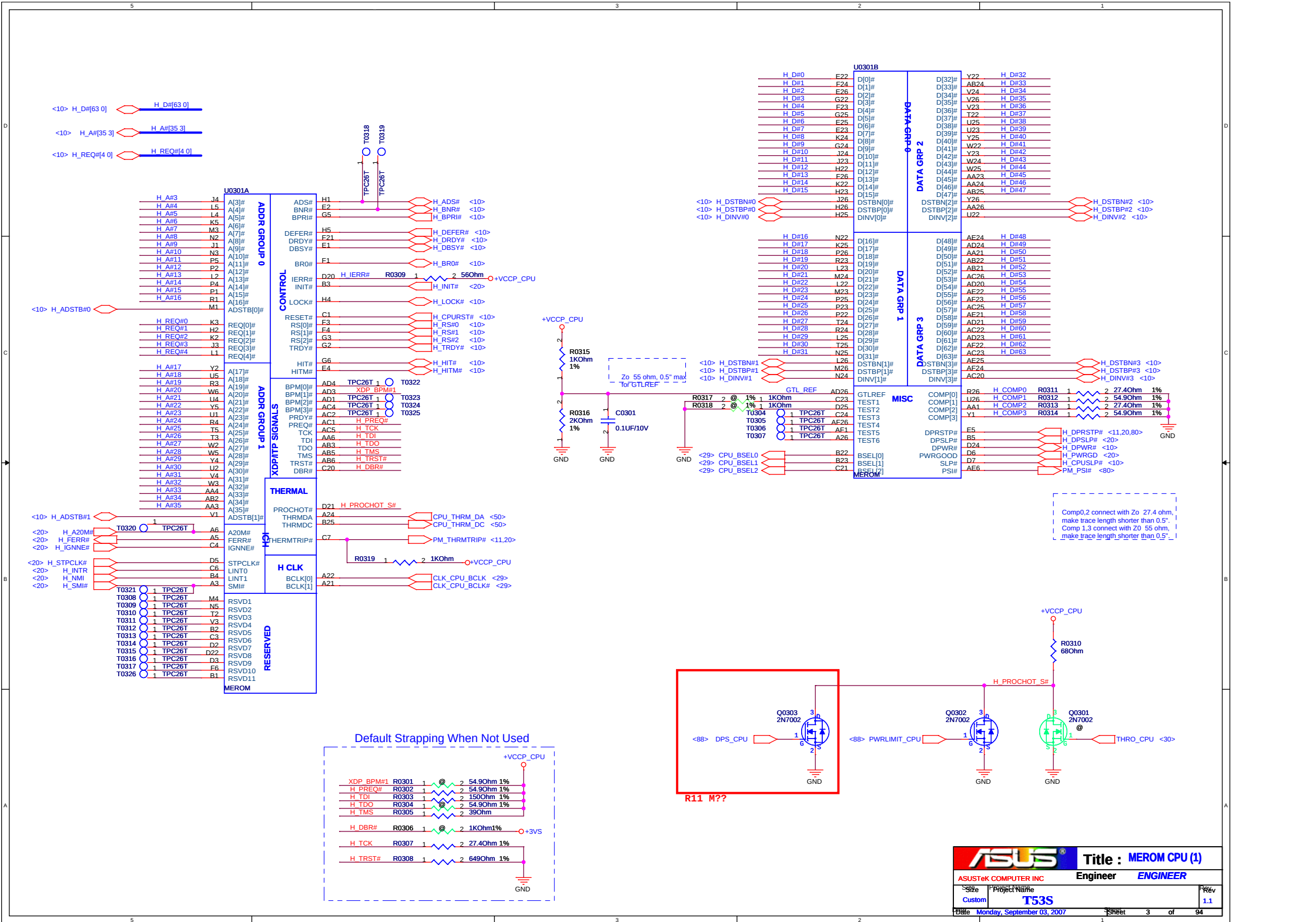
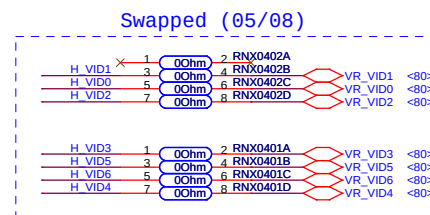
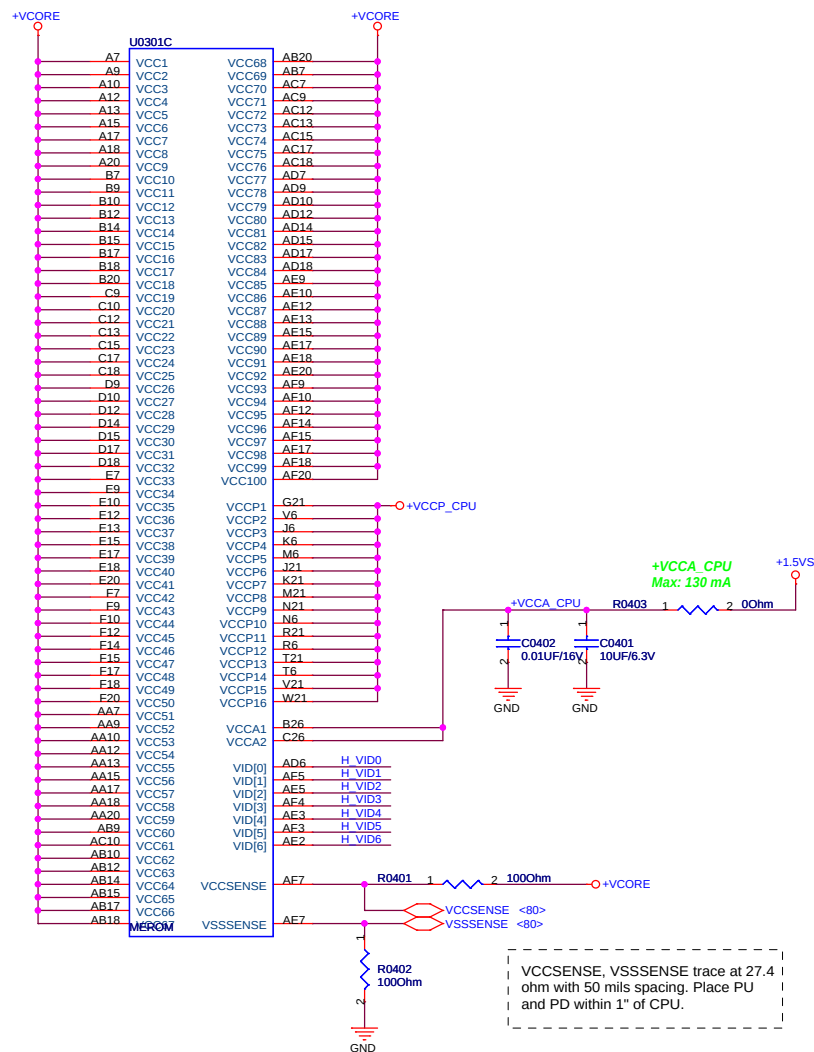
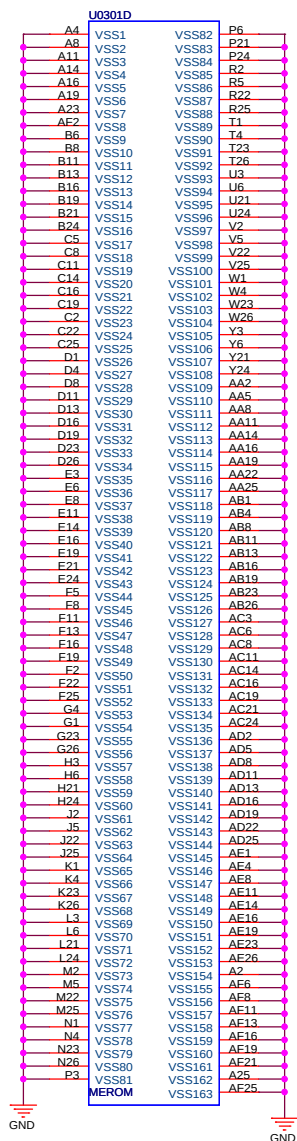
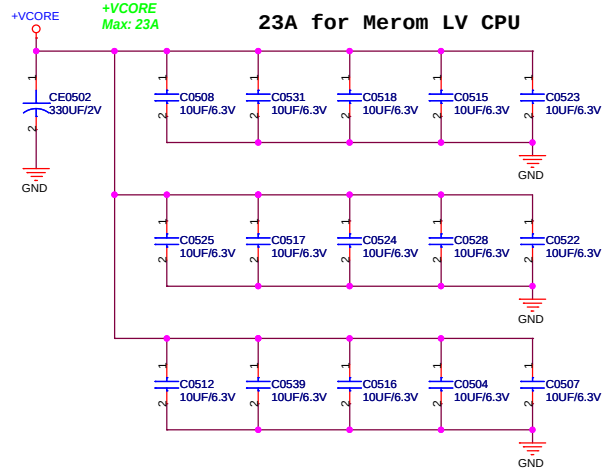




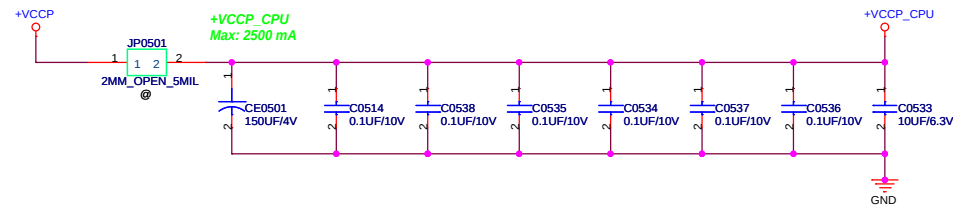






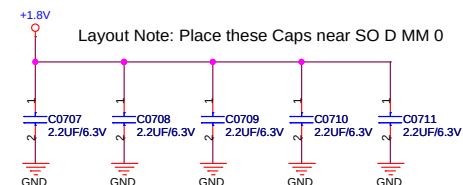
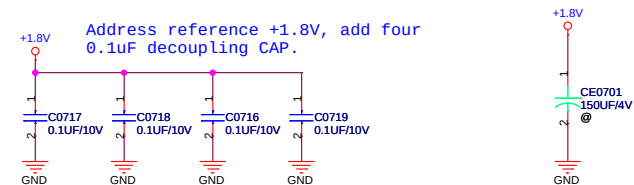
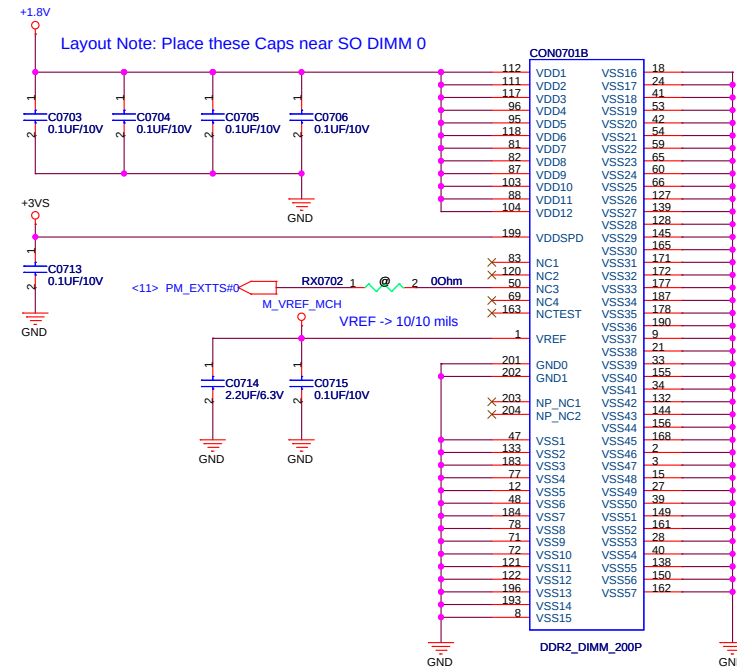
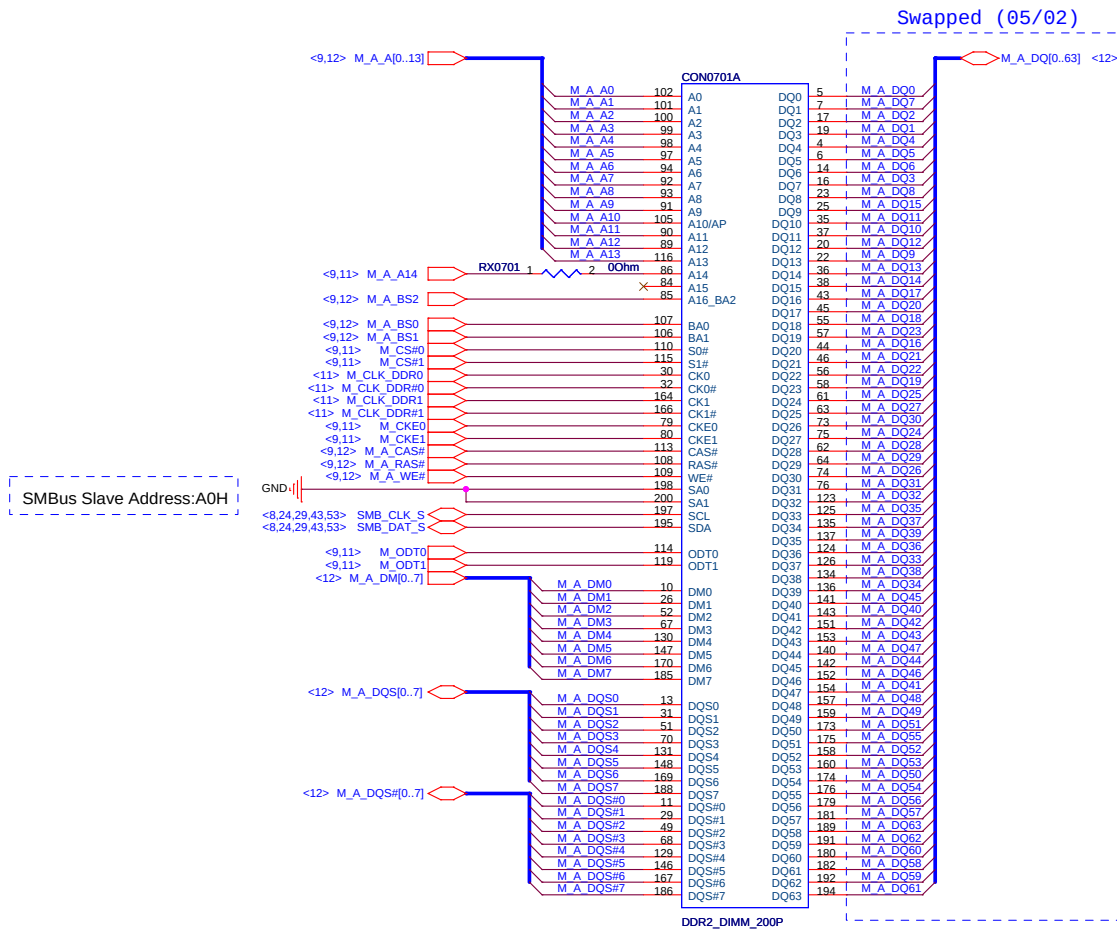
Decoupling guide from INTEL

VCORE 22uF/10V * 32pcs
 330uF/2V * 6pcs
 VCCP 0.1uF * 6pcs for CPU
 150uF * 1pcs for CPU



DIMM0 Tyco (H:4.0mm)

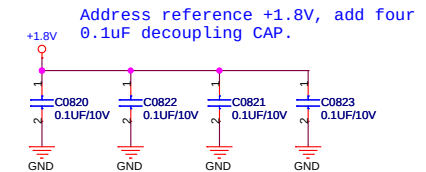
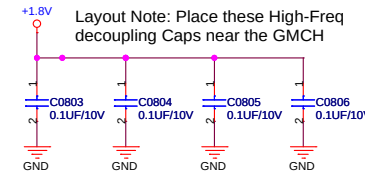
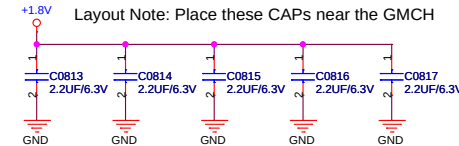
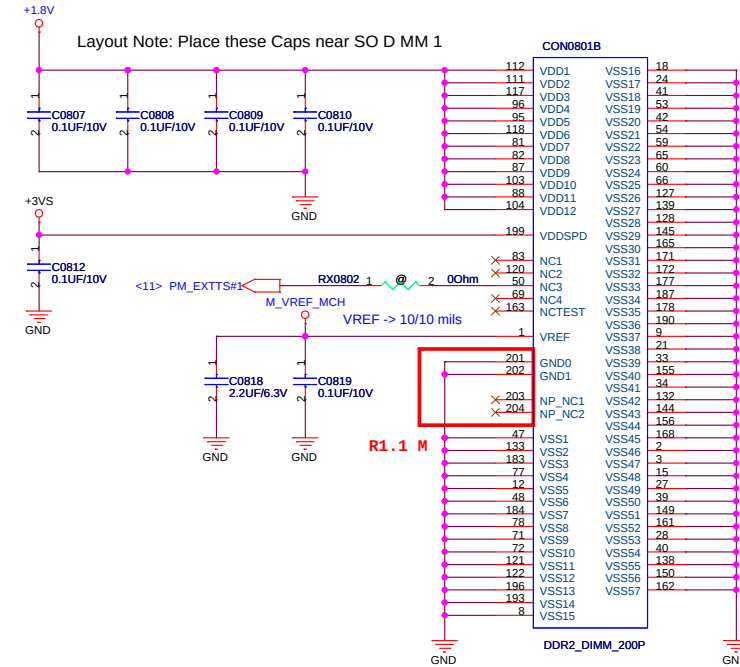
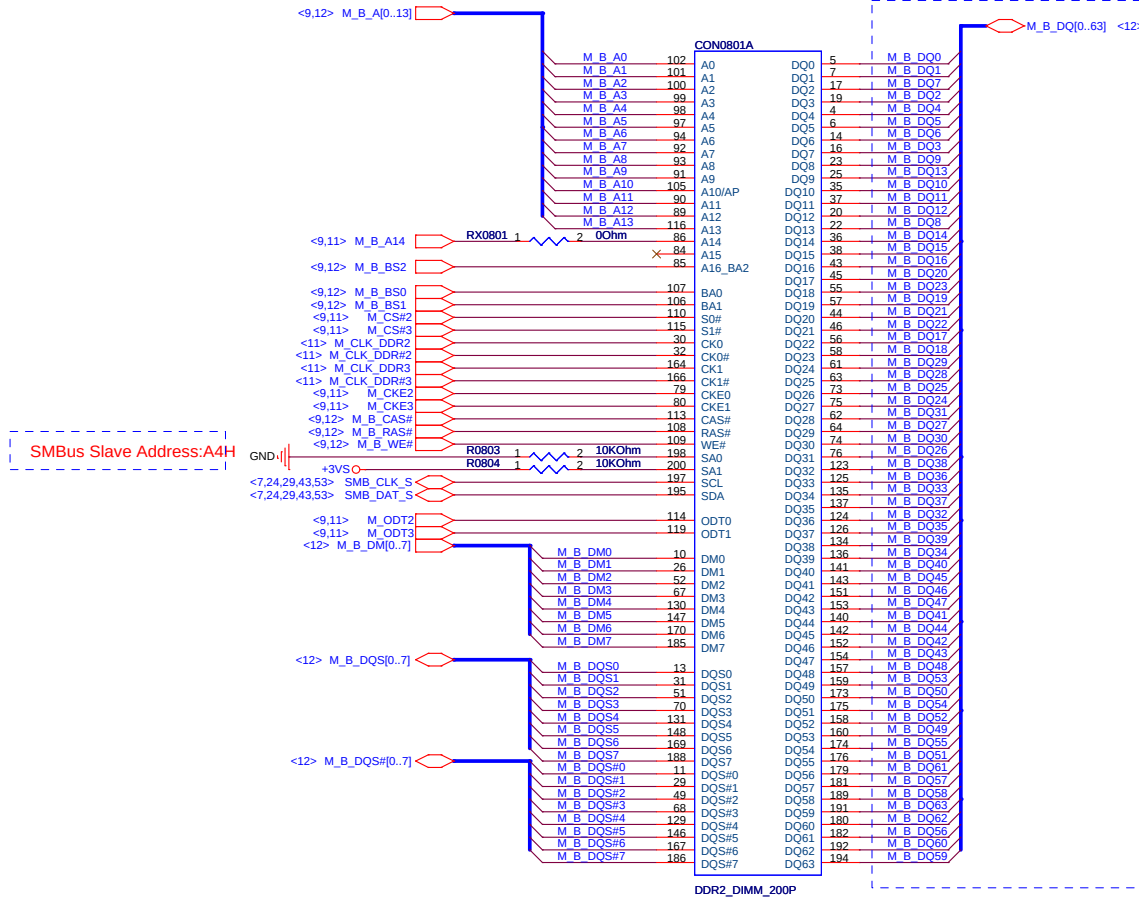
P/N: 12G02502200E
DIMM H=4mm Conn (T53S)



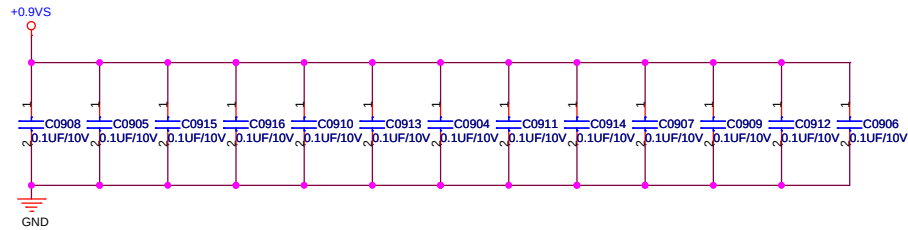
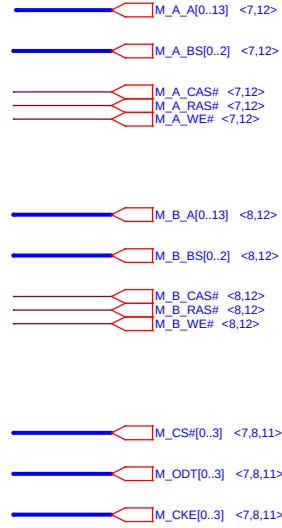
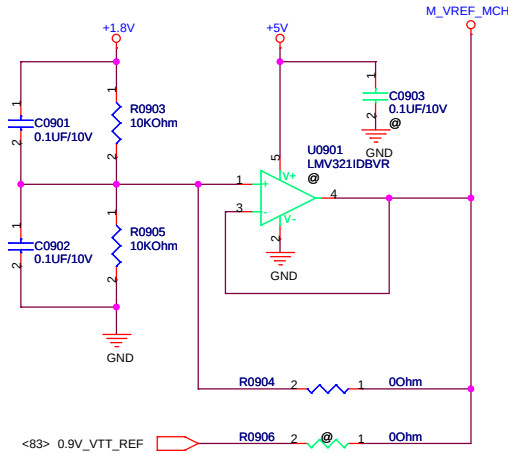
DIMM0 Tyco (H:8.0mm)

P/N: 12G025M22000
DIMM H=8mm Conn (T53S)

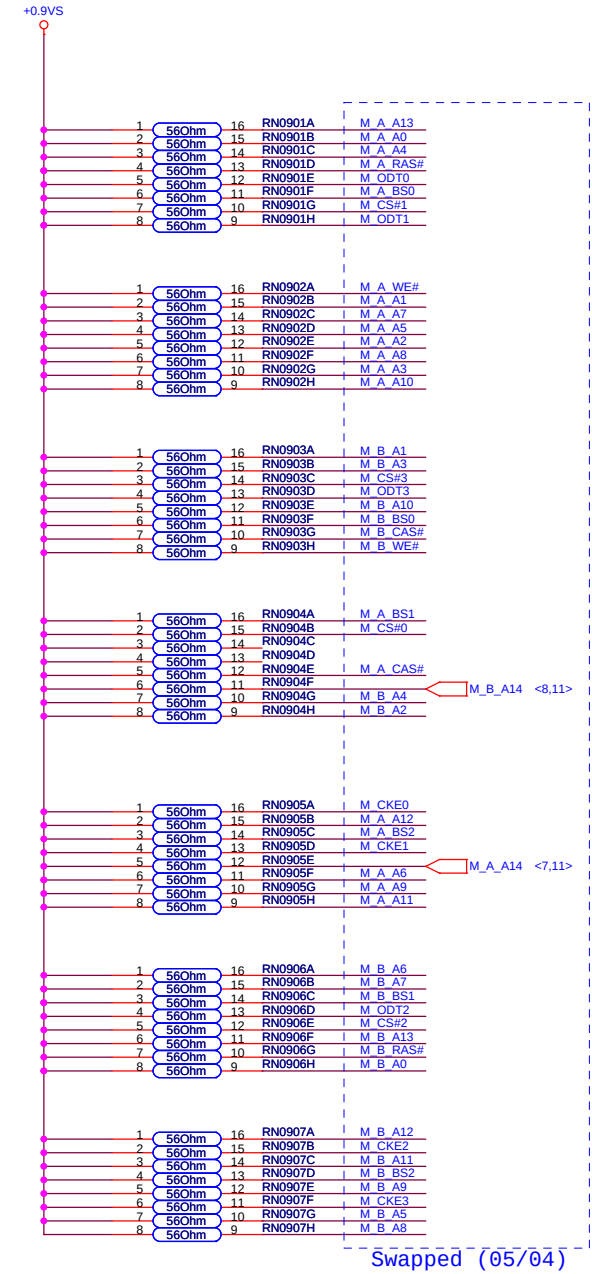
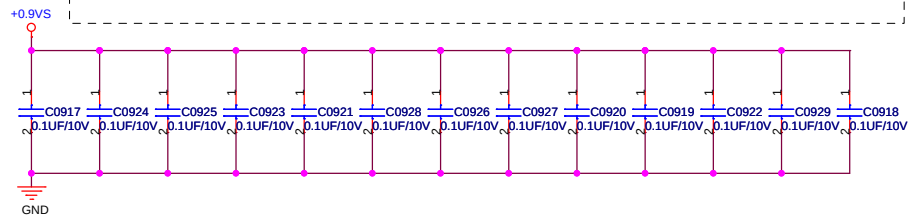
Swapped (05/02)



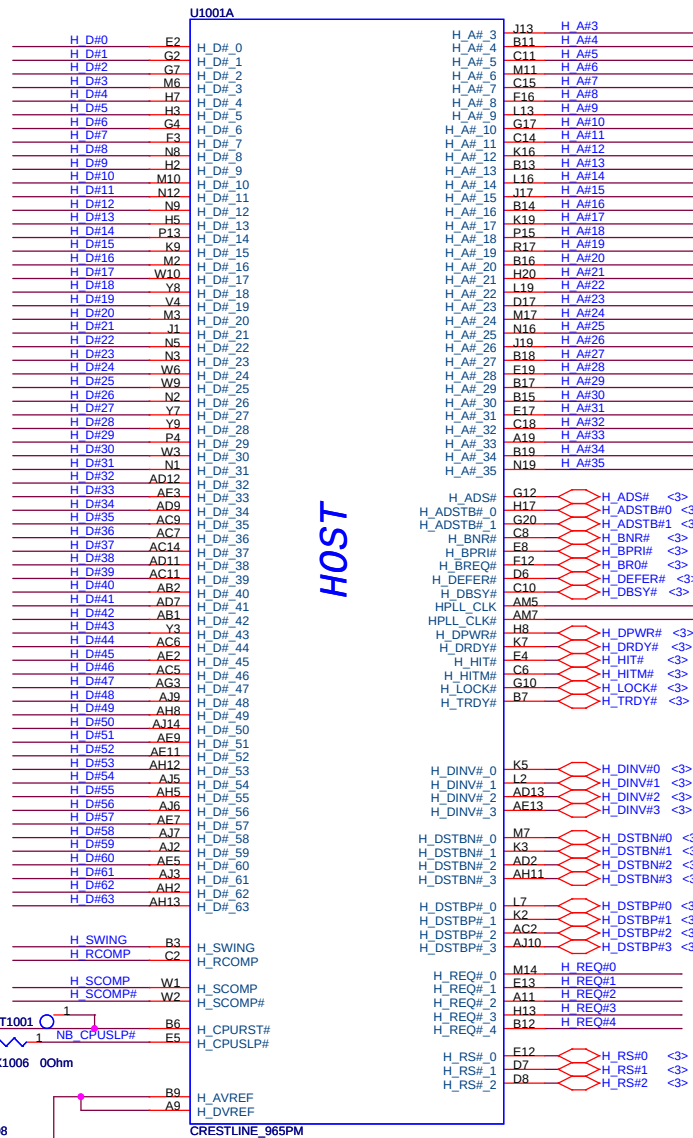
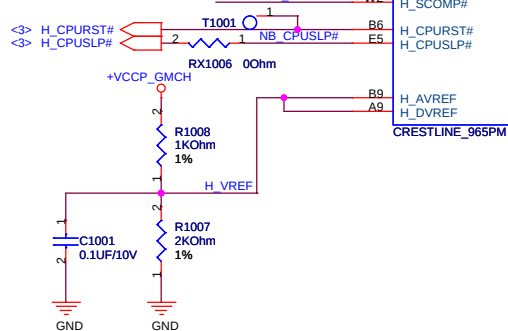
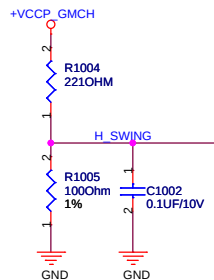
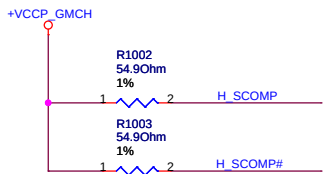
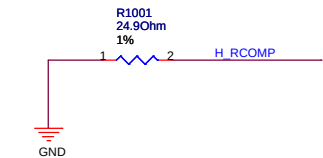
+0.9VS can be controlled to 0.9V or 0.9VS.
Remind PWR EE to reserve SUSB# and SUSC#



Layout note: Place one cap close to every 2 pullup resistors terminated to +0.9VS



Swapped (05/04)

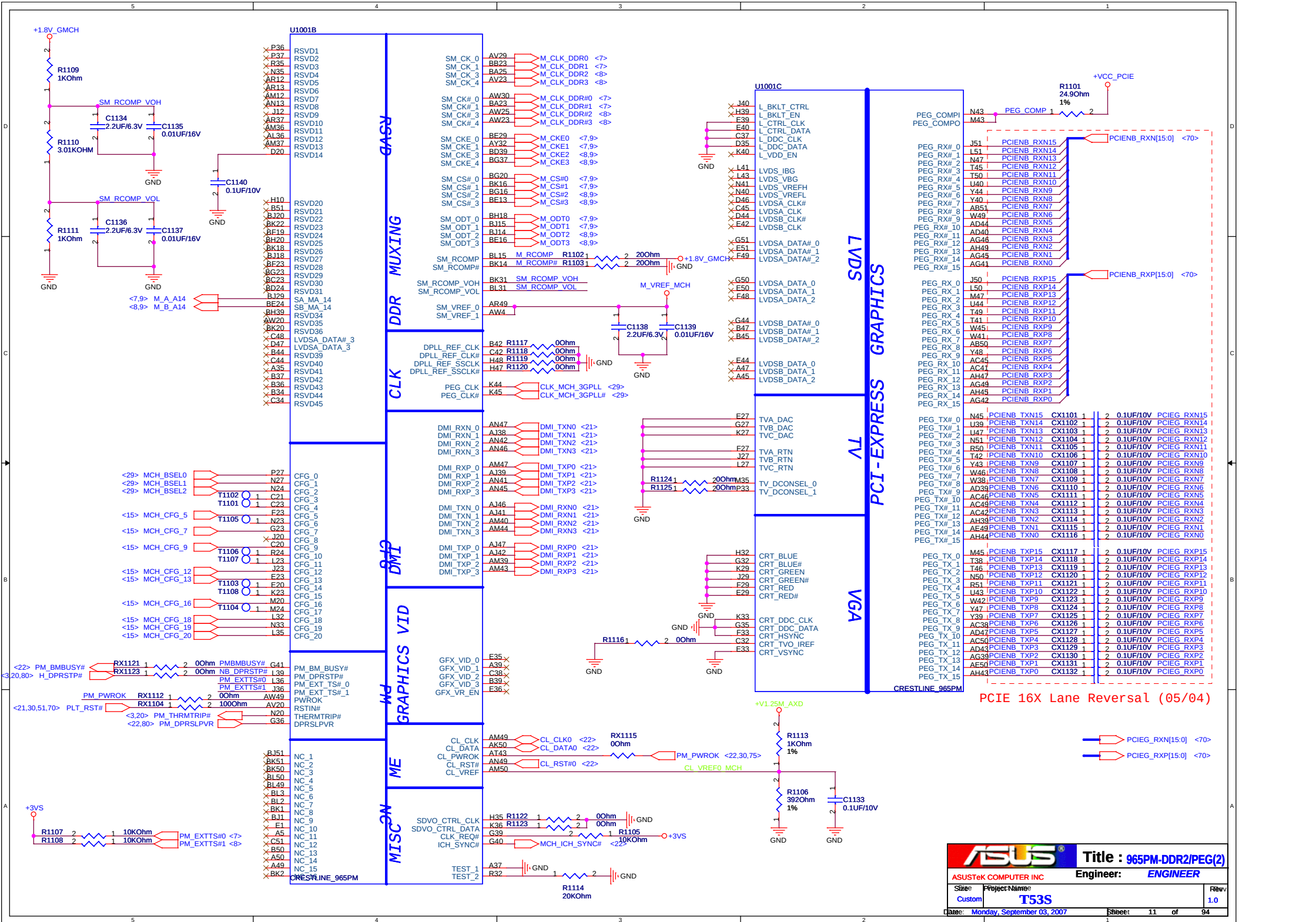


<3> H_A#[35:3] H_A#[35:3]

<3> H_REQ#[4:0] H_REQ#[4:0]

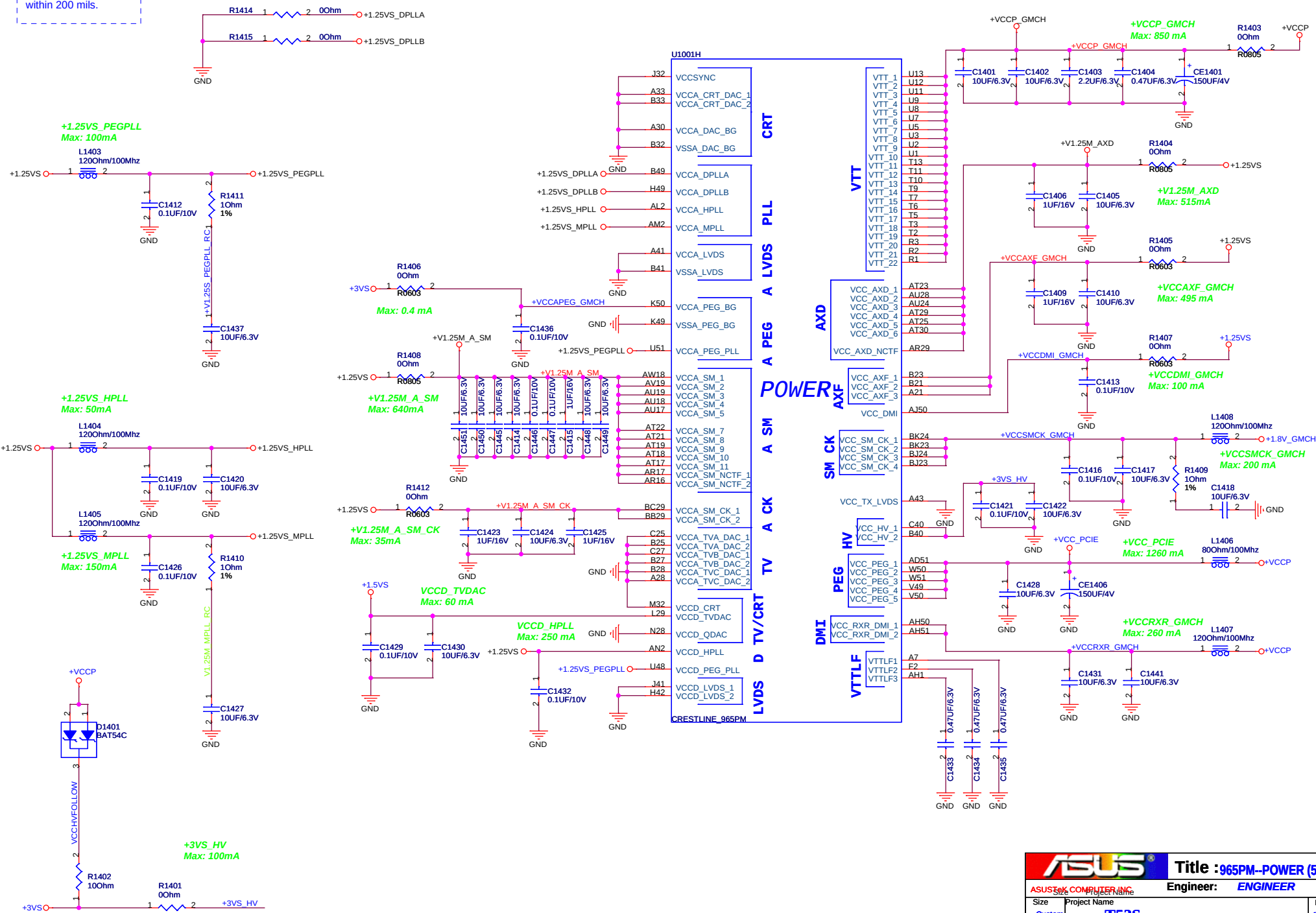
<3> H_D#[63:0] H_D#[63:0]

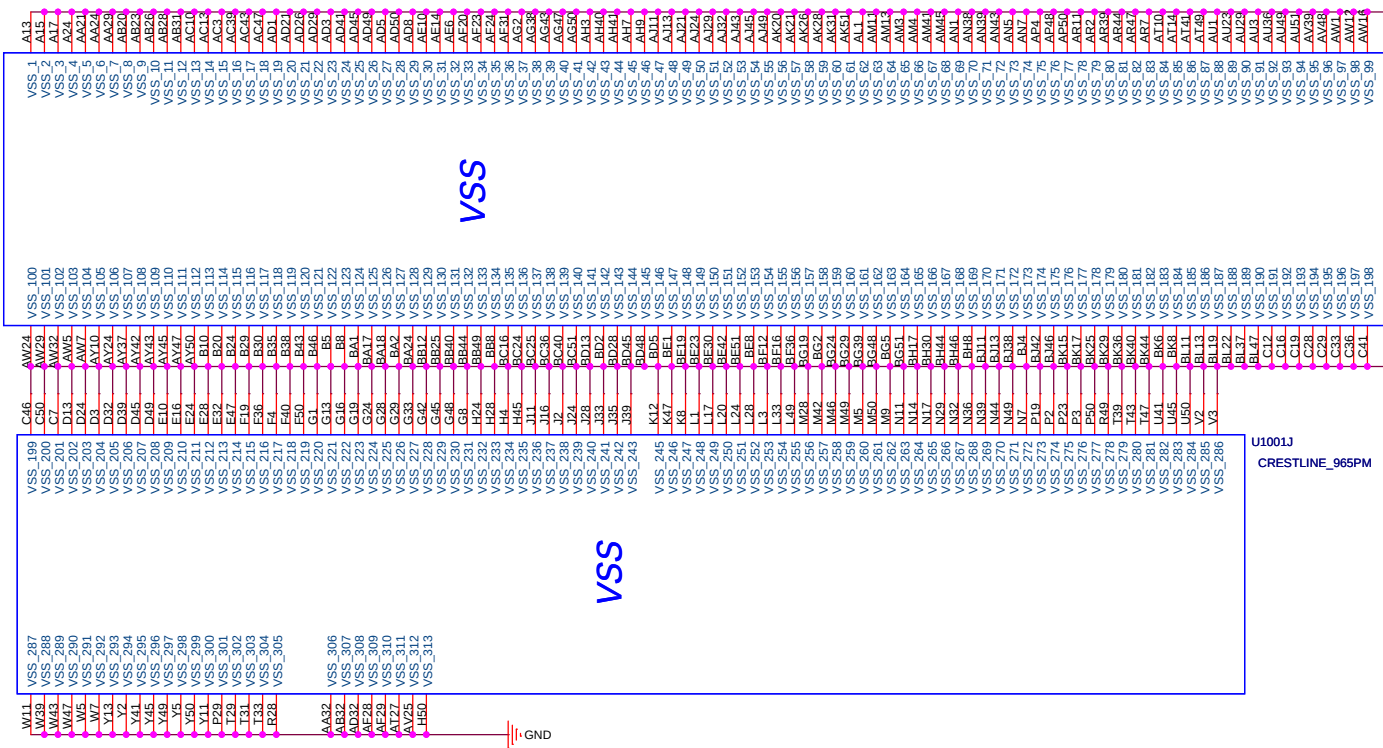
CLK_MCH_BCLK <29>
CLK_MCH_BCLK <29>





NOTE 0.1uF caps in 1.5VS_XPLL need to be located as edge caps within 200 mils.





CFG [13:12] : XOR/ALL-Z
00 = Reserved
01= XOR Mode Enabled
10= All-Z Mode Enabled
11= Normal Operation (Default)

<11> MCH_CFG_5 R1502 4.02KOHM @
CFG5 : DMI STRAP
LOW = DMI X 2
HIGH = DMI X 4 (Default)

<11> MCH_CFG_7 R1504 4.02KOHM @
CFG7 : CPU STRAP
LOW = Reserved
HIGH = Mobile CPU (Default)

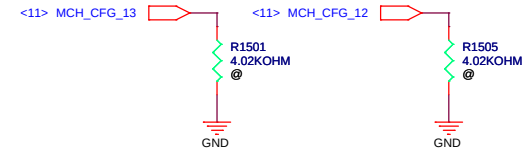
<11> MCH_CFG_9 R1507 4.02KOHM @
CFG9 : PCIE GRAPHIC LANE
LOW = REVERSE LANE
HIGH = Normal Operation (Default)
(05/04)

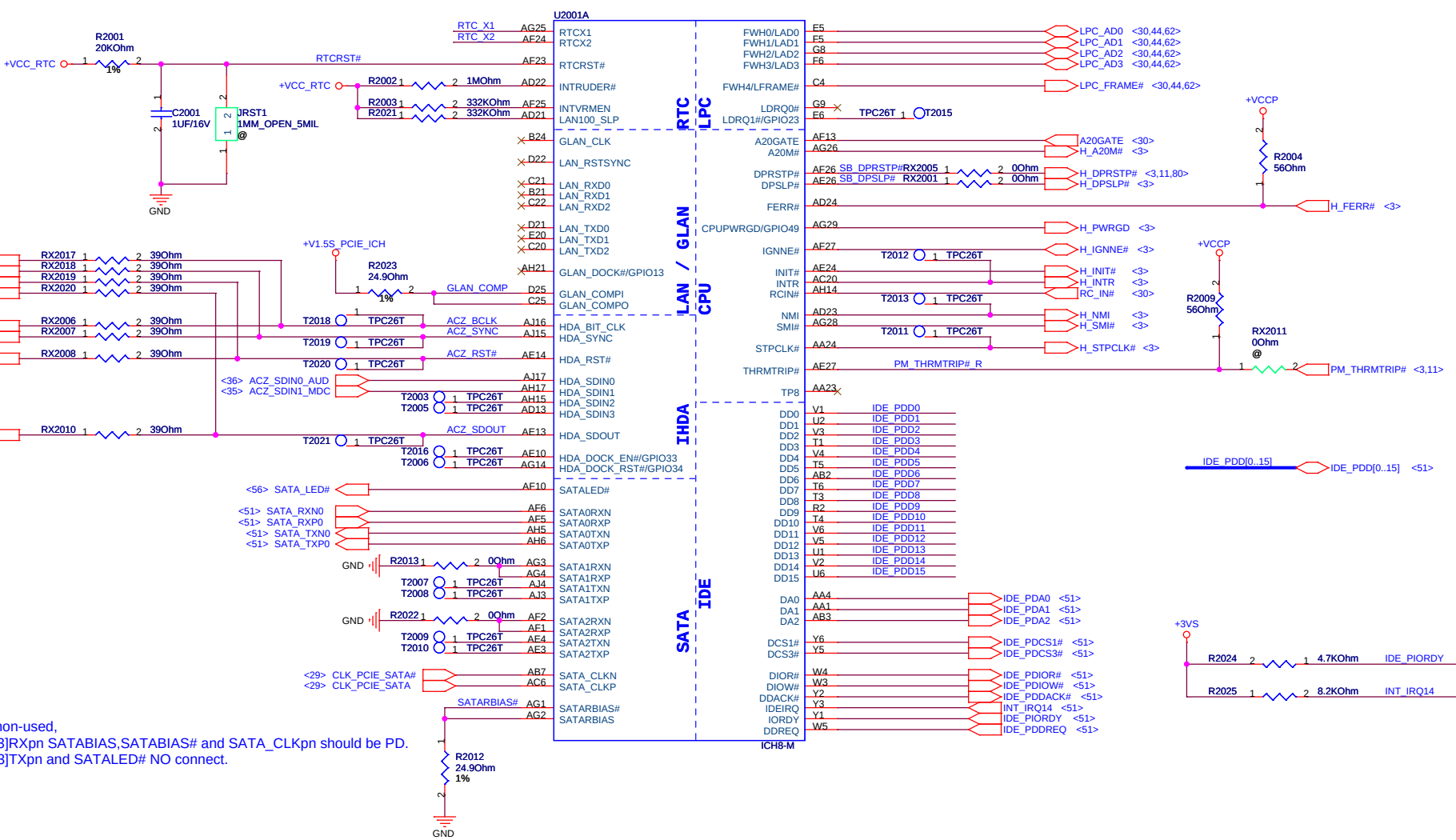
<11> MCH_CFG_16 R1508 4.02KOHM @
CFG16 : Dynamic ODT STRAP
Low = Dynamic ODT Disabled
HIGH = Dyanamic ODT Enabled (default)

<11> MCH_CFG_18 R1519 4.02KOHM @
CFG18 : VCC SELECT
LOW = 1.05V (default)
HIGH = 1.5V

<11> MCH_CFG_19 R1503 4.02KOHM @
CFG19 : DMI LANE REVERSAL
LOW = NORMAL (default)
HIGH = LANES REVERSED

<11> MCH_CFG_20 R1506 4.02KOHM @
CFG20 : SDVO/PCIE CONCURRENT MODE
LOW = ONLY SDVO or PCIE is Operational
HIGH = SDVO and PCIE are operating simultaneously via the PEG port

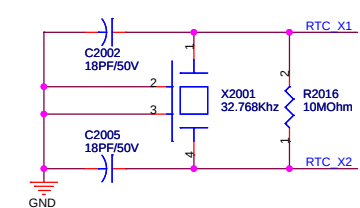
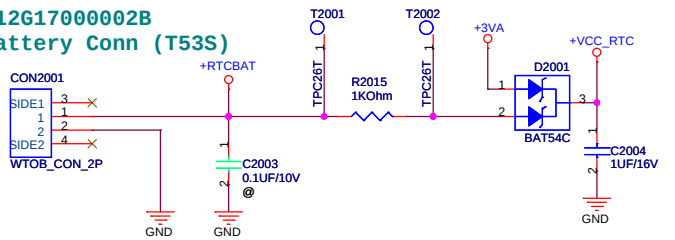


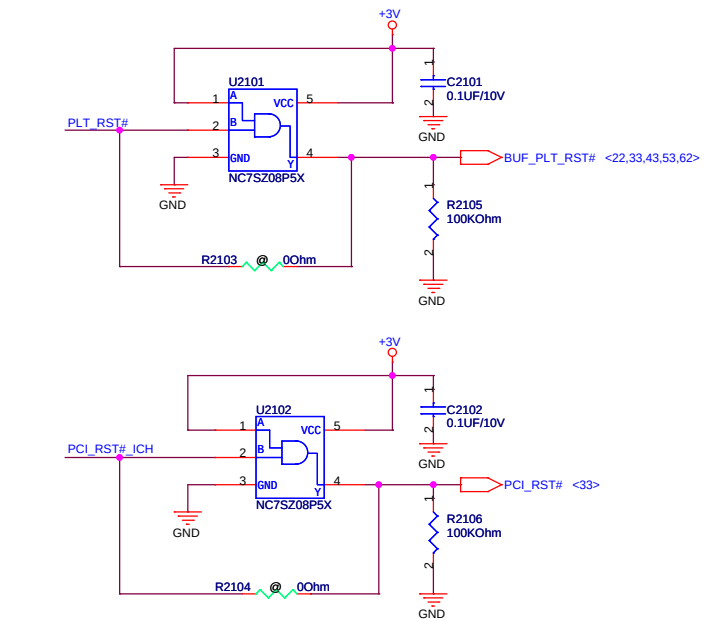
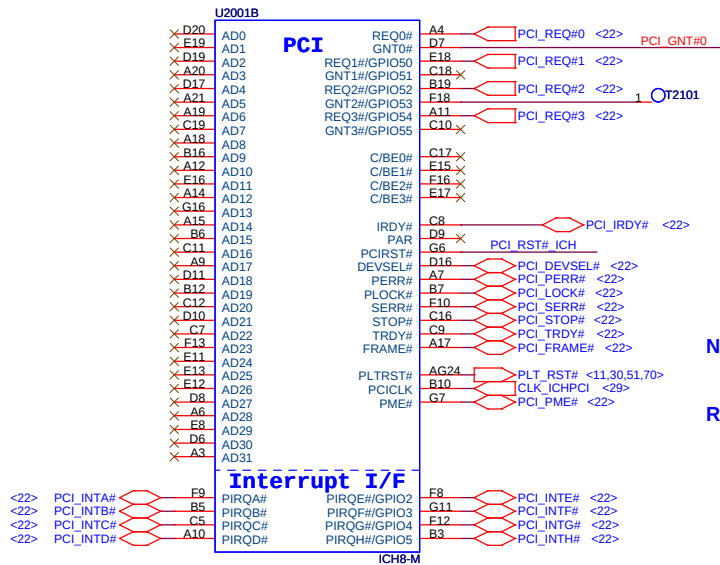


SATA if it non-used,
1)SATA[0:3]RXpn SATABIAS,SATABIAS# and SATA_CLKpn should be PD.
2)SATA[0:3]TXpn and SATALED# NO connect.

RTC BAT

P/N: 12G17000002B
RTC Battery Conn (T53S)





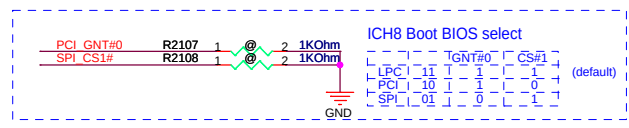
LAN

WLAN

NEWCARD

ROBSON

USB Devices	
Port 0	USB Port
Port 1	USB Port
Port 2	USB Port
Port 3	Card Reader
Port 4	NewCard
Port 5	BlueTooth
Port 6	CMOS Camara
Port 7	FingerPrint
Port 8	MiniCard (3G)
Port 9	N/A

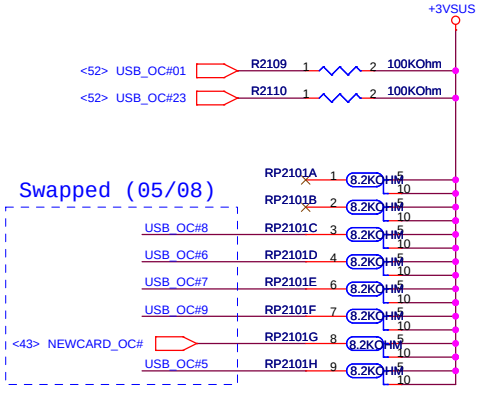
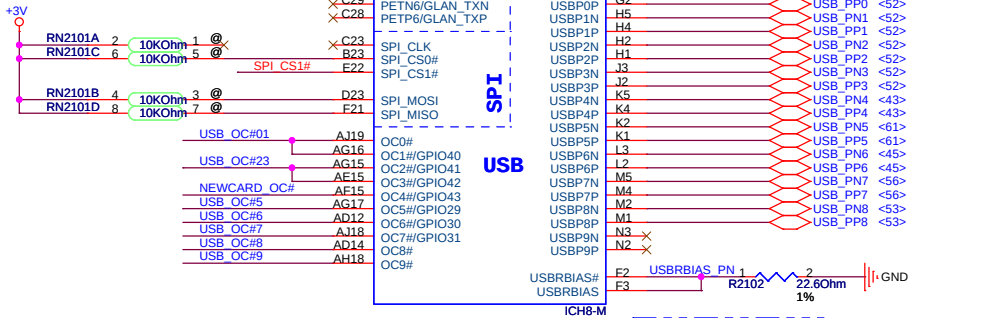


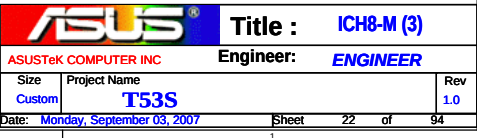
PCI-Express

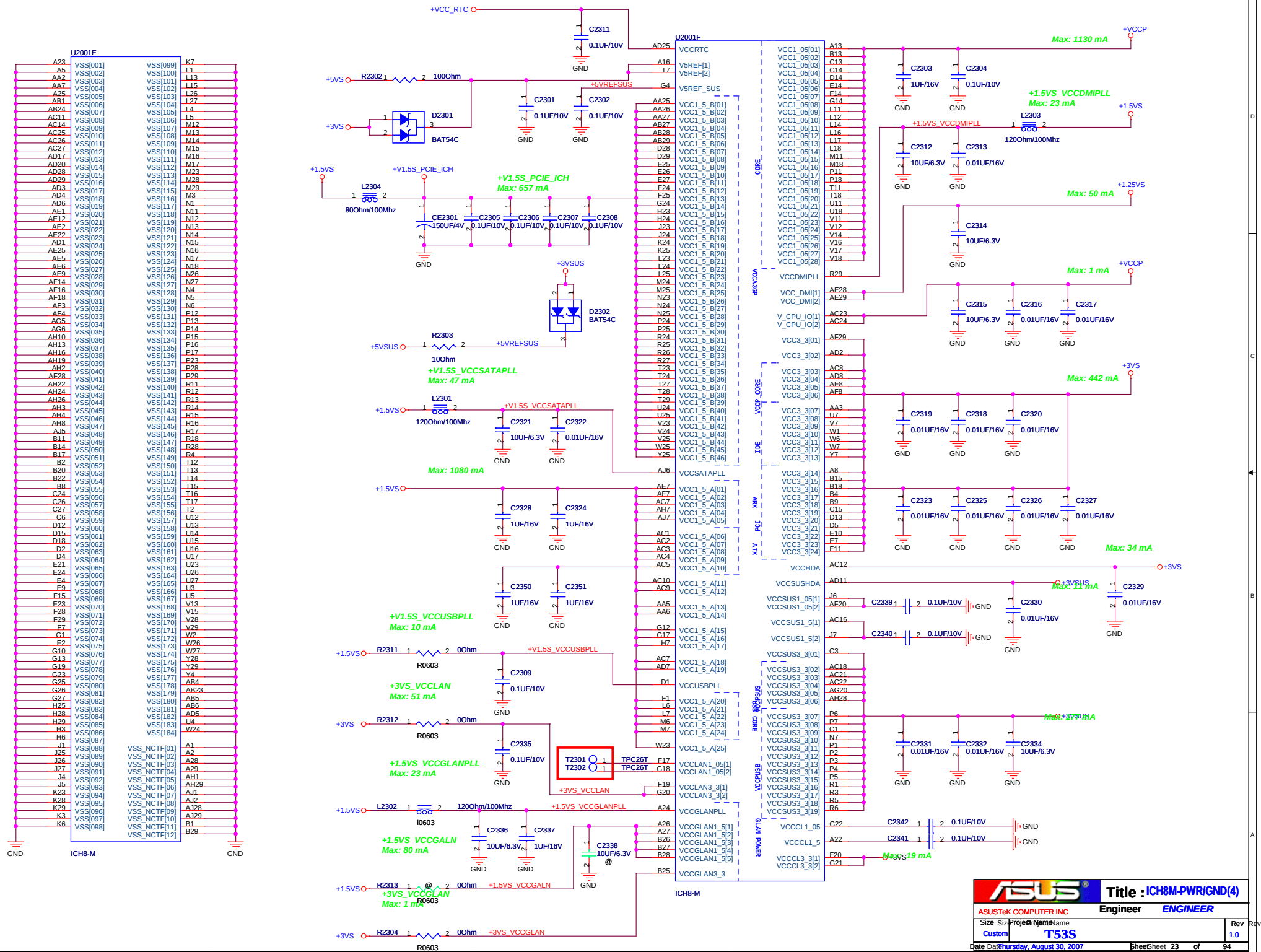
Direct Media Interface

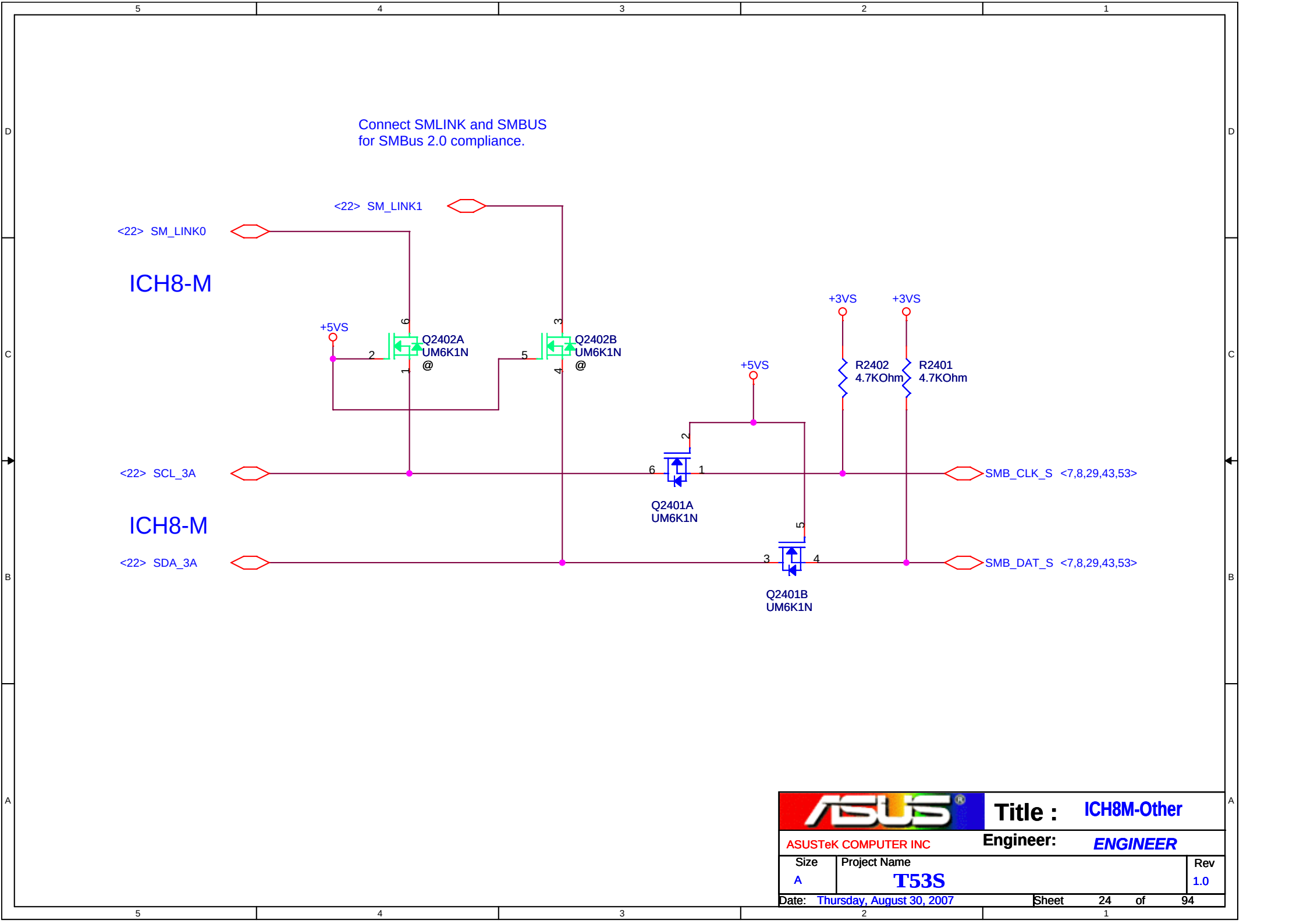
SPI

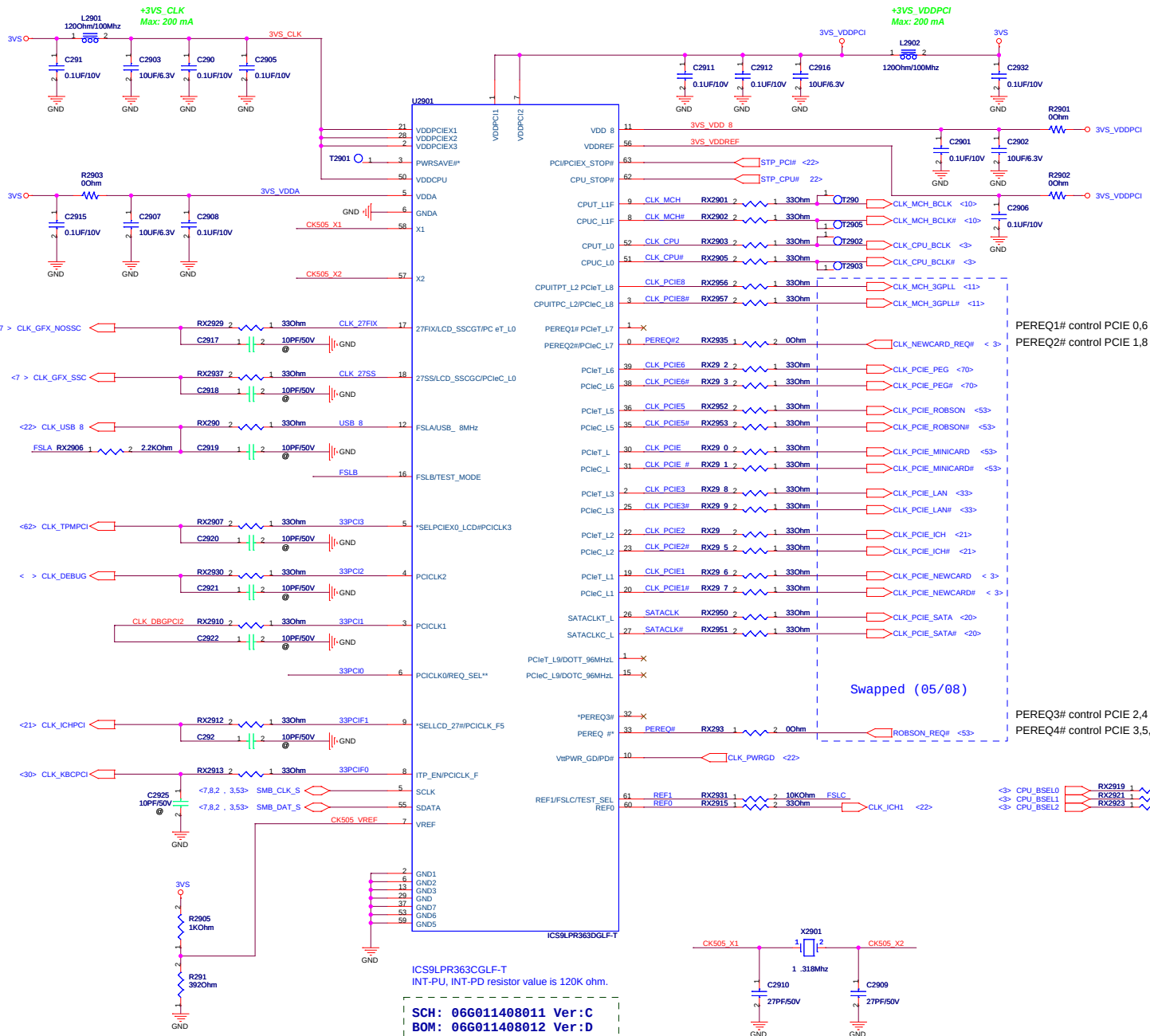
USB







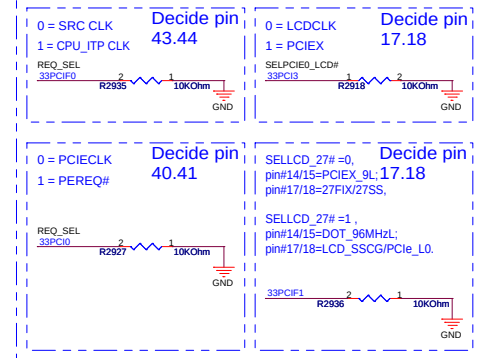




PEREQ#3
0 = Enable control PCIeX4/2 through I2C
1 = Disable PCIeX4/2 Controlled

PEREQ#4
0 = Enable control PCIeX7/5/3 through I2C
1 = Disable PCIeX7/5/3 Controlled

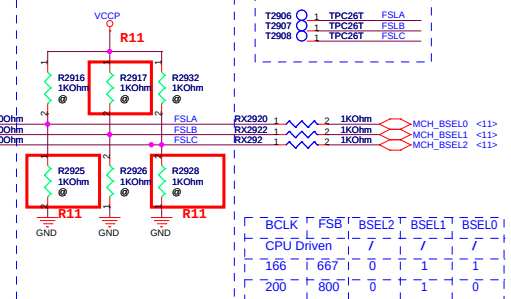
Latched Input Select



Swapped (05/08)

PEREQ3# control PCIE 2,4
PEREQ4# control PCIE 3,5,7

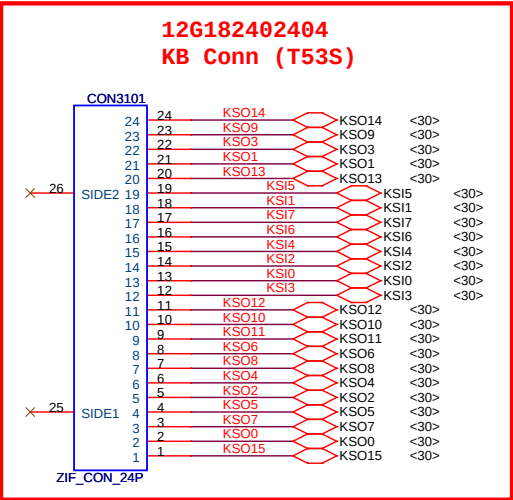
For ATS



SCH: 06G011408011 Ver:C
BOM: 06G011408012 Ver:D

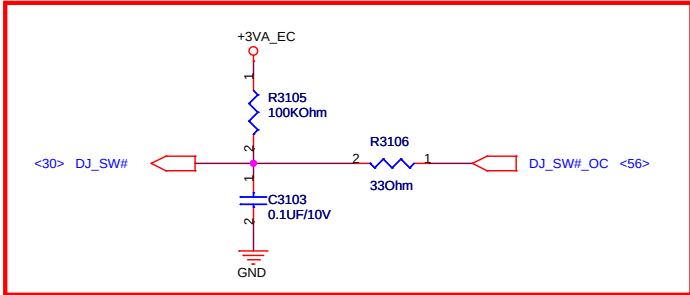
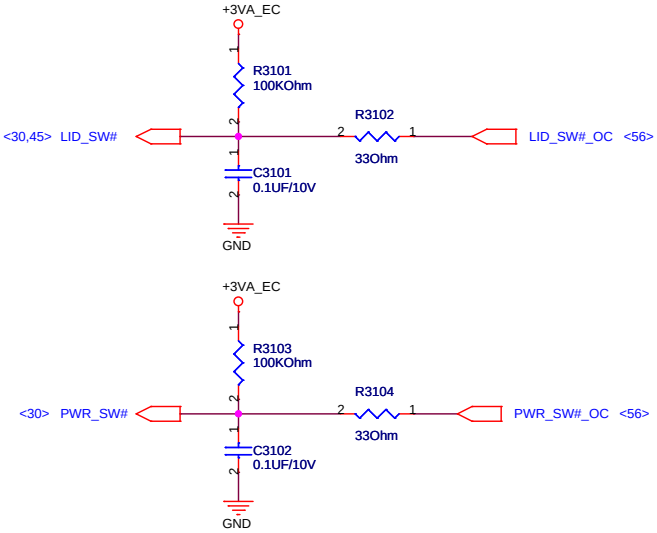
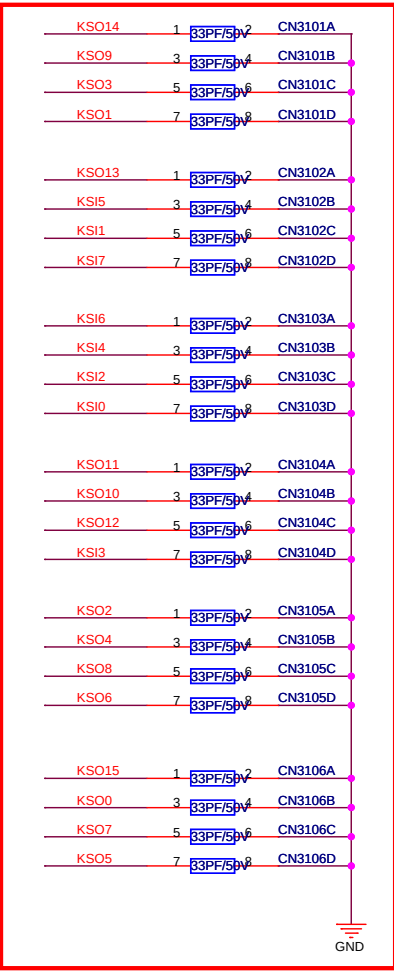
Keyboard Conn.

R11 M??

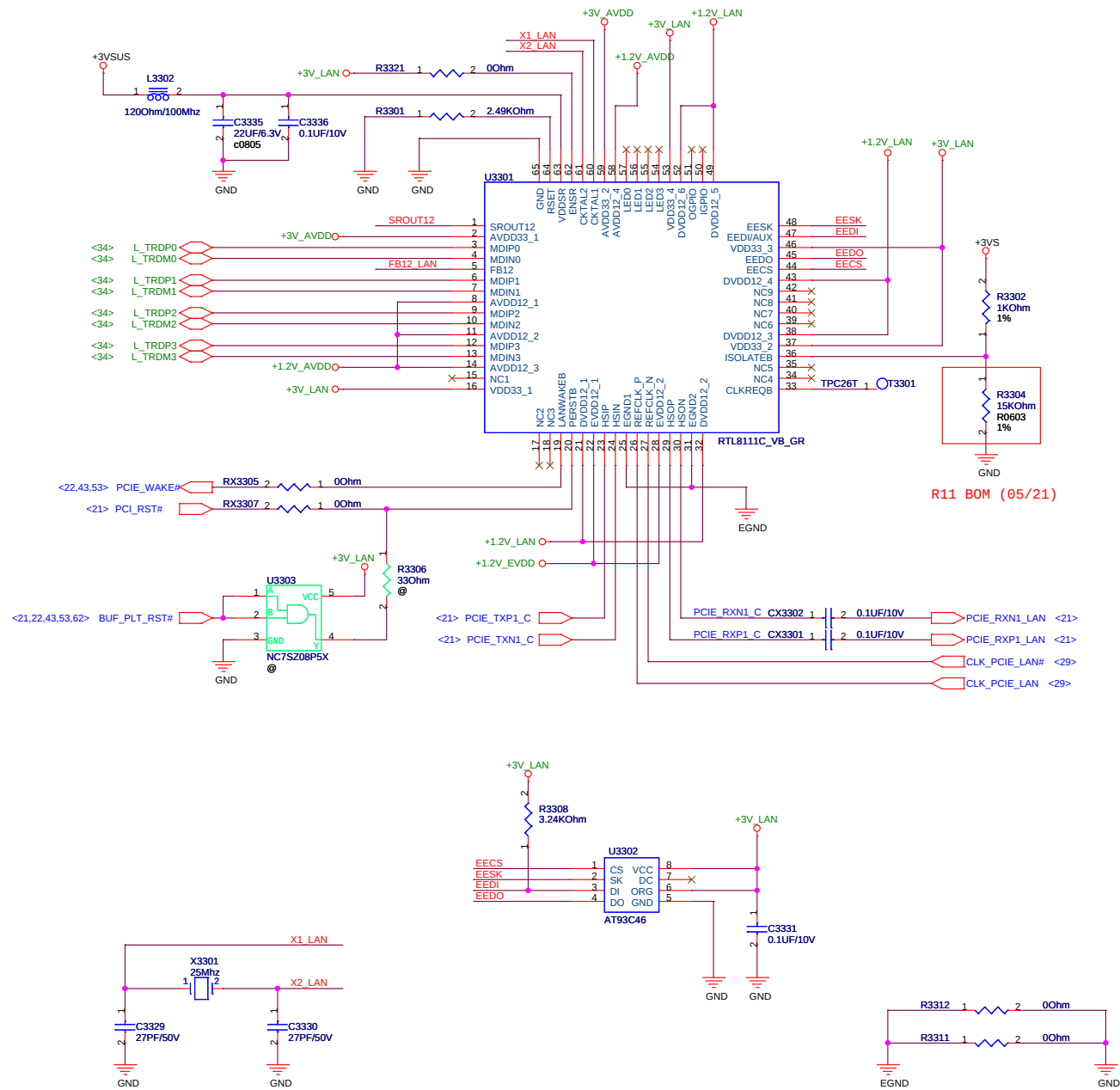


Follow ASUS Standard KB Matrix (T14)

R11 For 3G WWAN



R11 M??



0.1uF are for VDD33
pins 16, 37, 46 and 53.

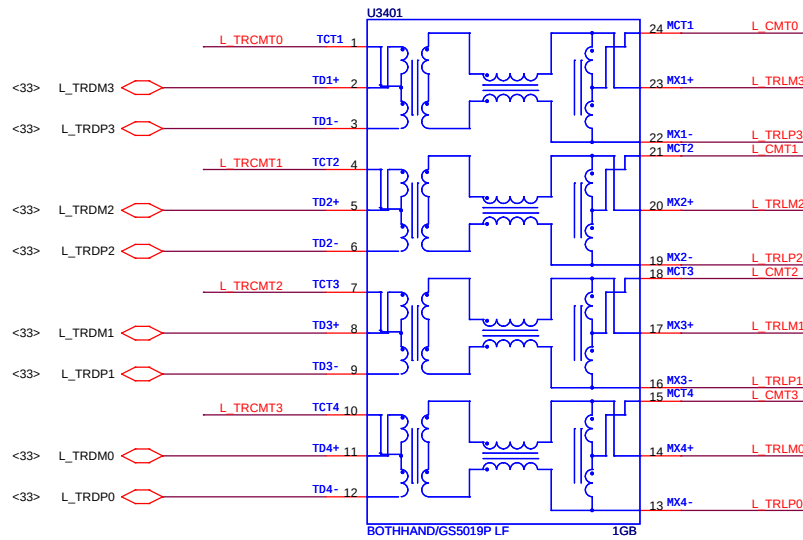
0.1uF are for AVDD33
pins 2 and 59.

0.1uF are for DVDD12
pins 21, 32, 38, 43, 49 and 52.

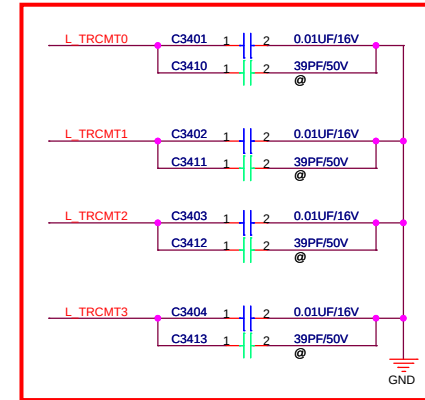
0.1uF are for AVDD12
pins 8, 11, 14 and 58

0.1uF are for EVDD12
pins 22 and 28

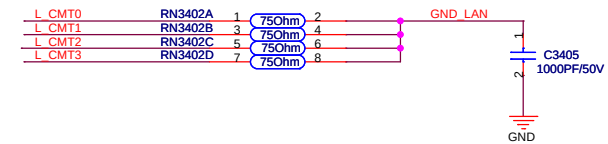
LAN Transformer 1GB (H:2.4mm)



L_TRLM2	1	00hm	2	RNX3401A	L_TRLM2
L_TRLP2	3	00hm	4	RNX3401B	L_TRLP2
L_TRLM0	5	00hm	6	RNX3401C	L_TRLM0
L_TRLP0	7	00hm	8	RNX3401D	L_TRLP0
L_TRLP1	1	00hm	2	RNX3403A	L_TRLP1
L_TRLM1	3	00hm	4	RNX3403B	L_TRLM1
L_TRLM3	5	00hm	6	RNX3403C	L_TRLM3
L_TRLP3	7	00hm	8	RNX3403D	L_TRLP3

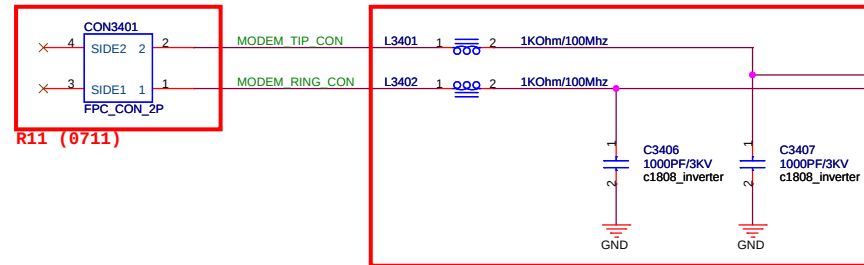


R1.1 For 3G WWAN



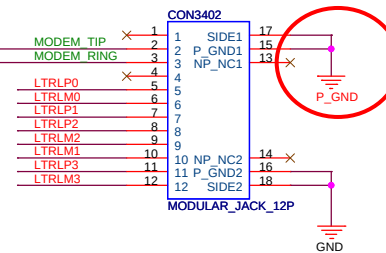
MDC Conn.

12G171010023
MDC Conn (T53S)

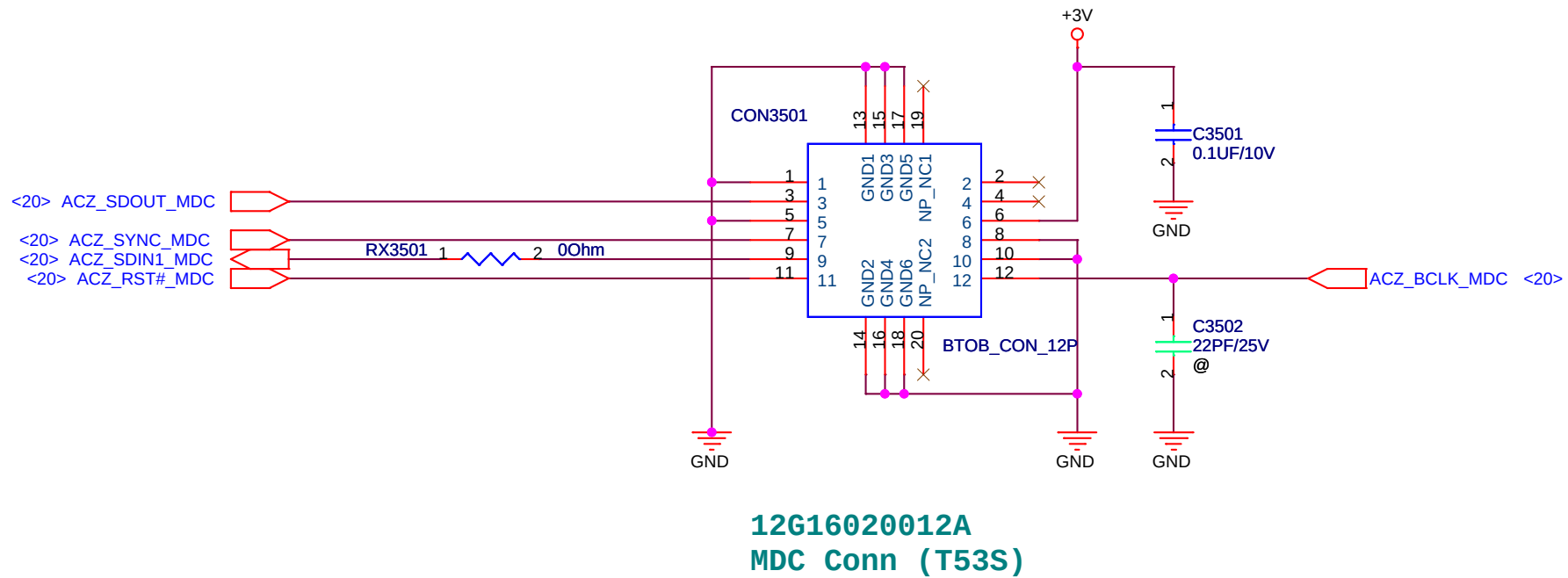


RJ11/RJ45 Jack

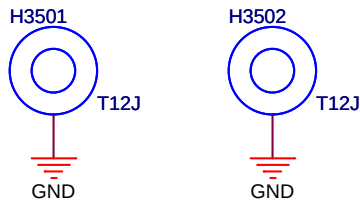
12G142111120
RJ11/RJ45 Conn (T53S)



MDC Connector (H:6mm)

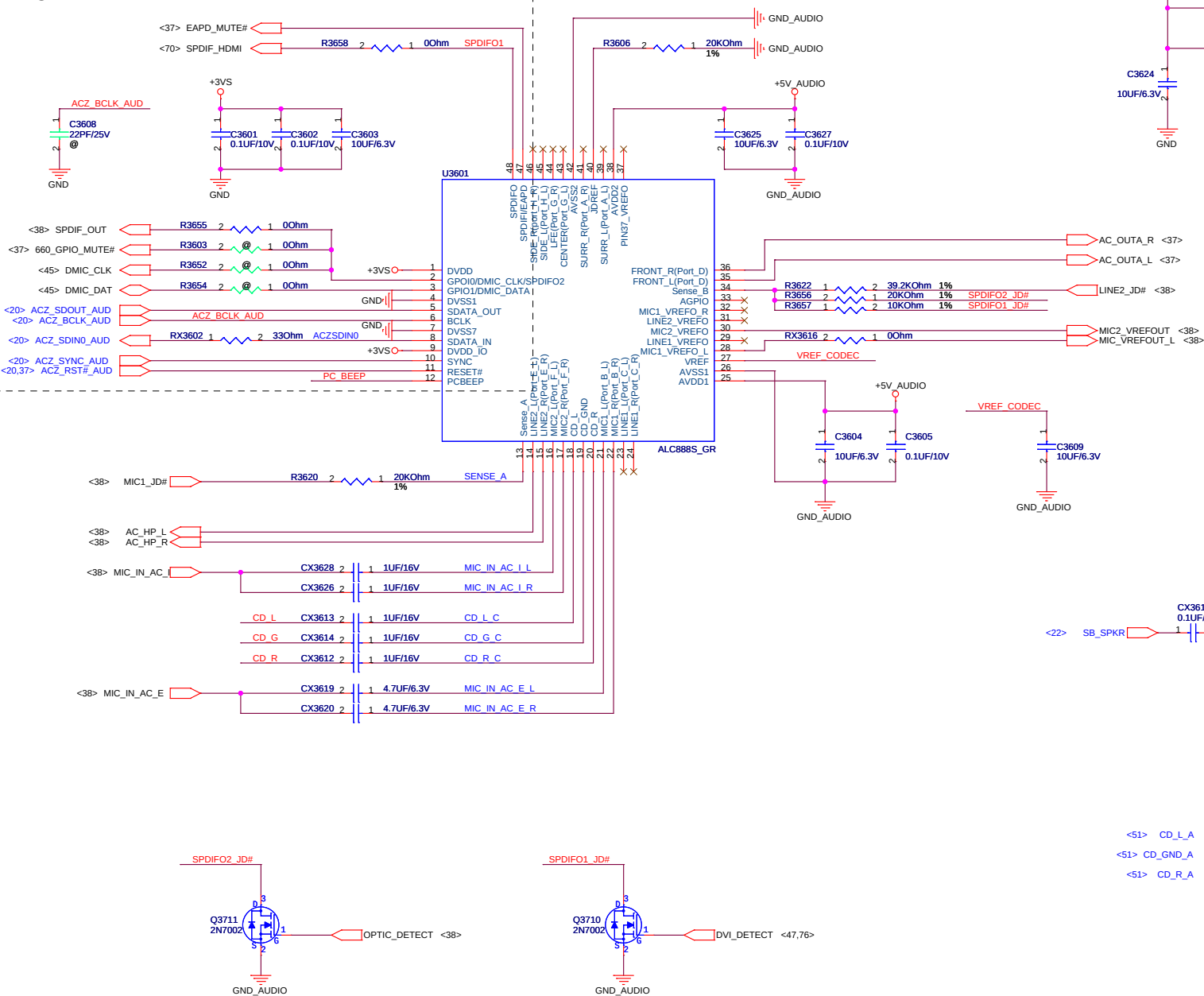


MDC Nut (H:6mm)

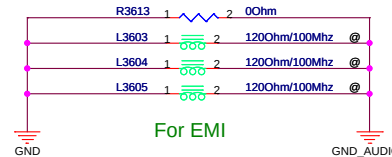
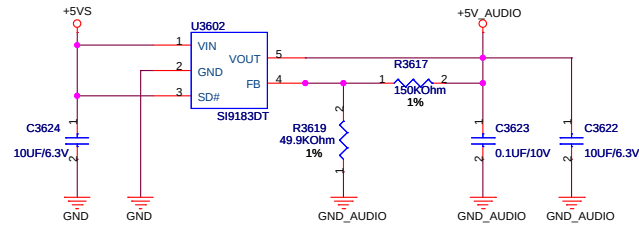


		Title : MDC	
ASUSTeK COMPUTER INC		Engineer: ENGINEER	
Size A	Project Name T53S		Rev 1.0
Date: Thursday, August 30, 2007		Sheet	35 of 94

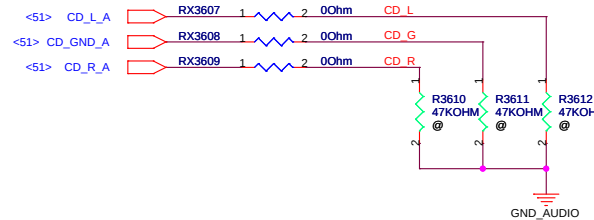
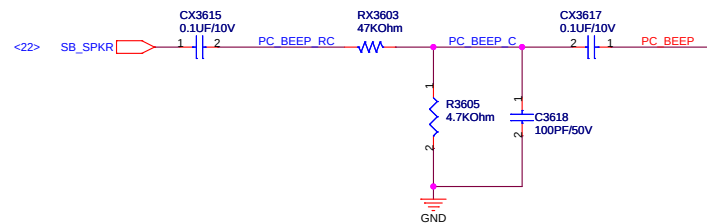
Digital



SI9183DT
 $V_{out} = 1.215 * (1 + (150K / 49.9K)) = 4.86V$

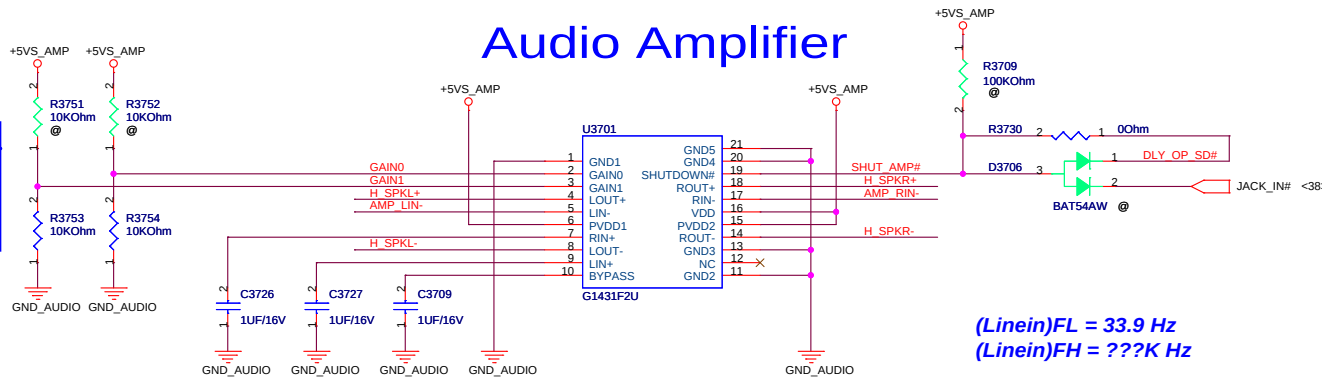


For EMI

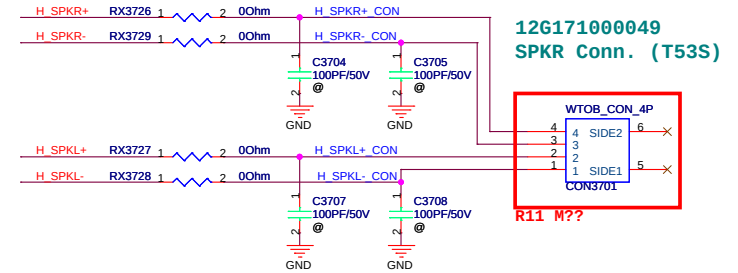
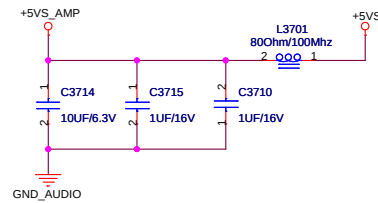
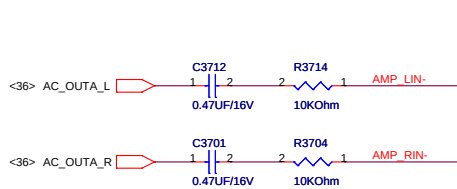


Audio Amplifier

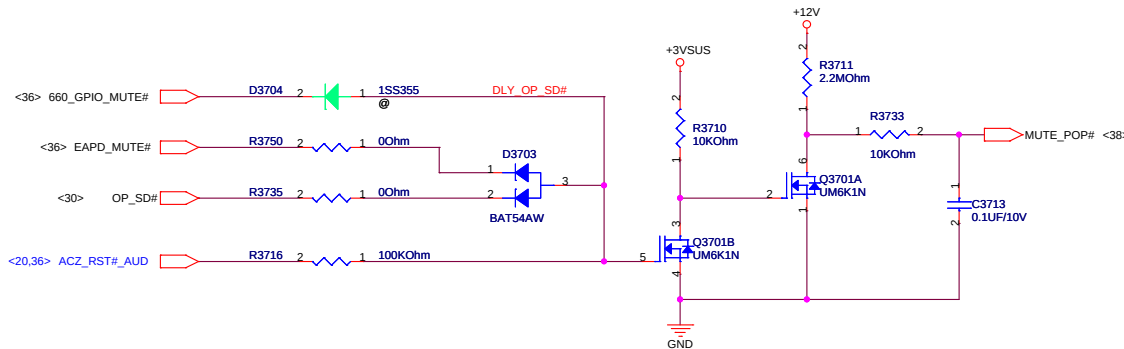
GAIN0	GAIN1	Av(Inv)
0	0	6 dB
0	1	10 dB
1	0	15.6 dB
1	1	21.6 dB



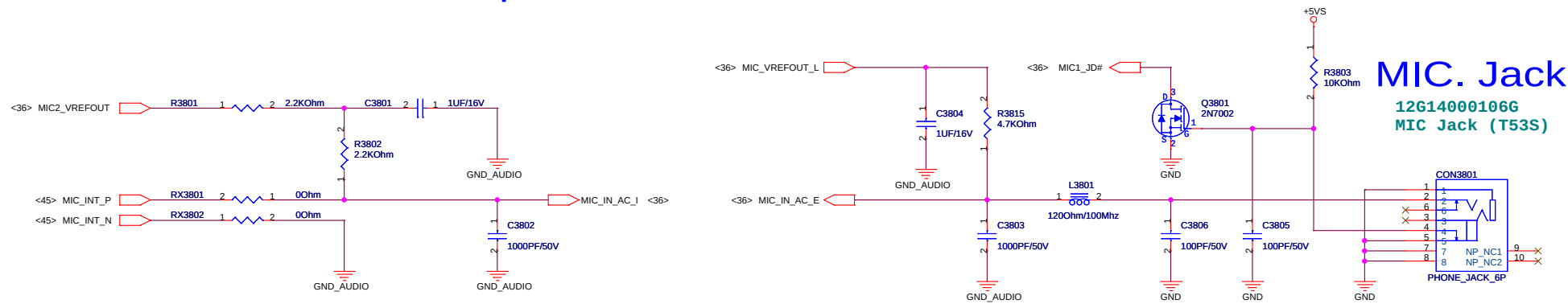
SPKR Conn.



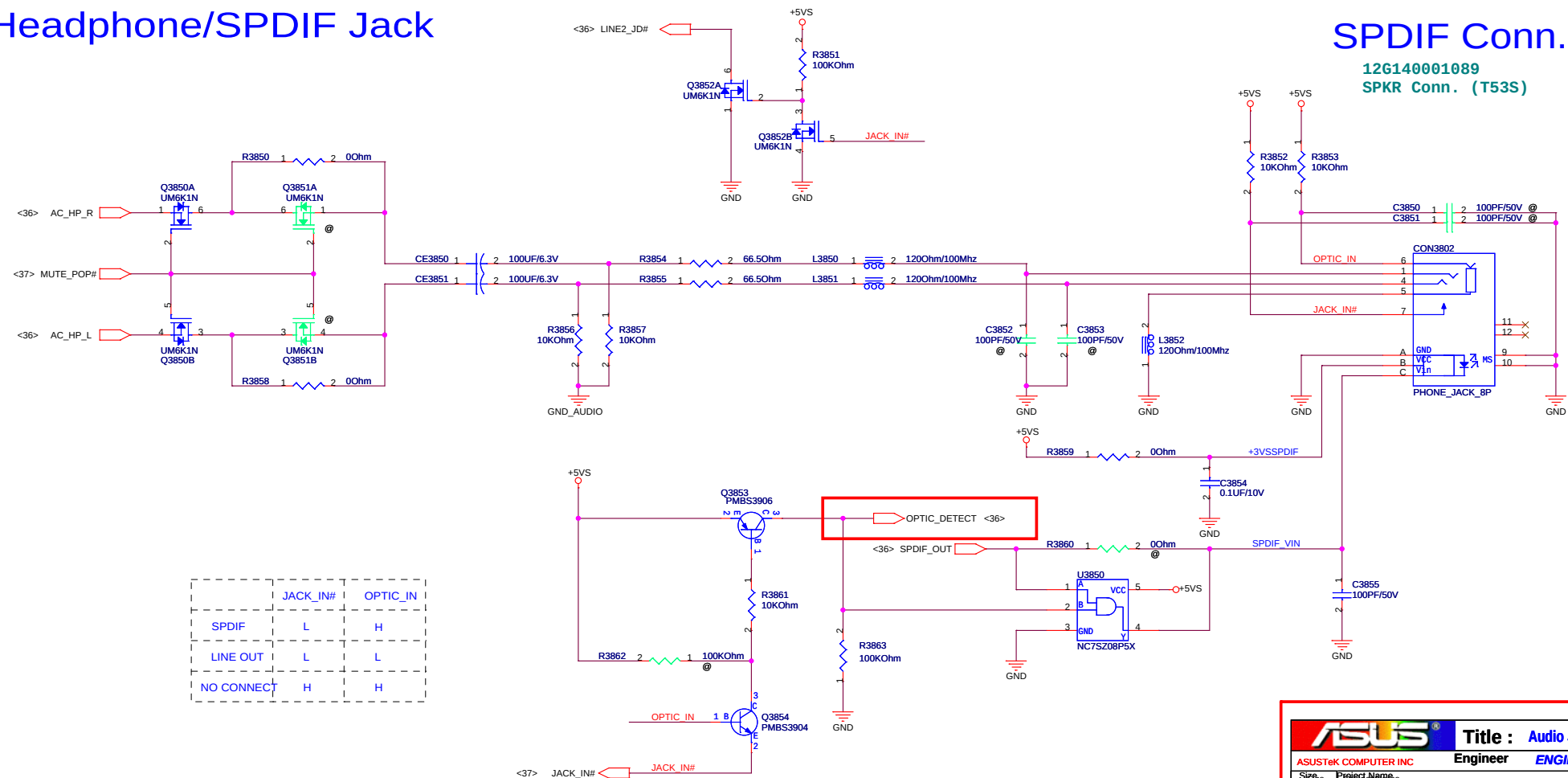
DePOP Circuit

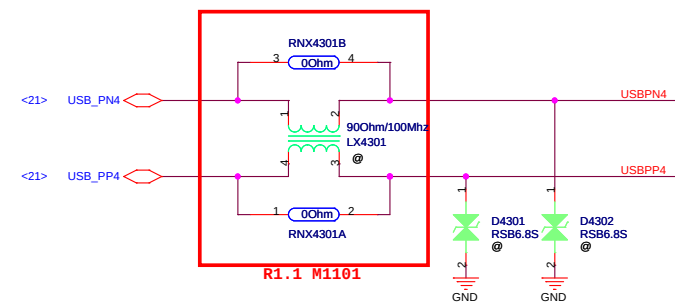
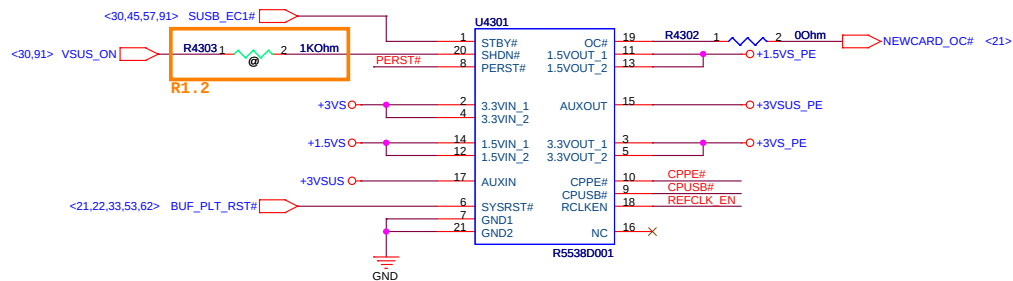


Internal & External Microphones

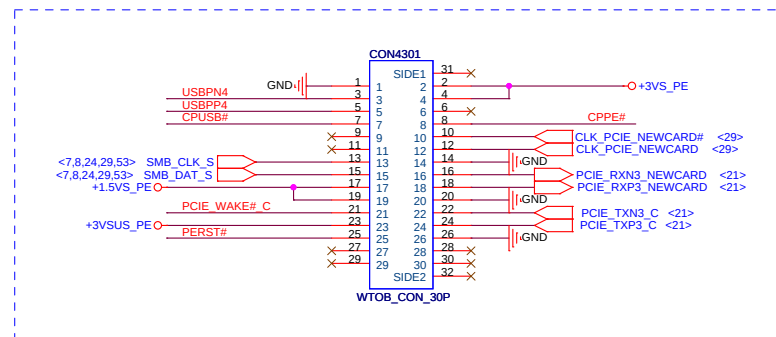


Headphone/SPDIF Jack



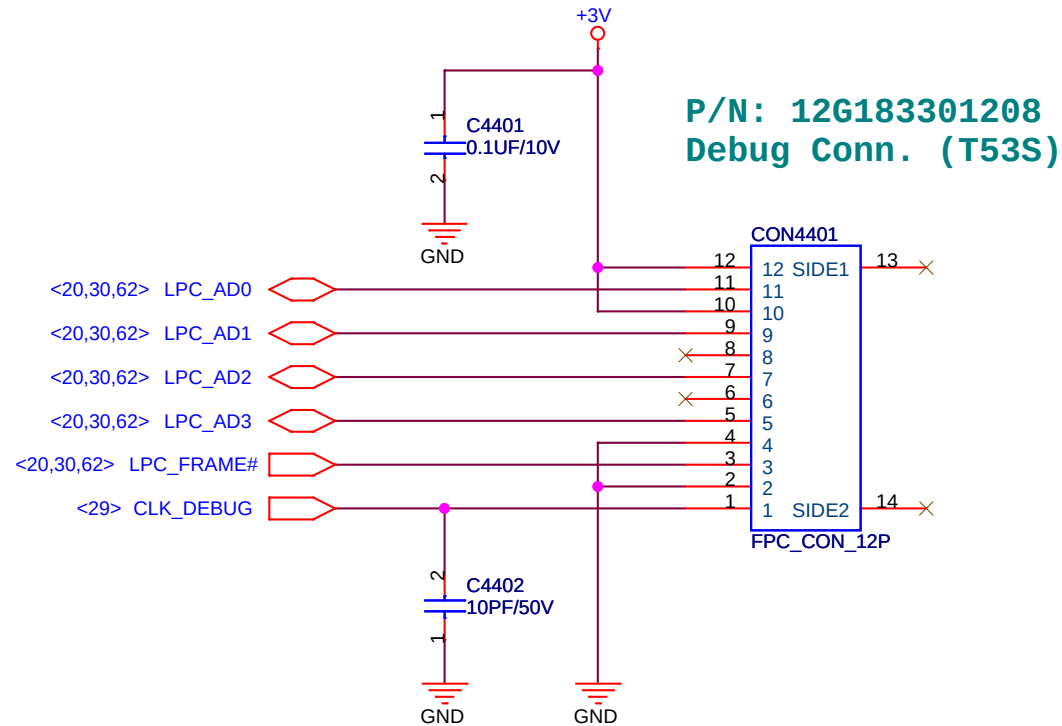


12G171010307
NewCard Conn. (T53S)

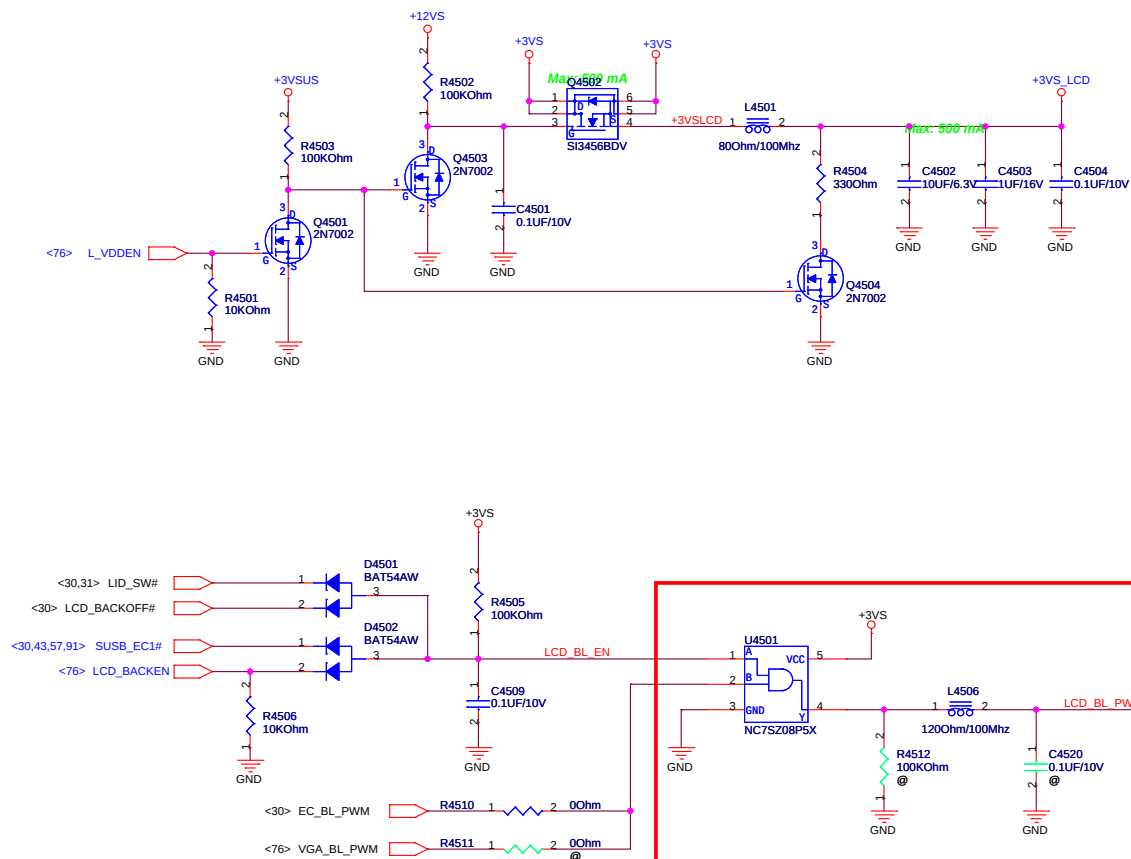


- Update NewCard Conn (05/09)

LPC Debug Port

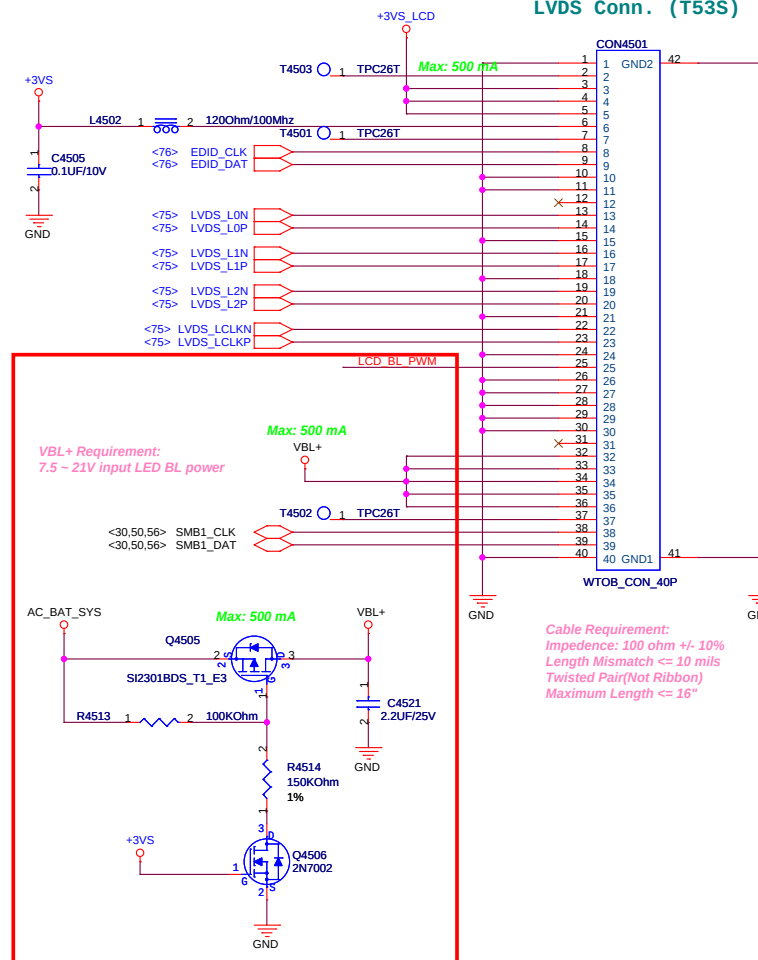


LCD Power

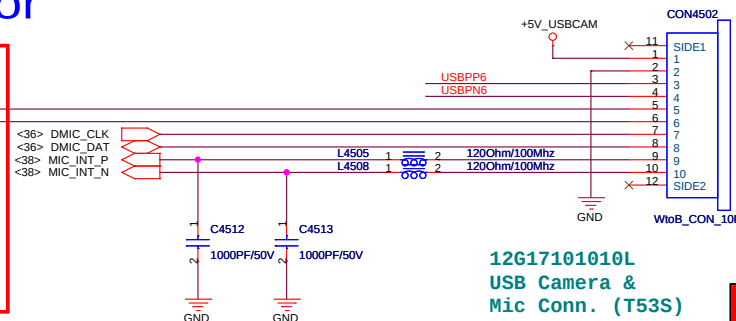
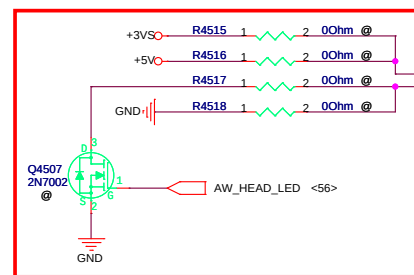
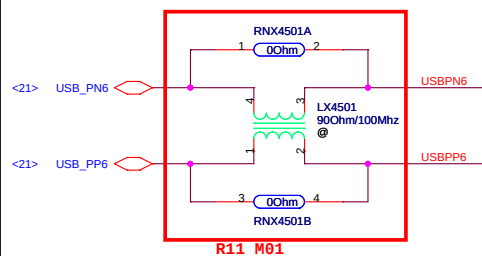


LVDS Connector

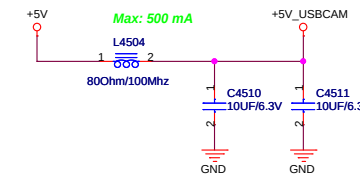
12G171190403
LVDS Conn. (T53S)

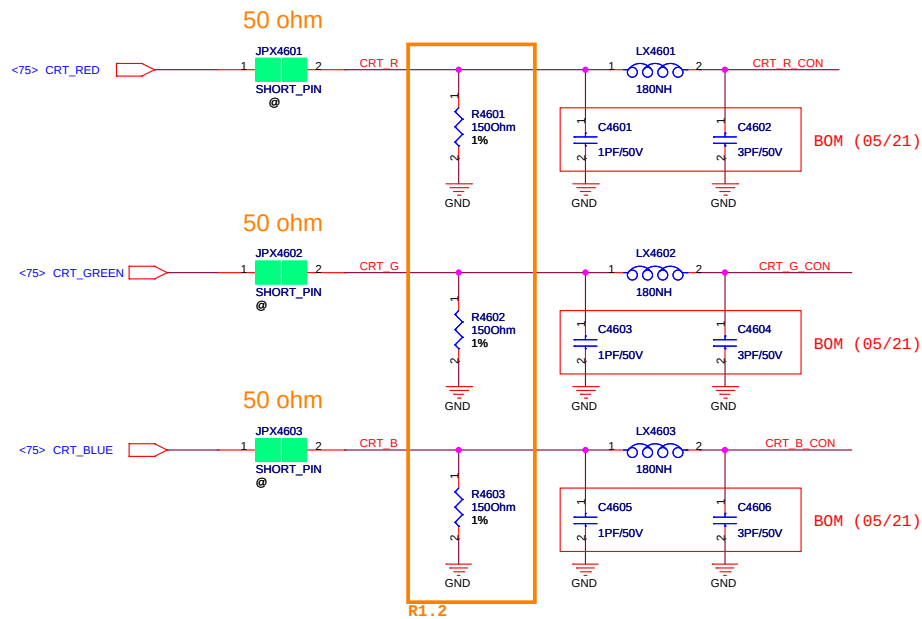


USB Camera & Mic Connector



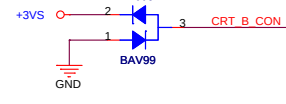
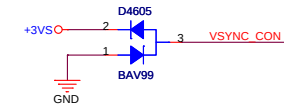
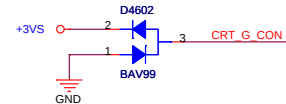
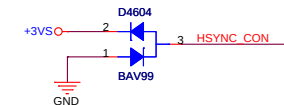
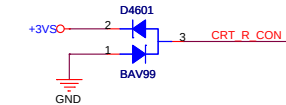
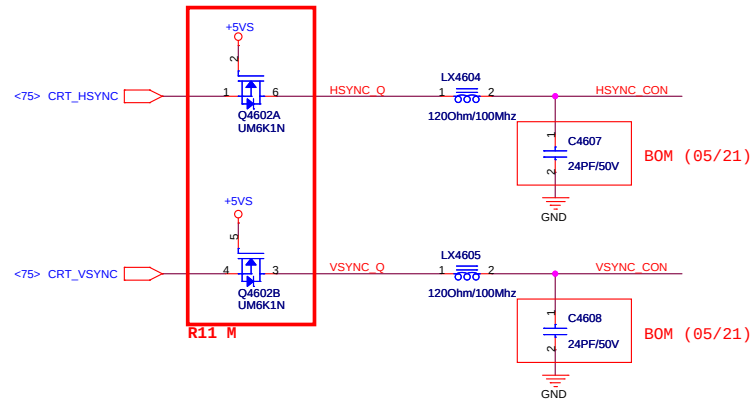
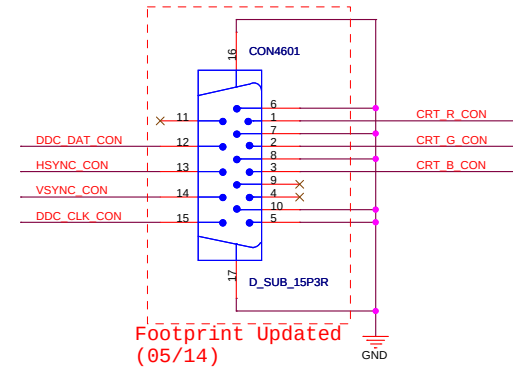
12G17101010L
USB Camera &
Mic Conn. (T53S)



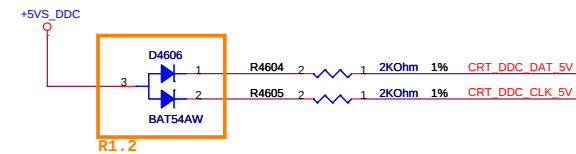
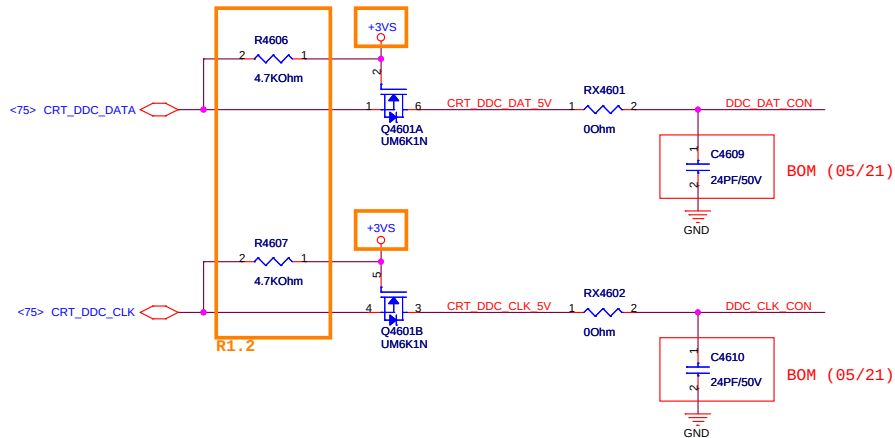


CRT Connector

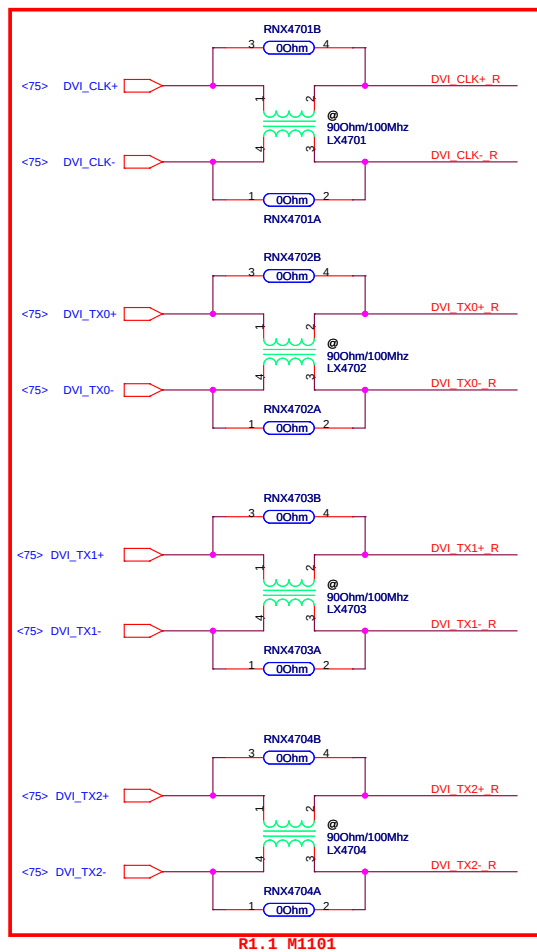
12G101102155
CRT Conn. (T53S)



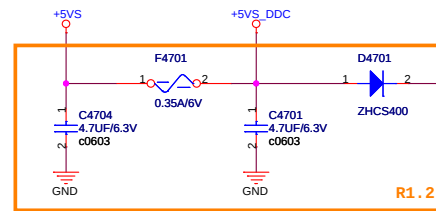
PLACE ESD Diodes near VGA port



ASUS		Title : CRT Port	
ASUSTeK COMPUTER INC		Engineer ENGINEER	
Size	Project Name	Rev	1.2
Custom	T53S		
Date	Monday, September 03, 2007	Sheet	46 of 94

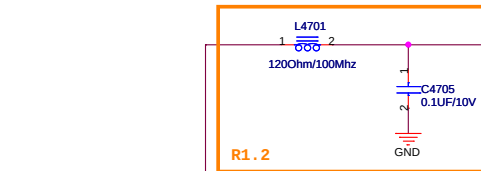


R1.1 M1101

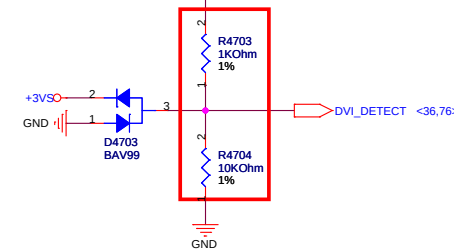


R1.2

R1.1 NB8x do not support HDMI CEC function

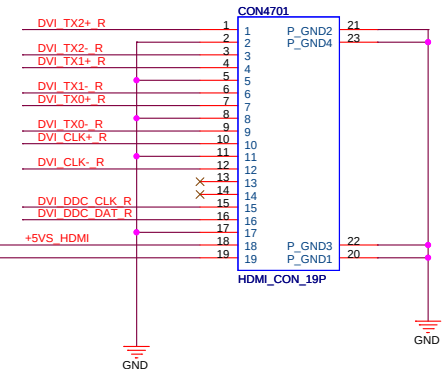


R1.2

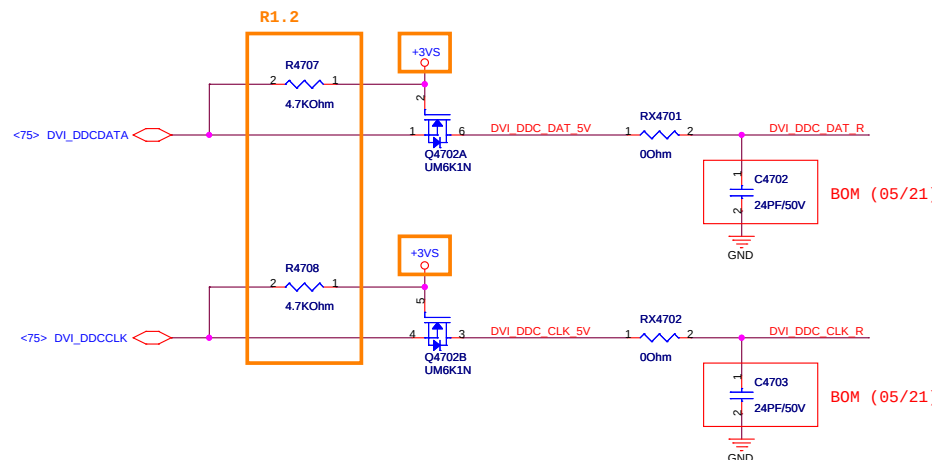


HDMI Connector

12G24110190X
HDMI Conn (T53S)



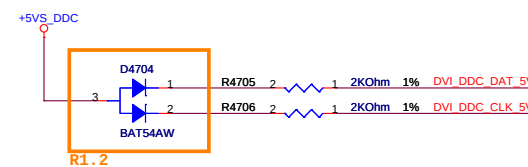
Pin 1 TMDS Data2+
Pin 2 TMDS Data2 Shield
Pin 3 TMDS Data2iV
Pin 4 TMDS Data1+
Pin 5 TMDS Data1 Shield
Pin 6 TMDS Data1iV
Pin 7 TMDS Data0+
Pin 8 TMDS Data0 Shield
Pin 9 TMDS Data0iV
Pin 10 TMDS Clock+
Pin 11 TMDS Clock Shield
Pin 12 TMDS ClockiV
Pin 13 CEC
Pin 14 Reserved (N.C. on device)
Pin 15 SCL
Pin 16 SDA
Pin 17 DDC/CEC Ground
Pin 18 +5 V Power
Pin 19 Hot Plug Detect



R1.2

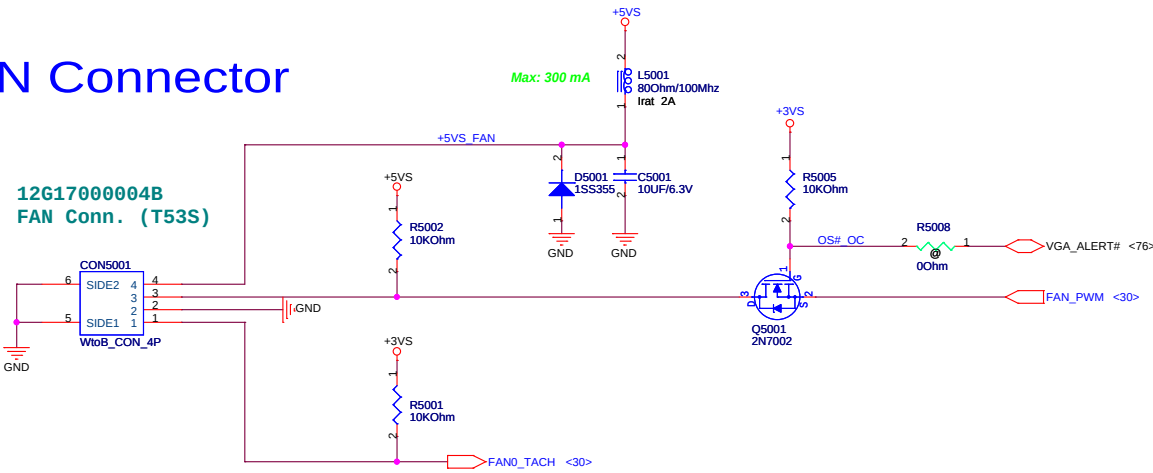
BOM (05/21)

BOM (05/21)

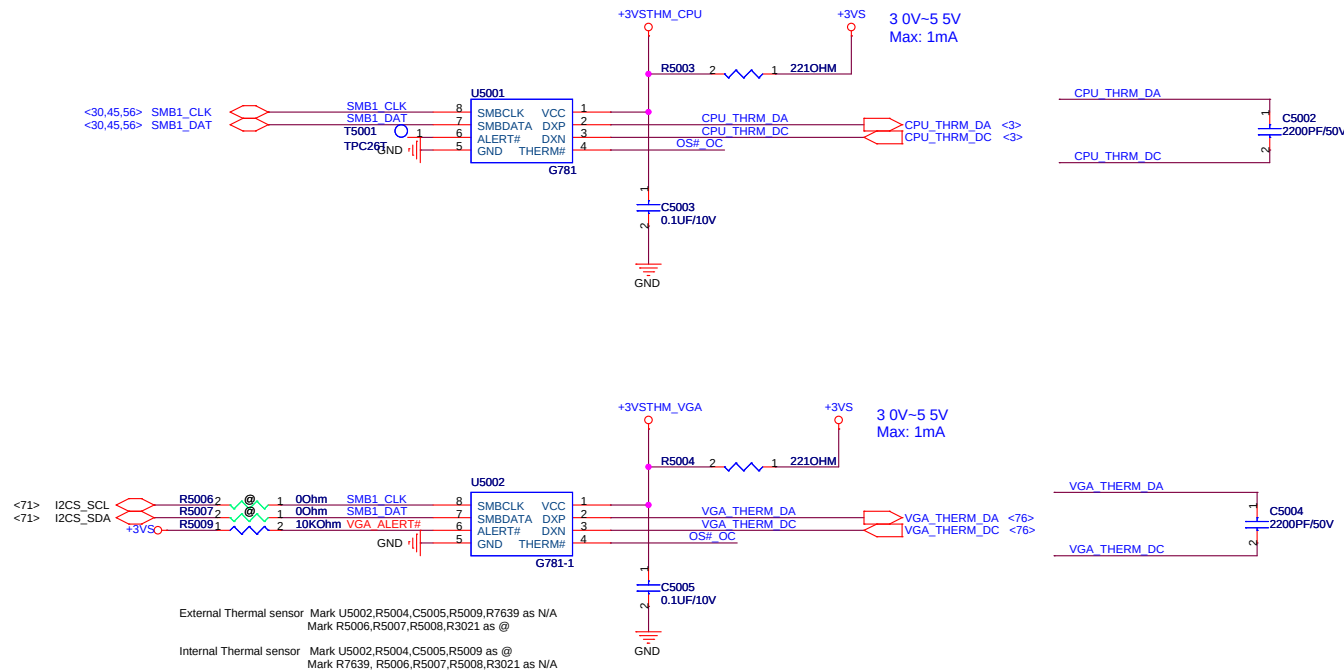


R1.2

FAN Connector



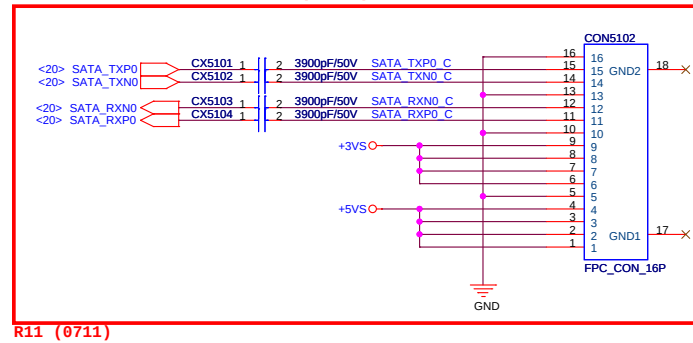
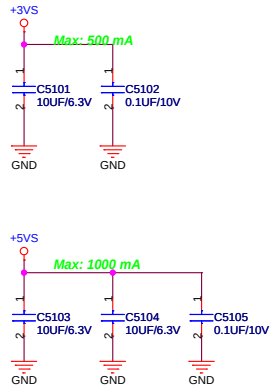
Temp Sensor



SATA HDD

HDD FPC Connector

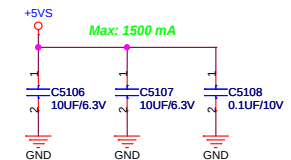
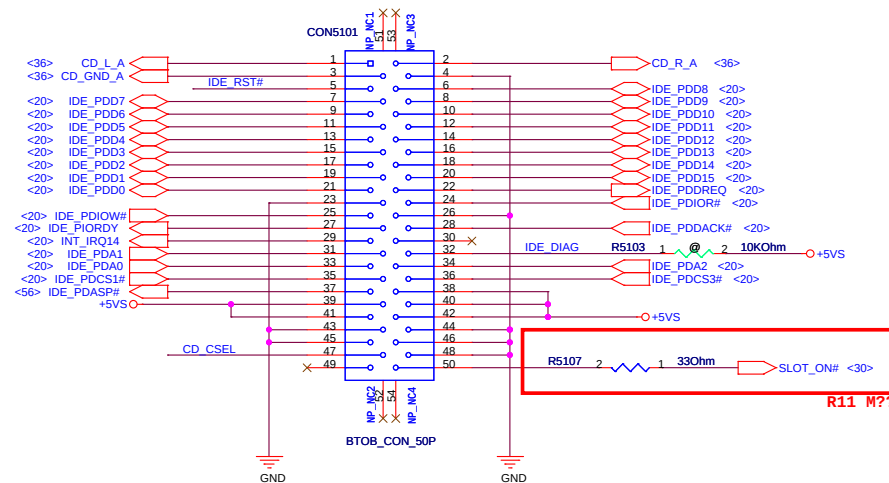
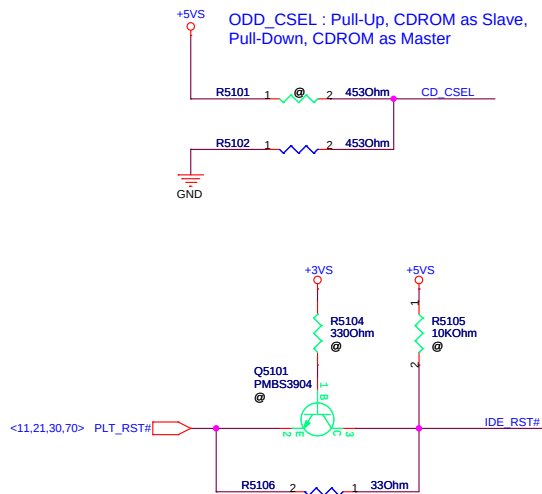
12G182101604
HDD FPC Conn. (T53S)



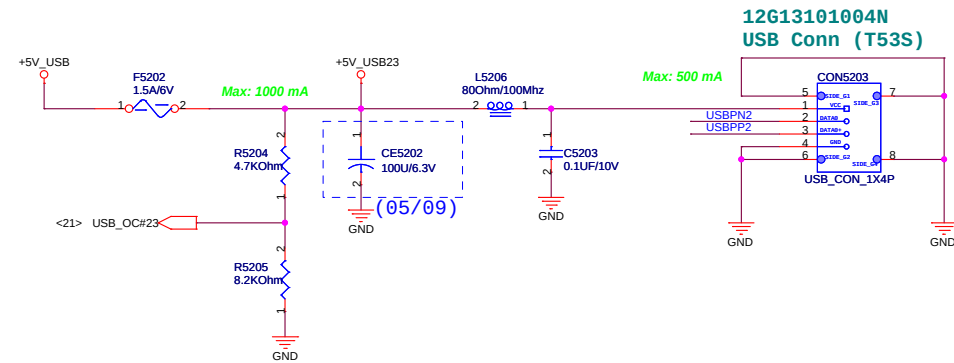
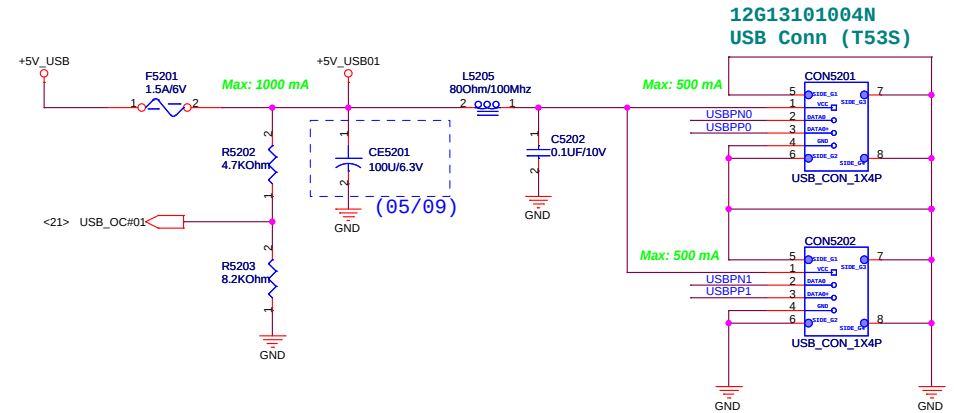
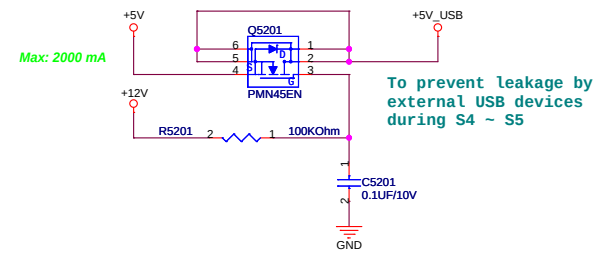
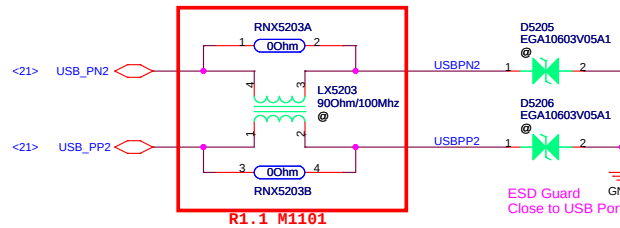
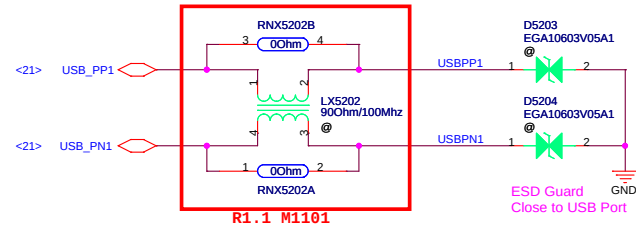
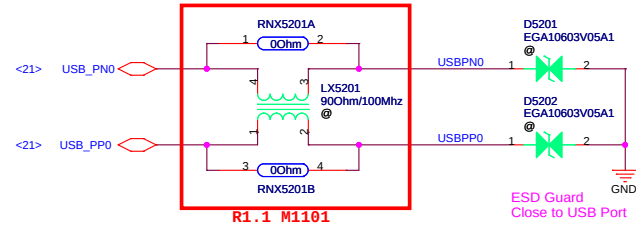
IDE CD-ROM

ODD Connector

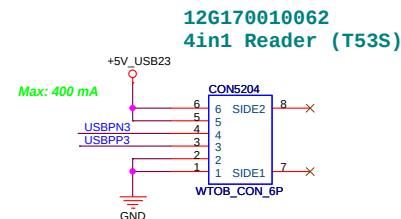
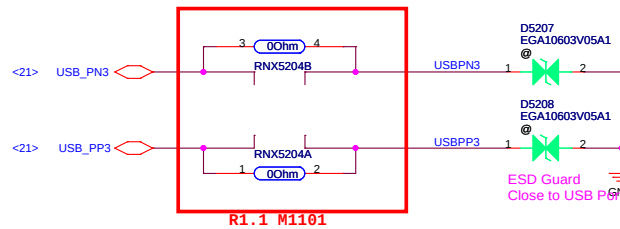
12G161240507
ODD Conn. (T53S)



3 * USB Connectors



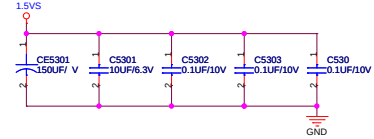
4 in 1 USB Card Reader



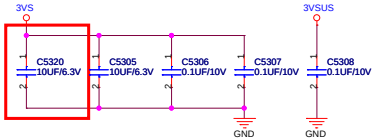
Intel WLAN 4965ABG

Stand-by Power Consumption:
150 mW
Thermal Design Power:
1.5 W

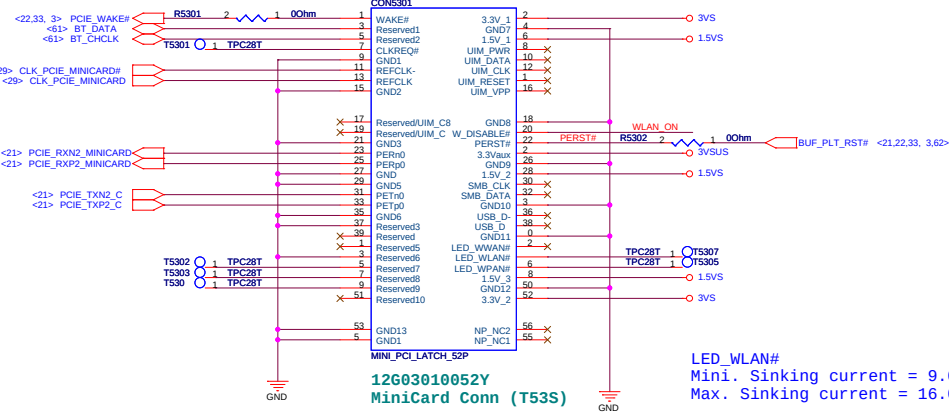
+1.5VS Total
Max. 500 mA
Avg. 375 mA



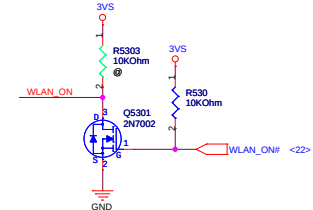
+3VS Total
Max. 1000 mA
Avg. 750mA



+3VSUS
Max. 275 mA
D3 5 mA

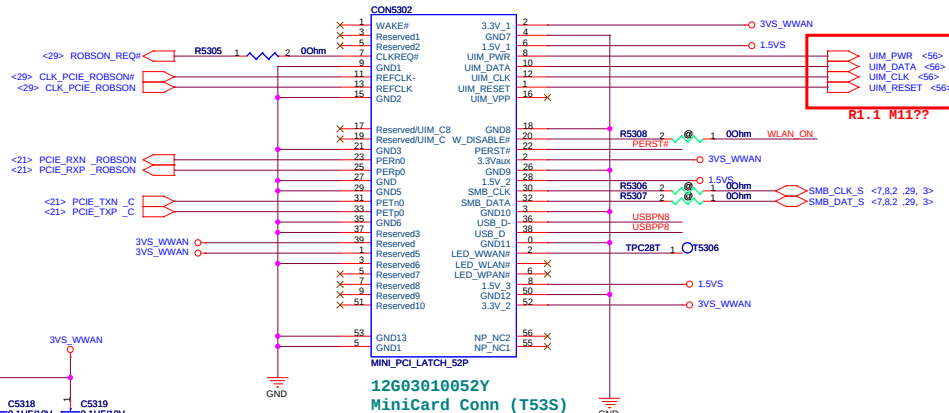
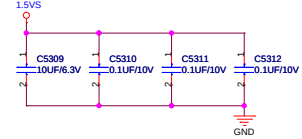


LED_WLAN#
Mini. Sinking current = 9.0 mA
Max. Sinking current = 16.0 mA



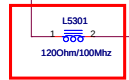
Robson or 3G WWAN Card (Option / Nvate)

+1.5VS Total
Max. 500 mA
Avg. 375 mA



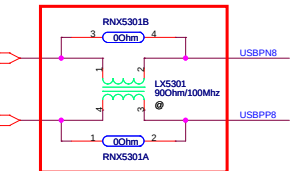
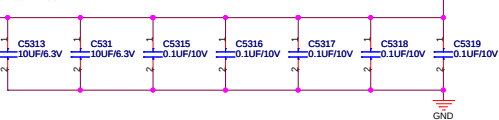
R1.1 M11??

3VS



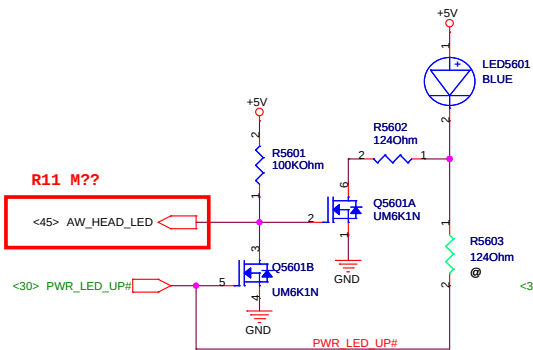
R1.1 For
3G WWAN

+3VS Total
Max. 2750 mA

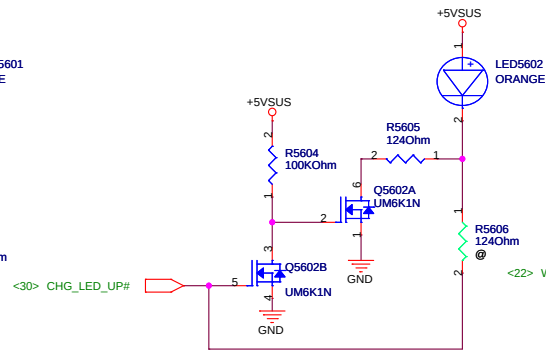


R1.1 M1101

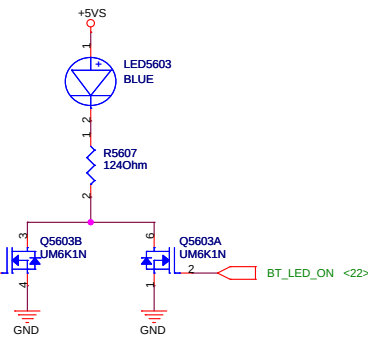
POWER LED (BLUE)



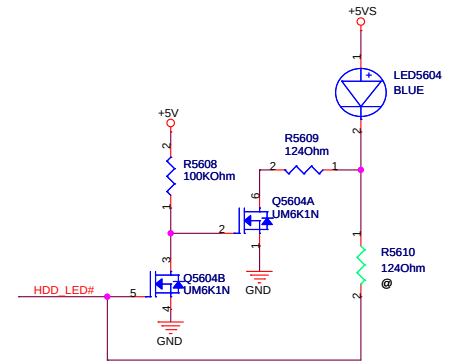
CHARGE LED (ORANGE)



RF LED (BLUE)

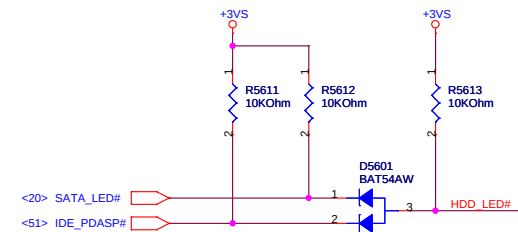
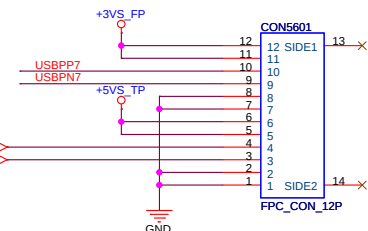
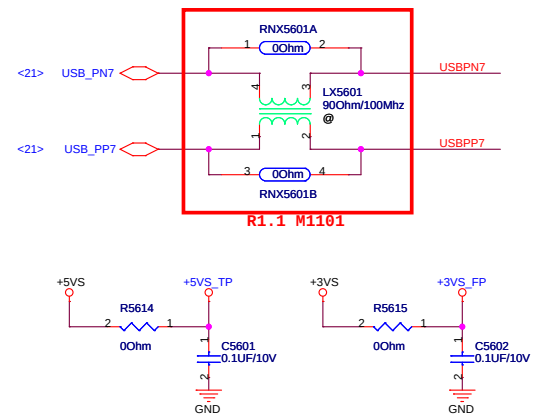


HDD LED (BLUE)



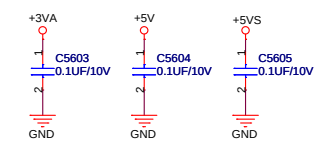
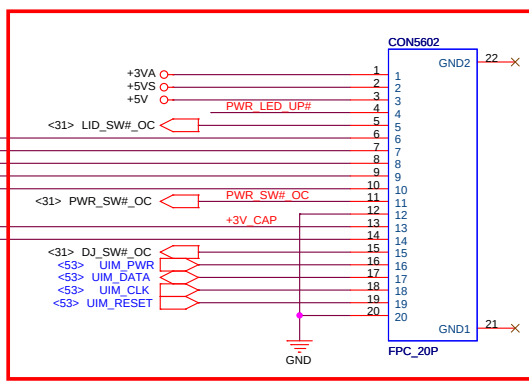
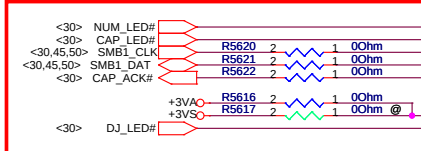
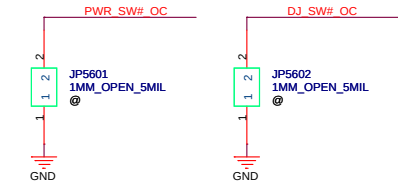
T/P & FP BD.

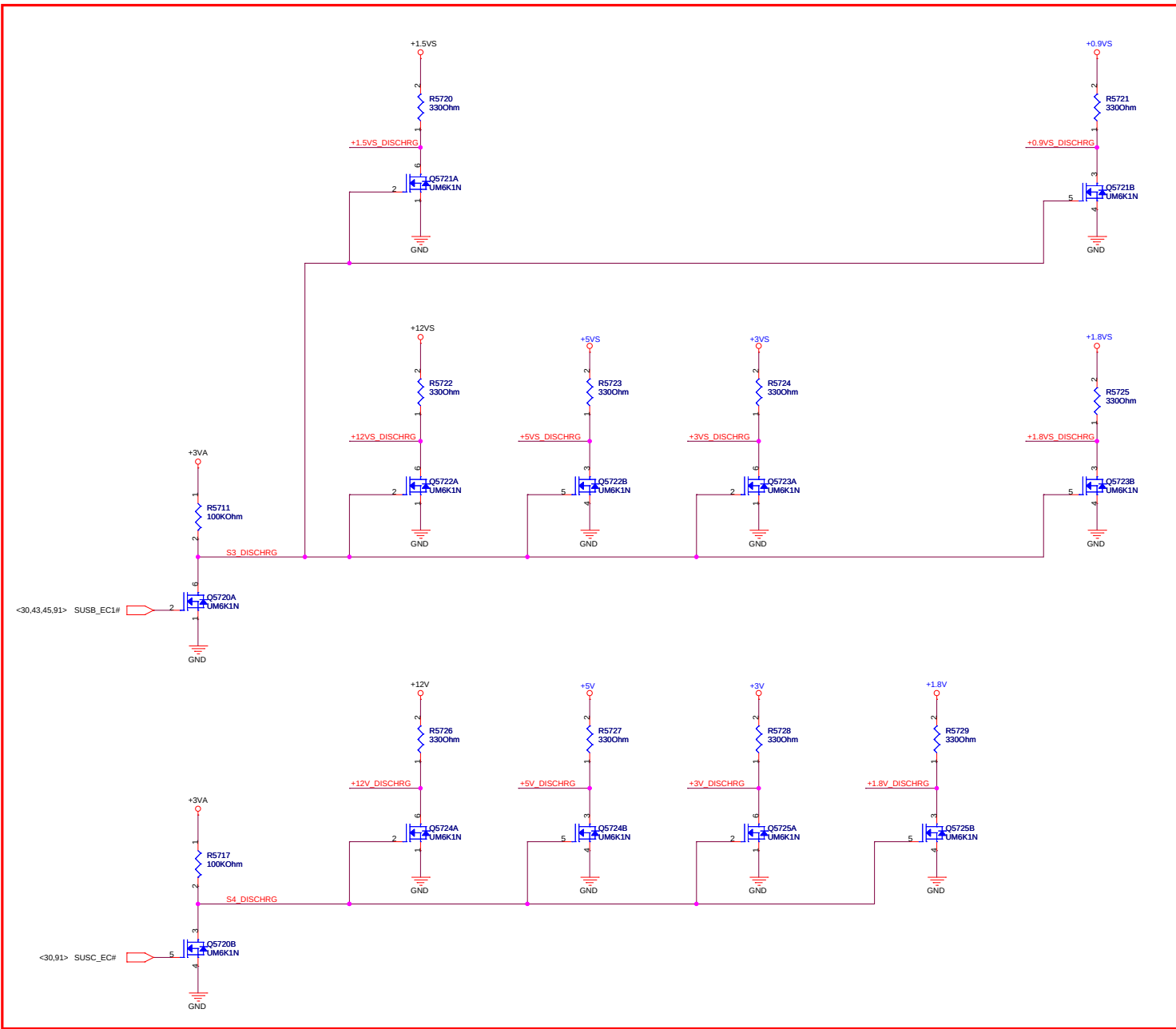
P/N: 12G183301208
T/P & FP Conn. (T53S)



Launch BD.

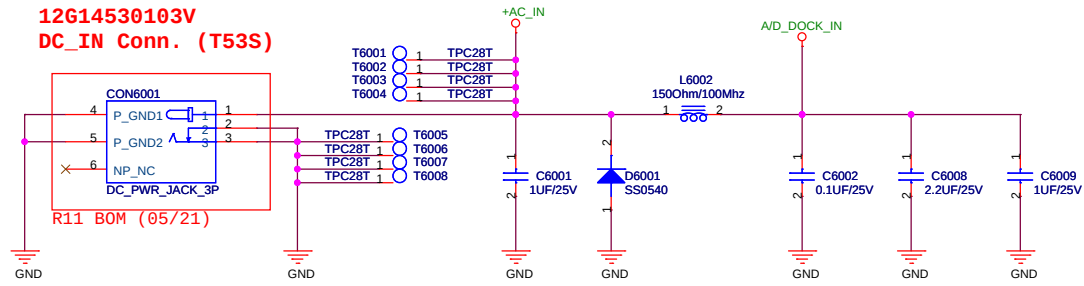
P/N: 12G183402008
Launch BD. Conn. (T53S)





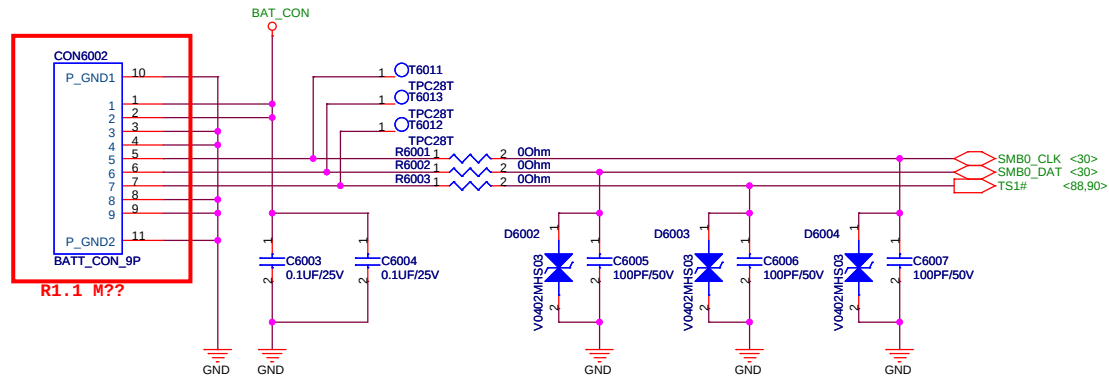
DC_IN Connector

12G14530103V
DC_IN Conn. (T53S)

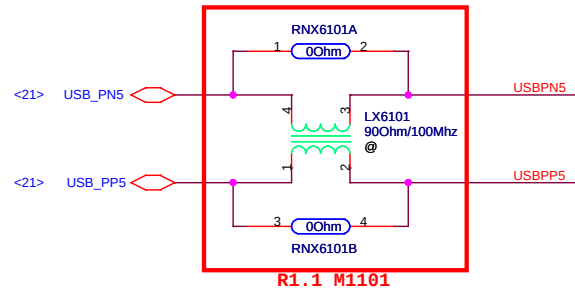
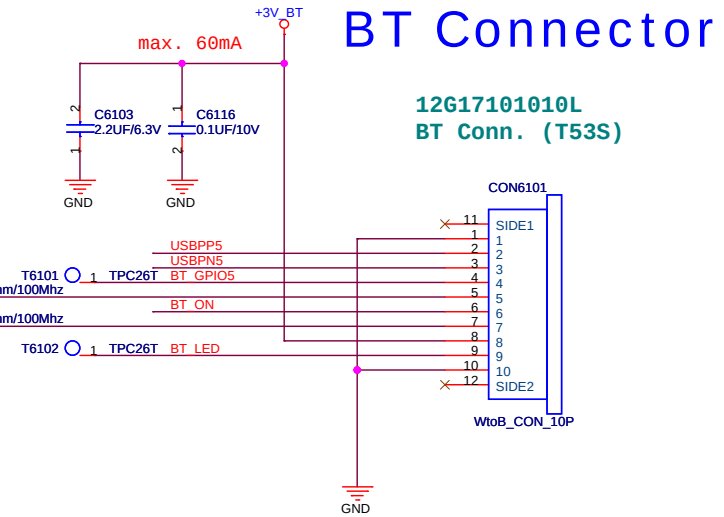
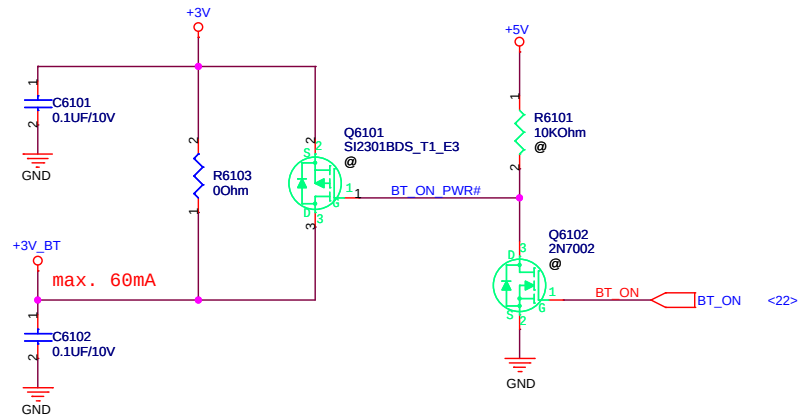


Battery Connector

12G200010914
Battery Conn. (T53S)



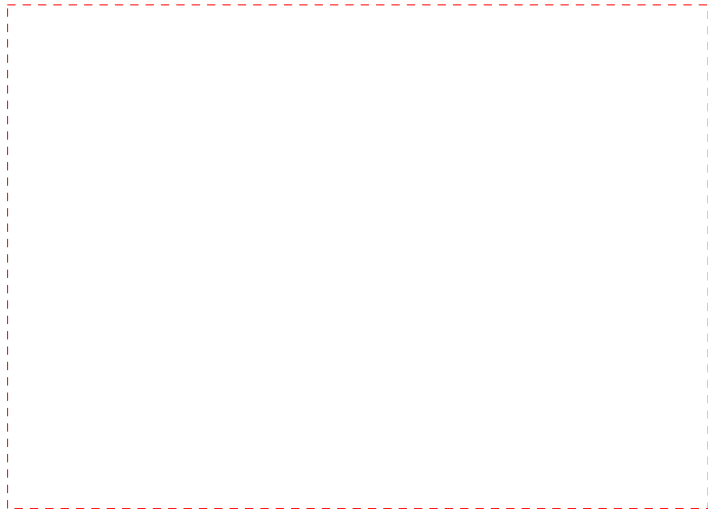
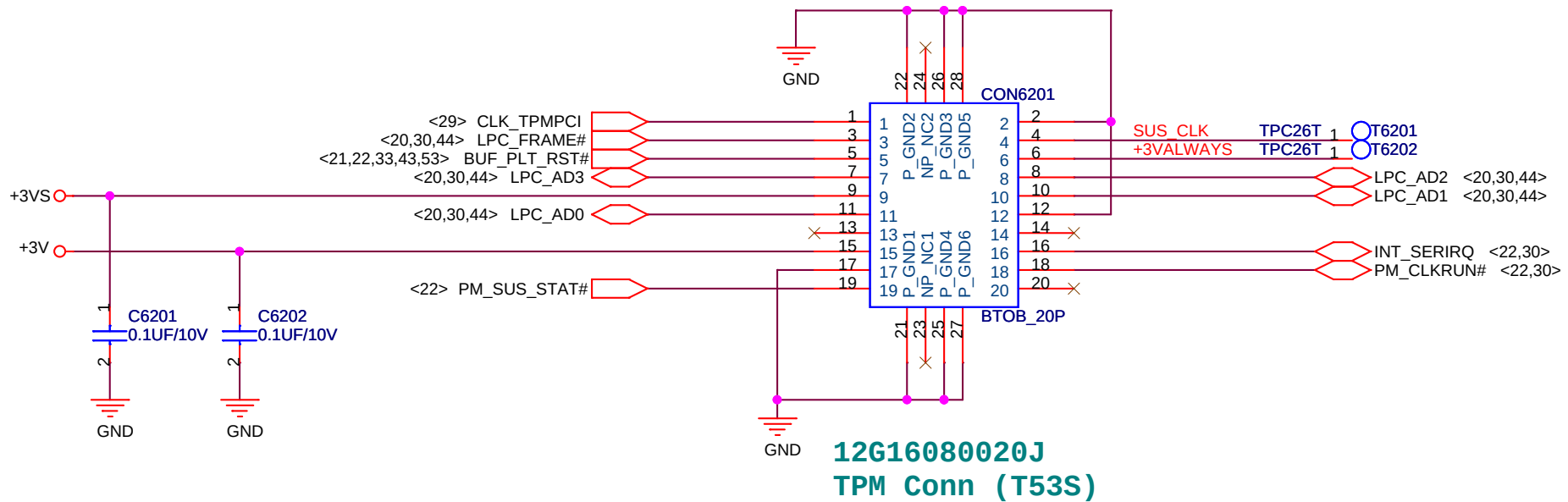
BT ON/OFF Control



12G17101010L
BT Conn. (T53S)

ASUS		Title : BLUETOOTH	
ASUSTeK COMPUTER INC		Engineer: ENGINEER	
Size B	Project Name T53S		Rev 1.1
Date: Thursday, August 30, 2007		Sheet	61 of 94

TPM Conn (H:3.0mm)



Delete TPM Nut (05/14)

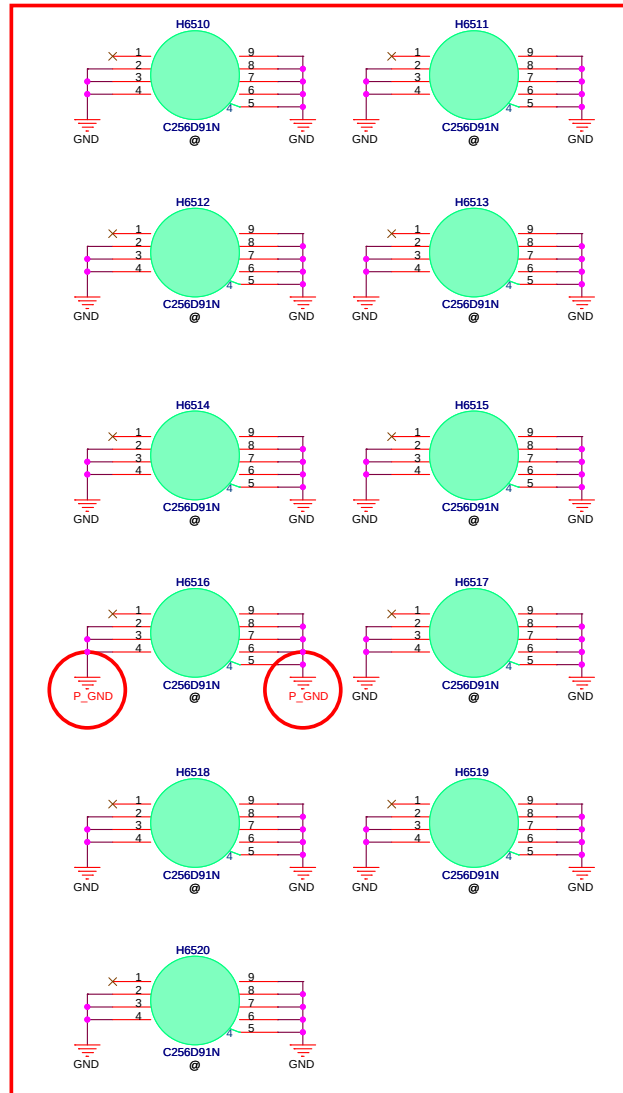
		Title : TPM	
ASUSTeK COMPUTER INC		Engineer: ENGINEER	
Size A	Project Name T53S		Rev 1.0
Date: Thursday, August 30, 2007		Sheet	62 of 94

PCI Device	IDSEL#	REQ/GNT#	Interrupts
Chipset (Host to PCI)	AD30 (Internal)		
VGA	AD16		

SM-Bus Device	SM-Bus Address
Clock Generator	1101001x (D2)
SO-DIMM 0	1010000x (A0)
SO-DIMM 1	1010010x (A4)
Thermal Sensor (CPU)	1001100x (98)
Thermal Sensor (VGA)	1001101x (9A)

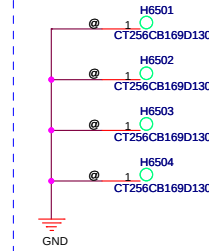
IC	REFERENCE	VENID	DEVID	VERSION	MARK
NB8P-GS			0x0407		

Screw NPTH

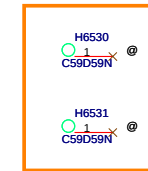
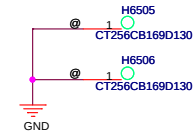


R11 M??

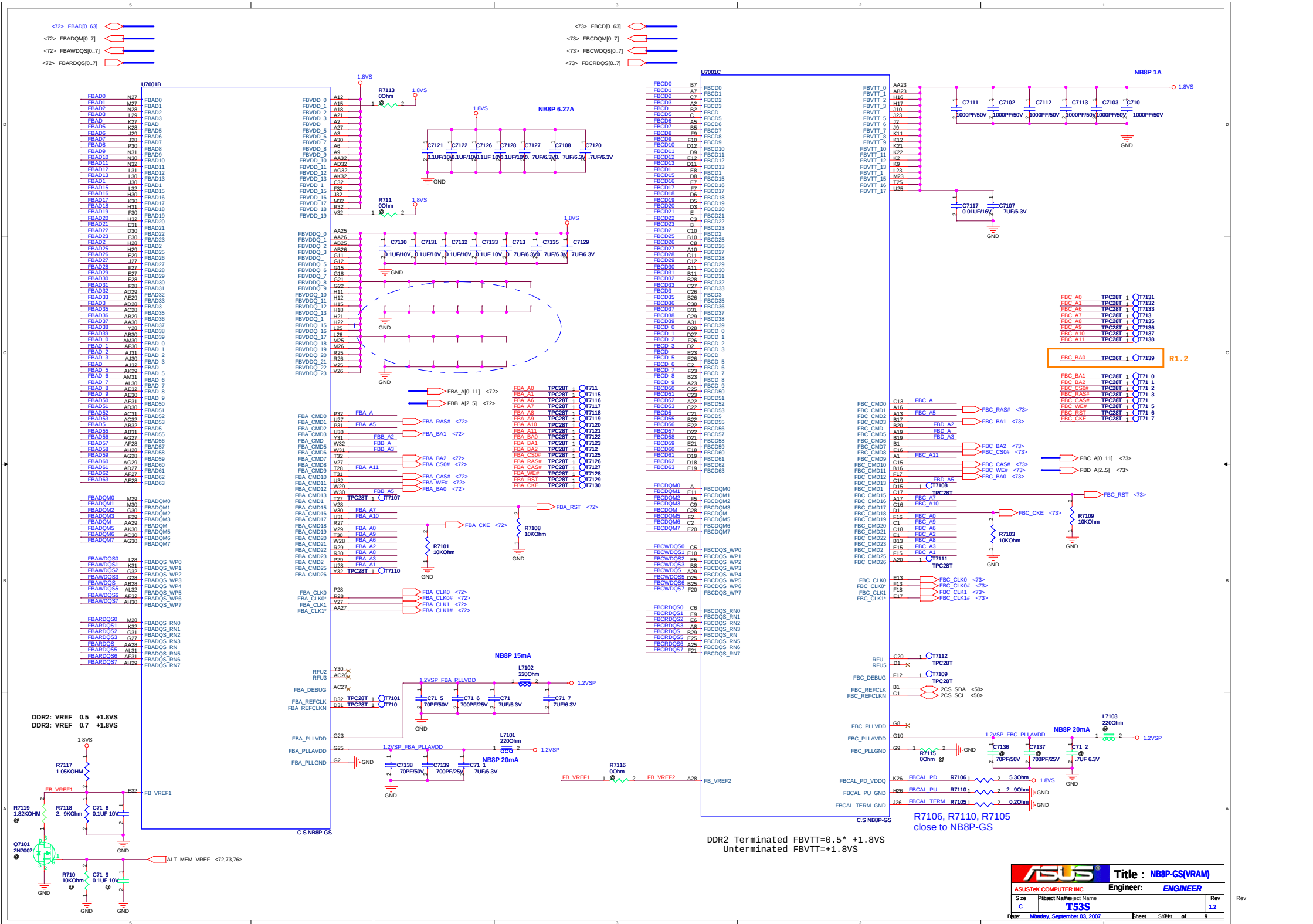
CPU PTH



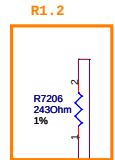
VGA PTH



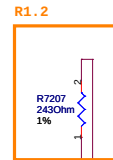
R1.2



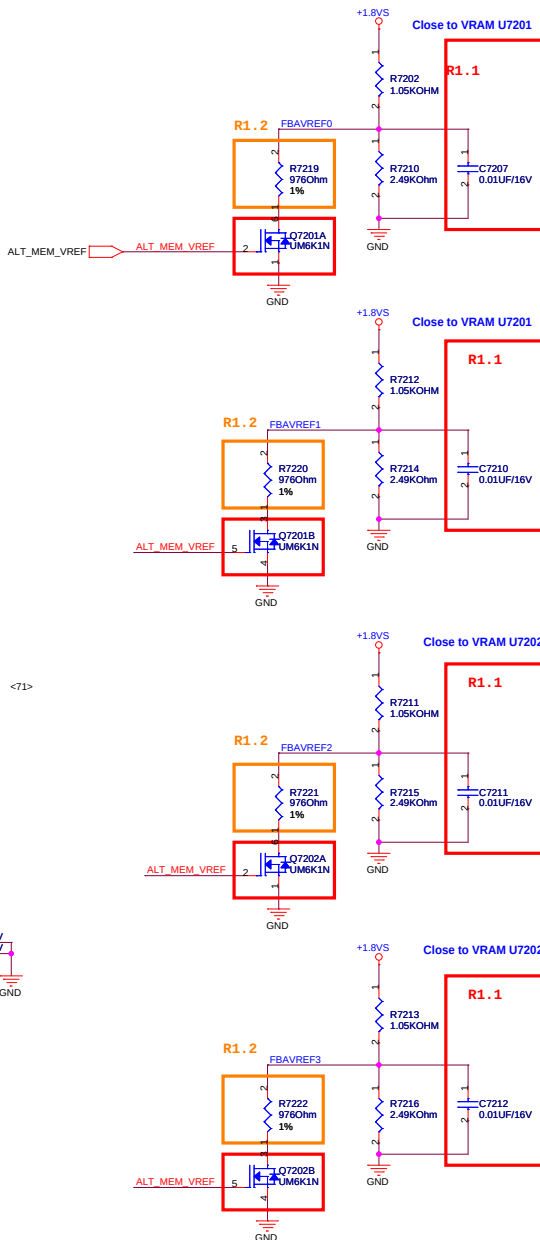
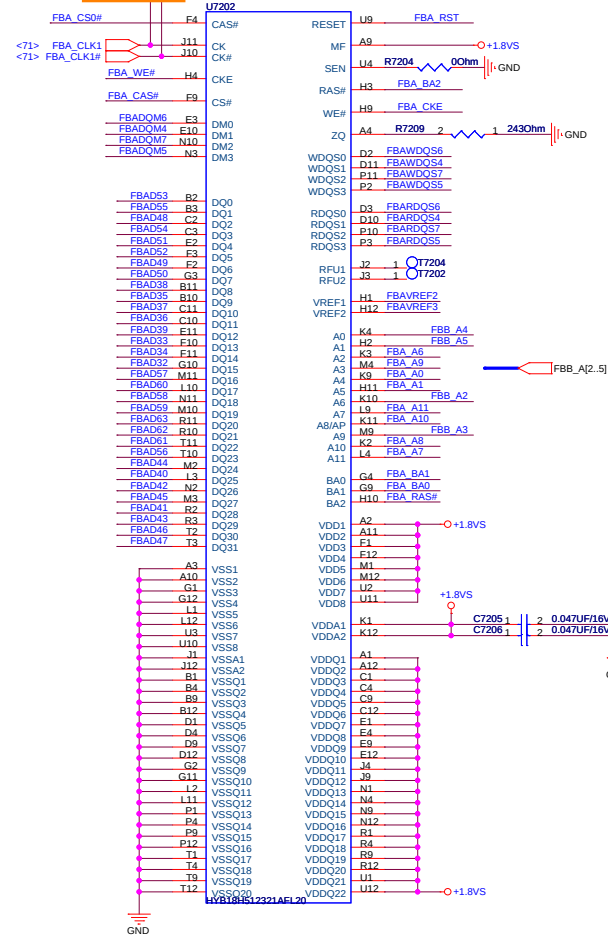
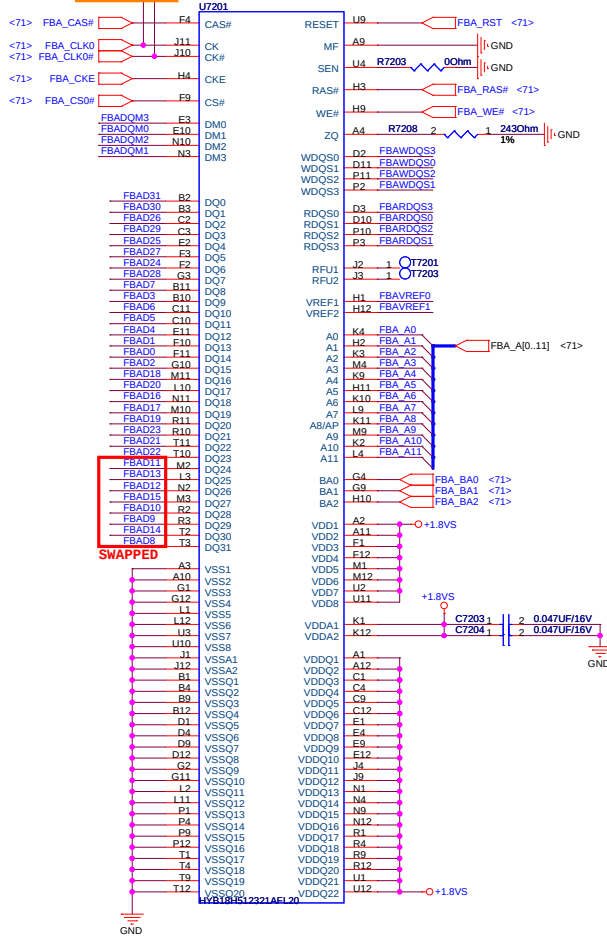
<71> FBAD[0..63] <71>
 <71> FBADQM[0..7] <71>
 <71> FBAWQSQ[0..7] <71>
 <71> FBARDQS[0..7] <71>



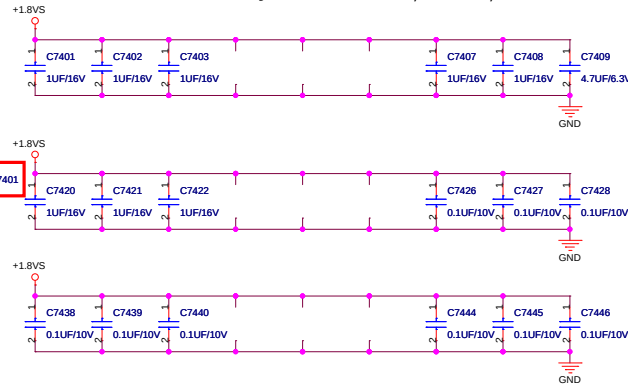
R7206 close to VRAM U7201



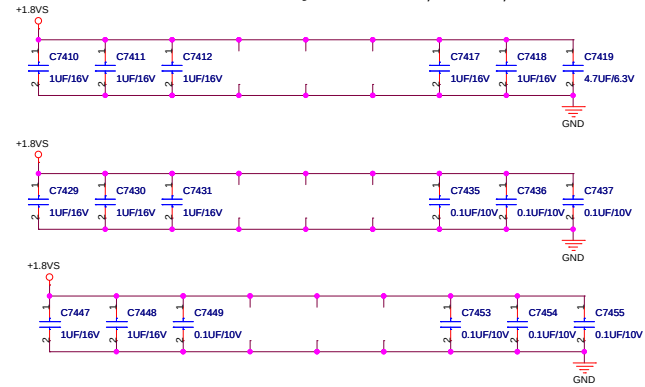
R7207 close to VRAM U7202



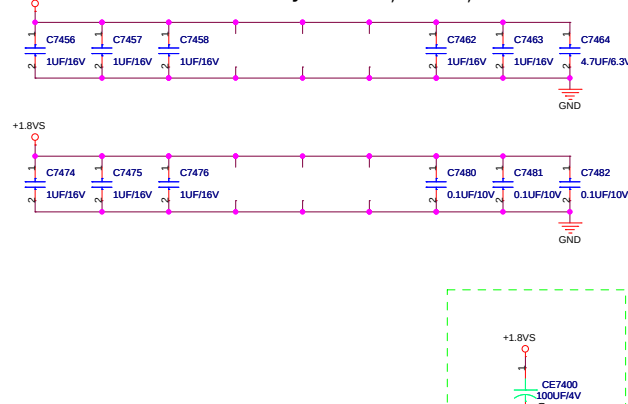
Decoupling for FBA. (DQ0-DQ31)
ACAP to memory 0.1Ux14, 1Ux12, 4.7Ux1



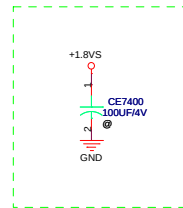
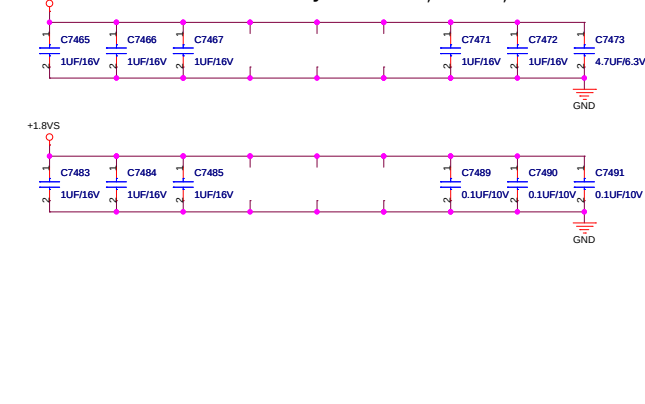
Decoupling for FBC. (DQ0-DQ31)
ACAP to memory 0.1Ux14, 1Ux12, 4.7Ux1



Decoupling for FBA. (DQ32-DQ63)
ACAP to memory 0.1Ux14, 1Ux12, 4.7Ux1



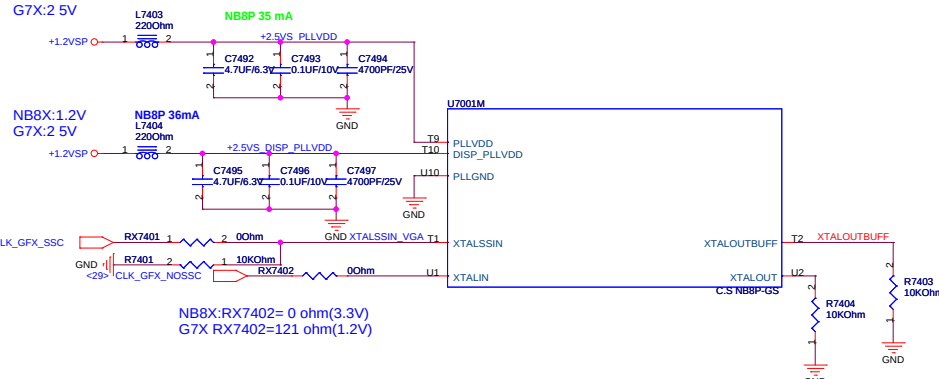
Decoupling for FBC. (DQ32-DQ63)
ACAP to memory 0.1Ux14, 1Ux12, 4.7Ux1

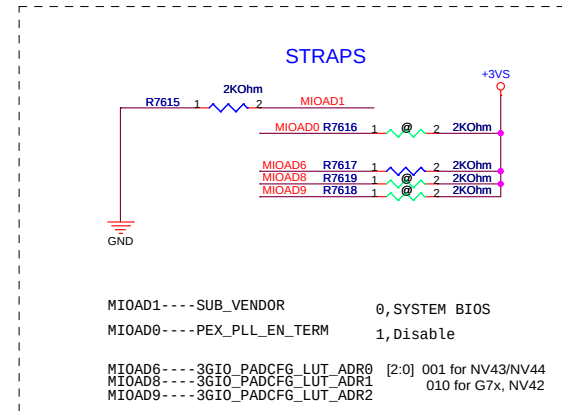
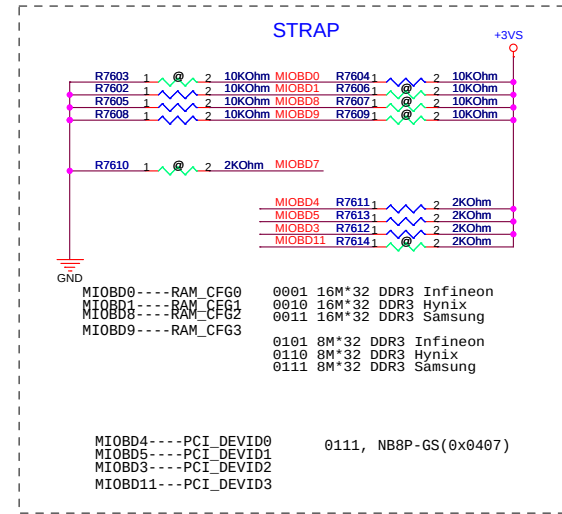
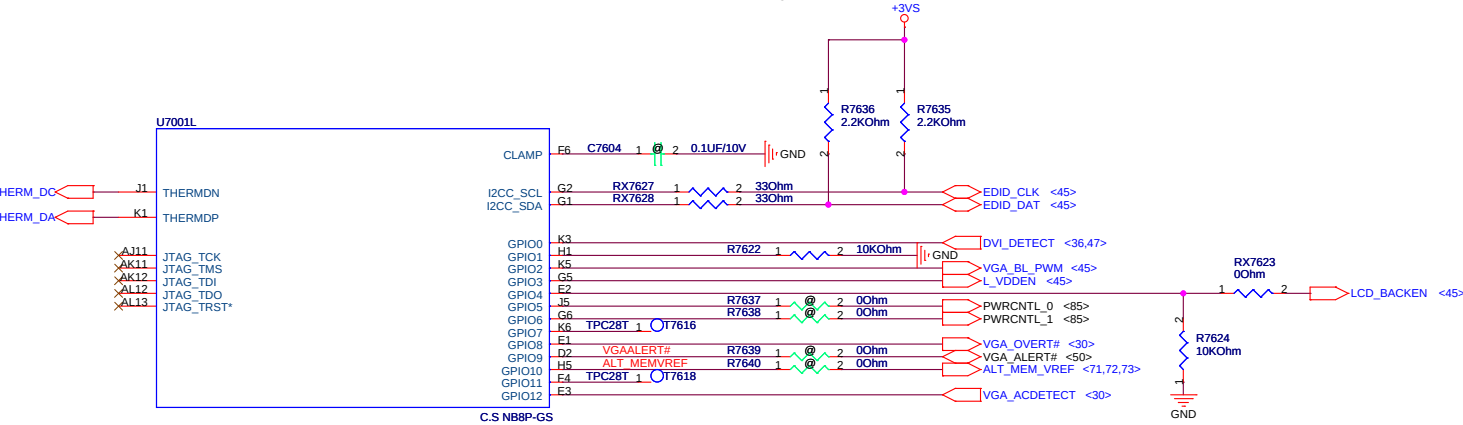
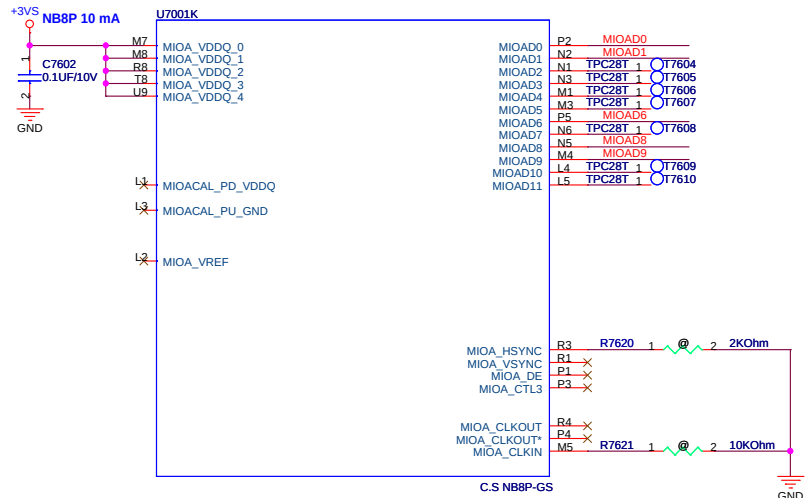
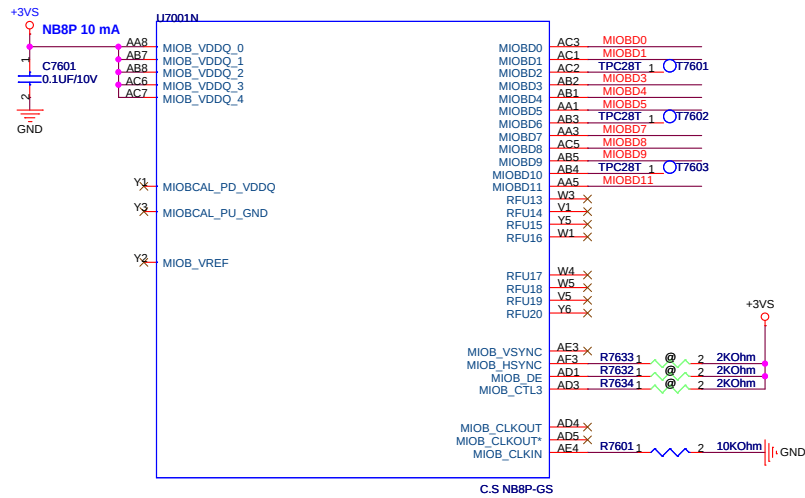


NB8X:1.2V
G7X:2.5V

NB8X:1.2V
G7X:2.5V

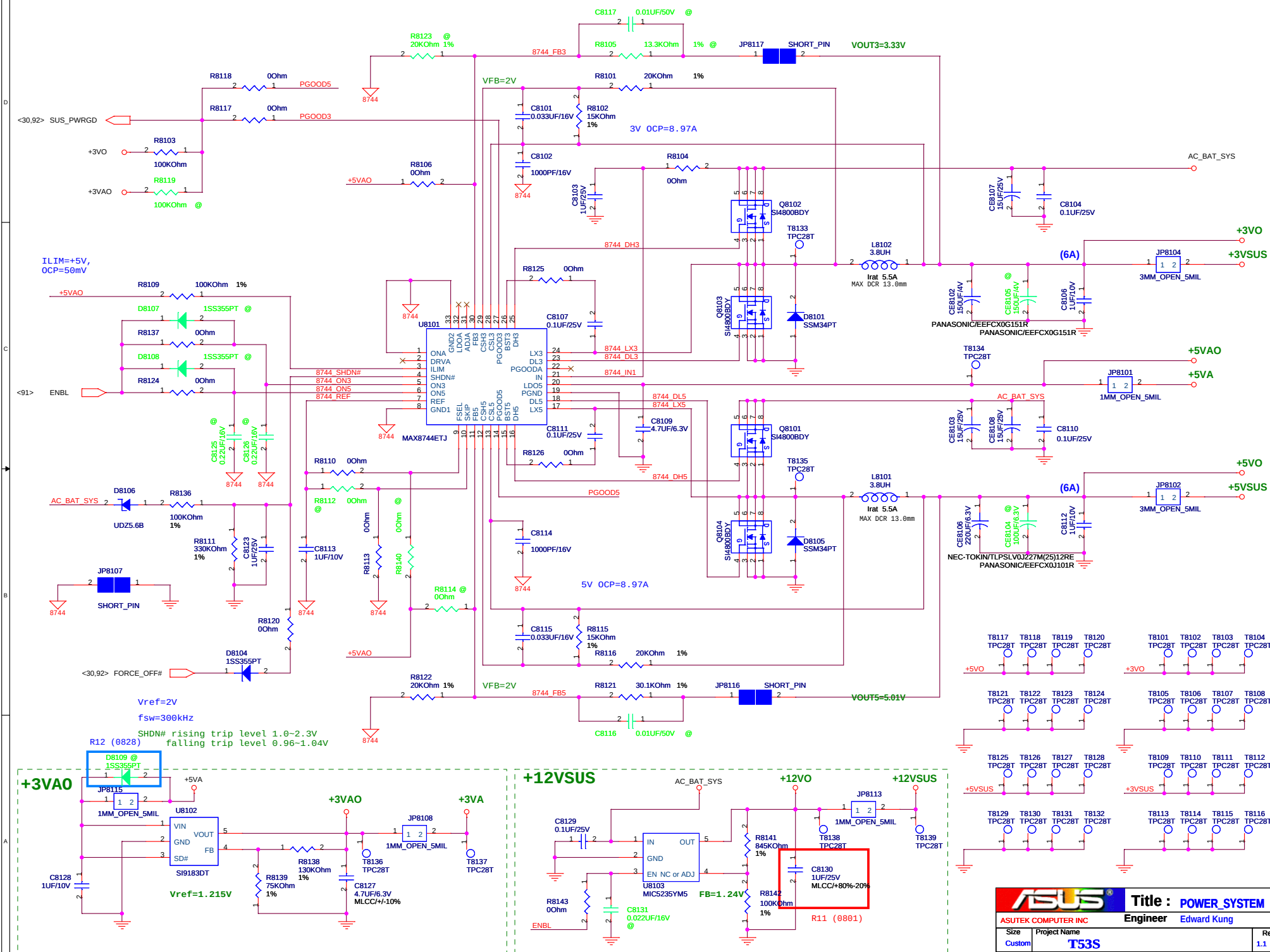
NB8X:RX7402= 0 ohm(3.3V)
G7X RX7402=121 ohm(1.2V)

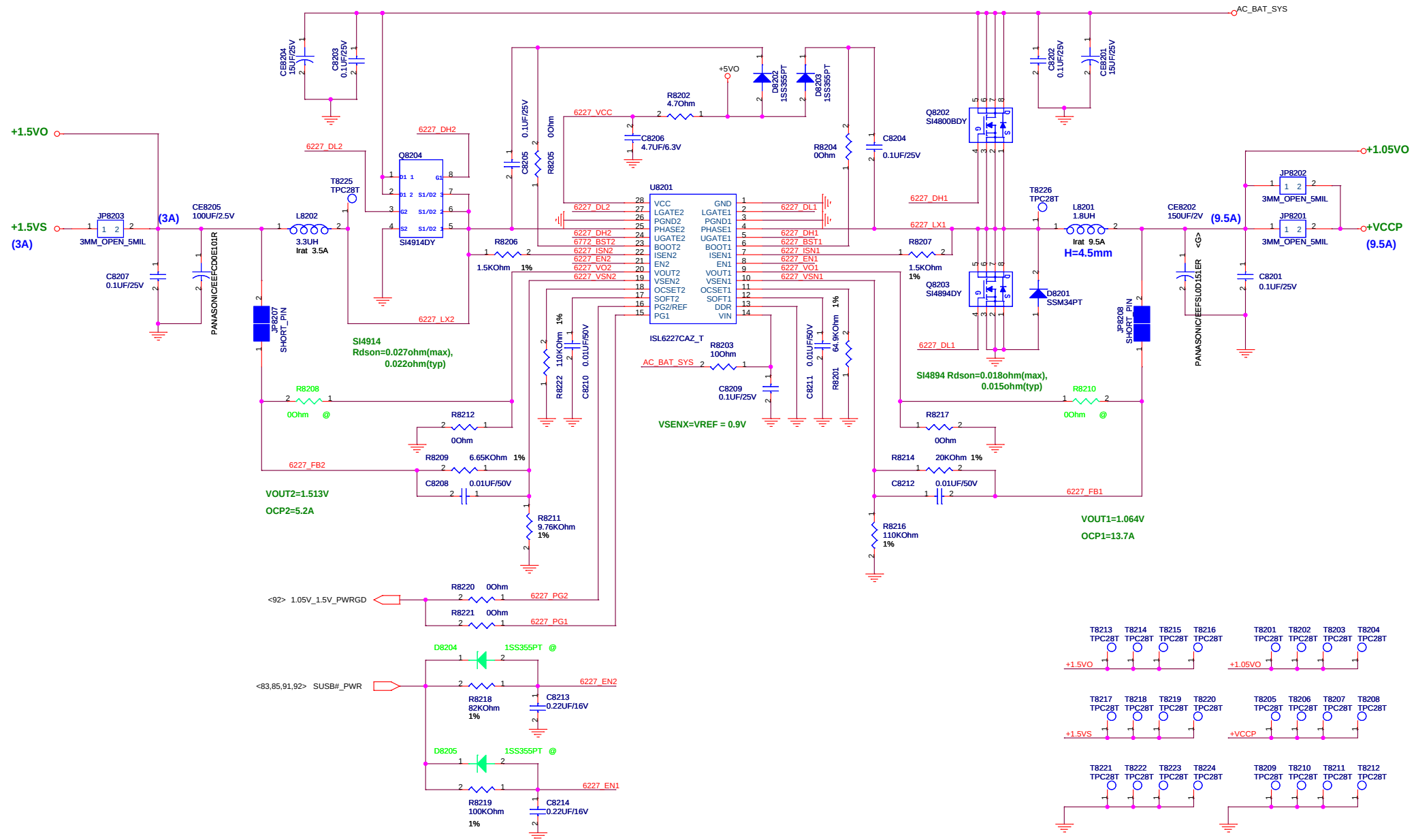




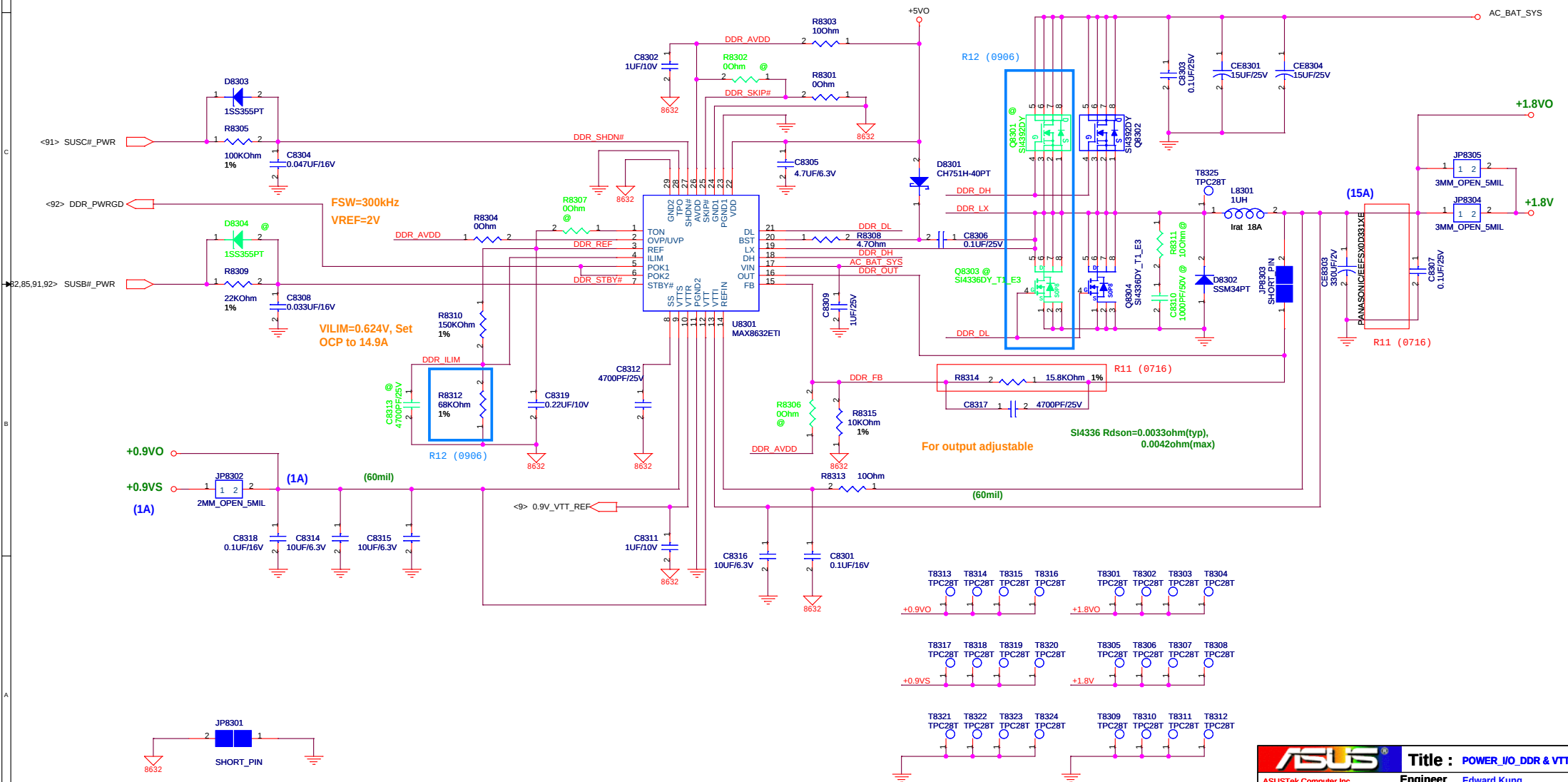
GPIO0, 1 -- Hot Plug Detect (For DVI)
GPIO2 -- Panel backlight brightness (PWM)
GPIO3 -- Panel power enable
GPIO4 -- Panel backlight ON/OFF
GPIO5 -- GPU VID0
GPIO6 -- GPU VID1
GPIO7 -- GPU VID2 or MEM VID
GPIO8 -- Thermal diode ALERT
GPIO9 -- Fan control
GPIO11 -- HDTV enable

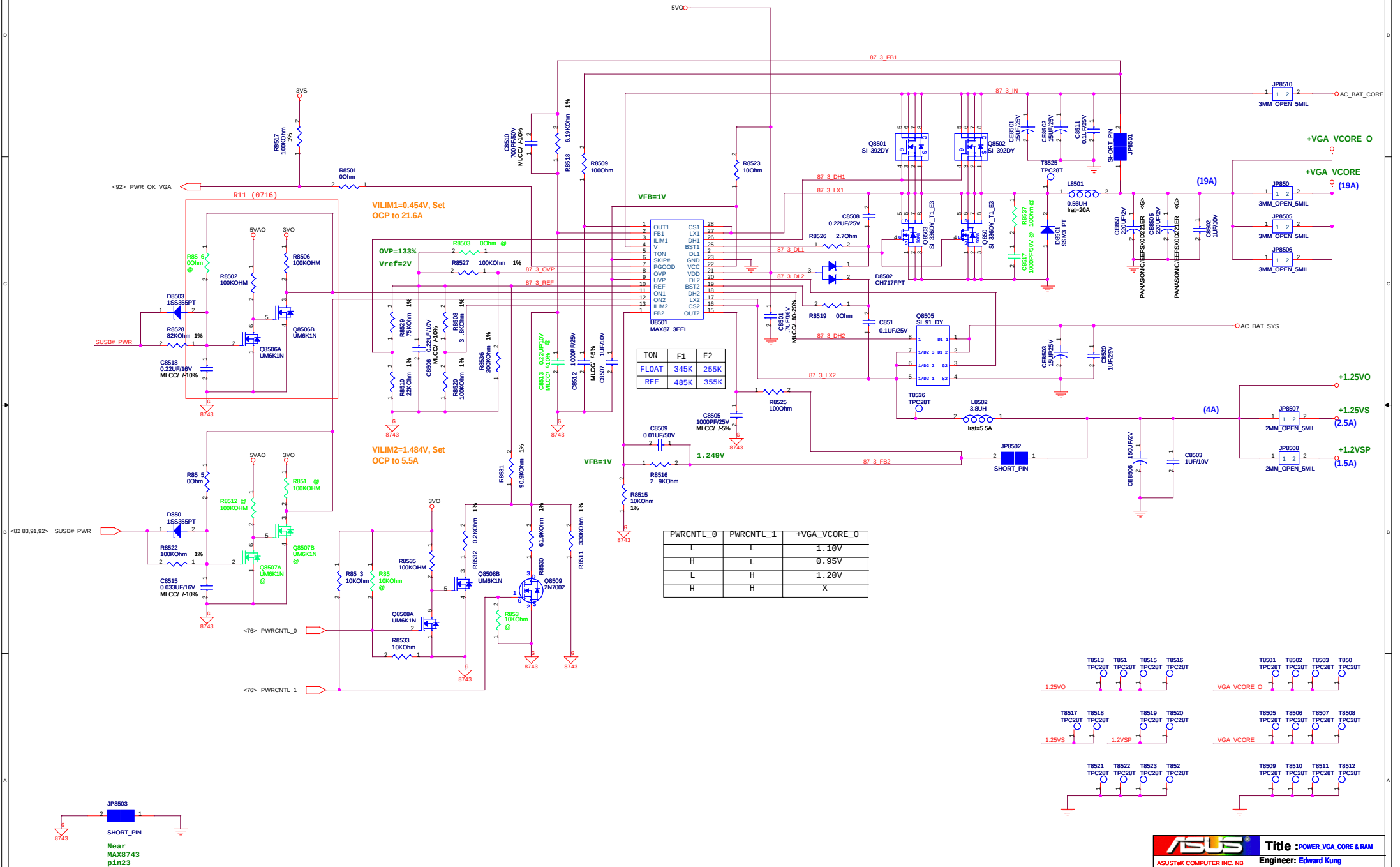
Internal Pull-down
GPIO0, 1, 2, 4, 5, 6, 7

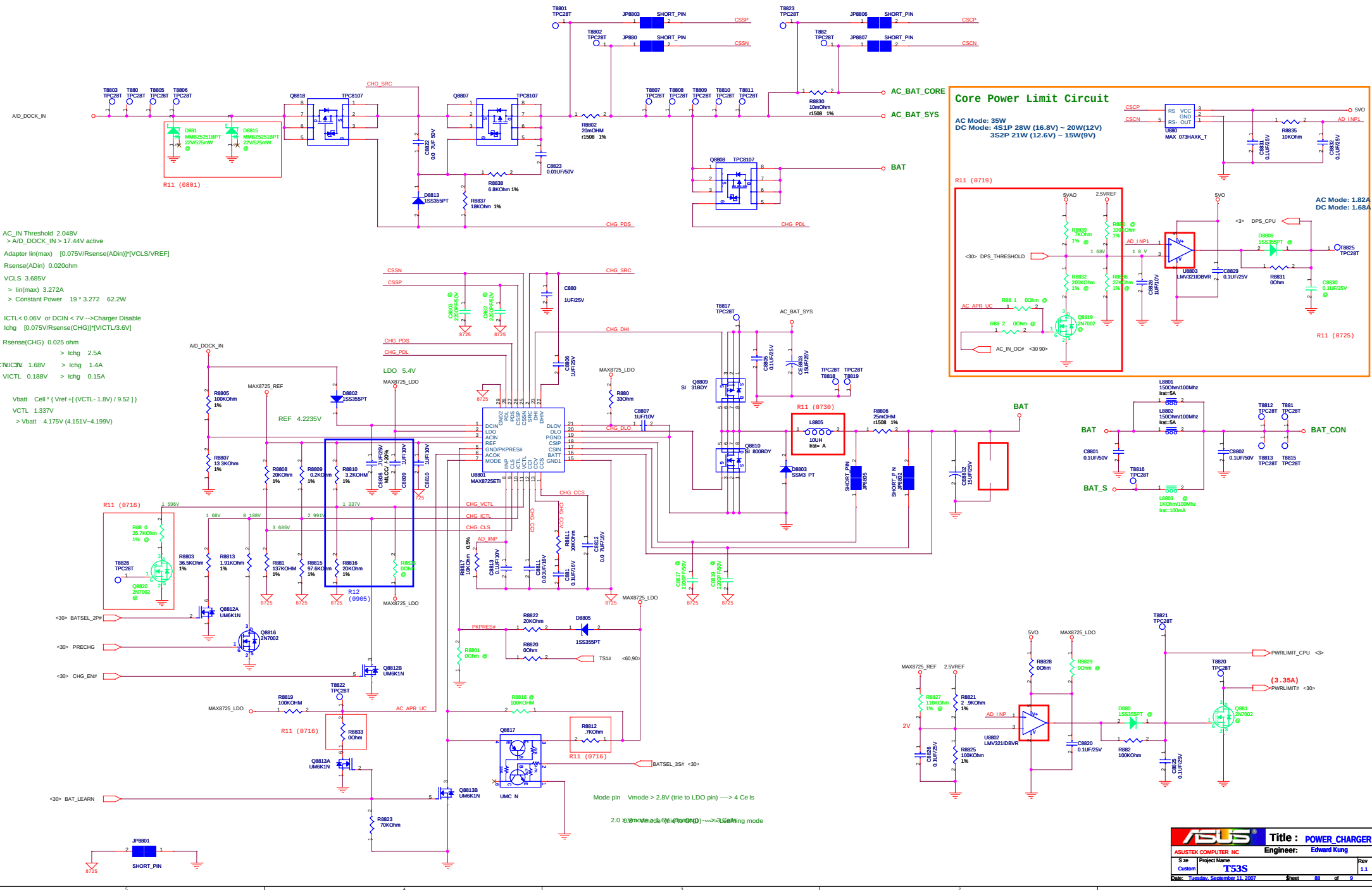




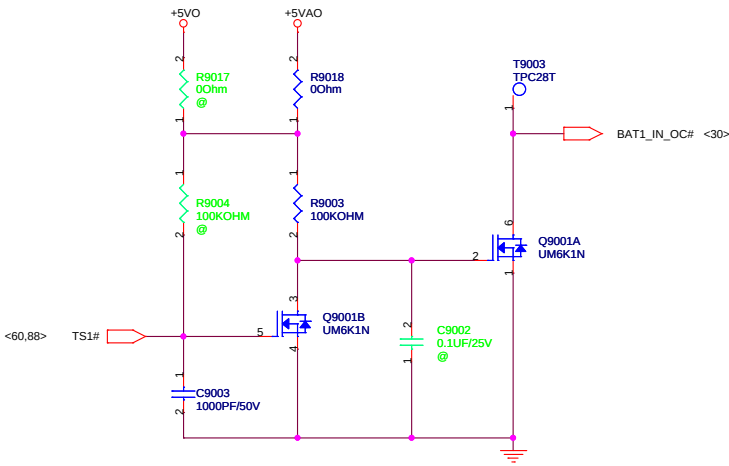
Status	SUSC# PWR (SHDN#)	SUSB# PWR (STBY#)	Output	
S4 / S5	LOW	LOW	OFF	OFF
S3	HIGH	LOW	ON	OFF
S0	HIGH	HIGH	ON	ON



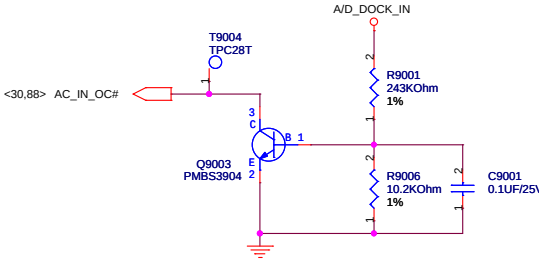




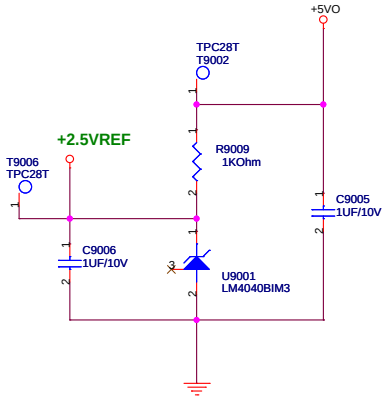
BATTERY IN DETECT



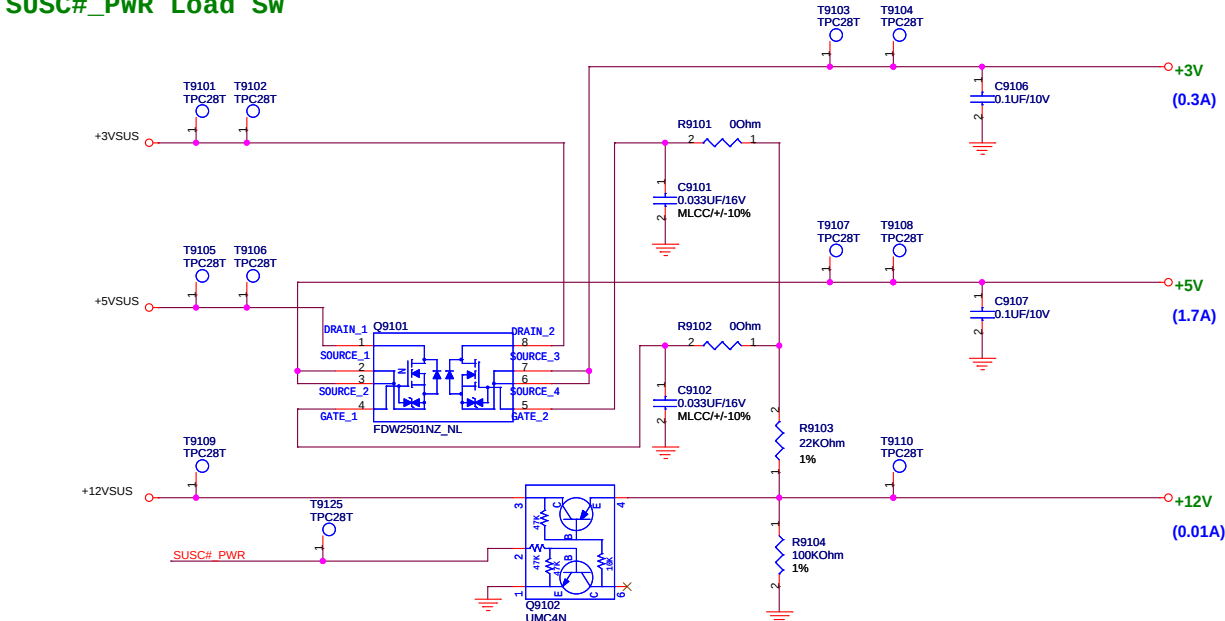
ADAPTER IN DETECT



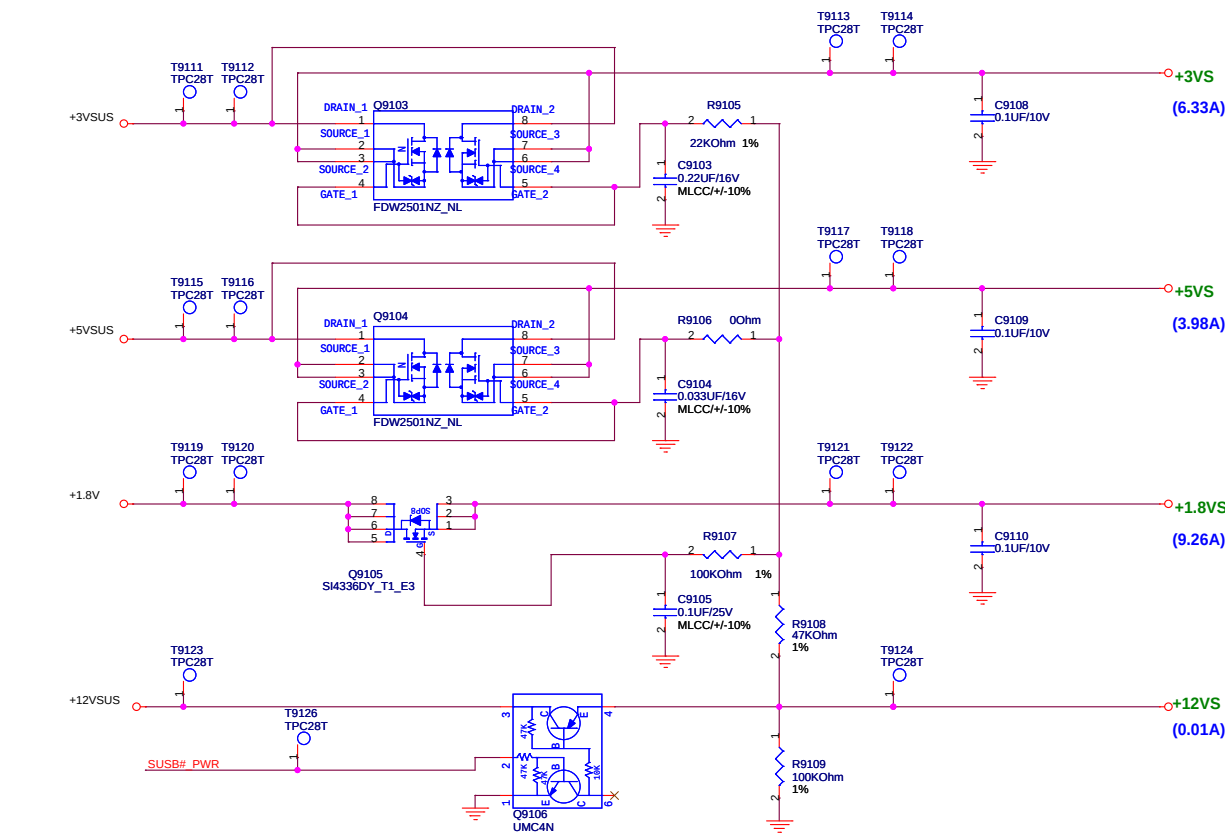
+2.5VREF



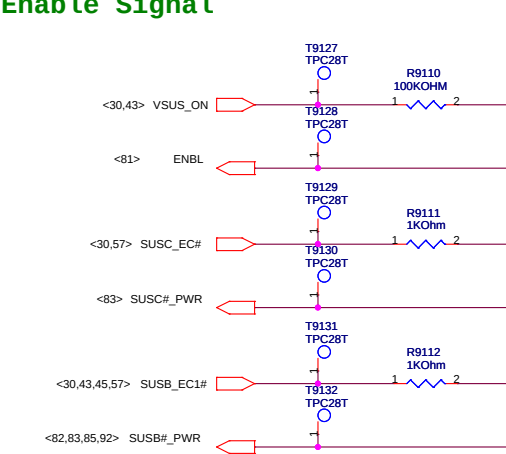
SUSC#_PWR Load SW



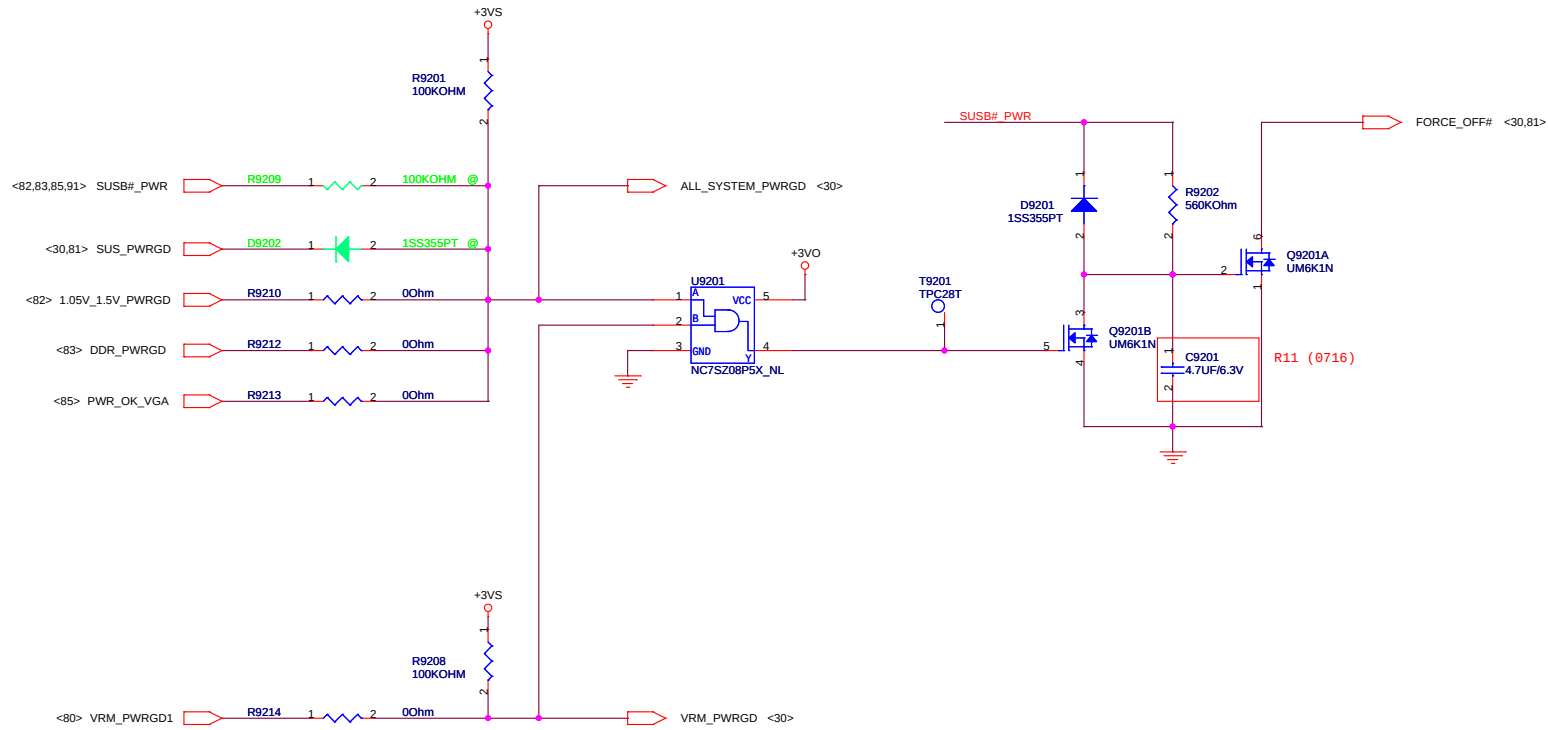
SUSB#_PWR Load SW



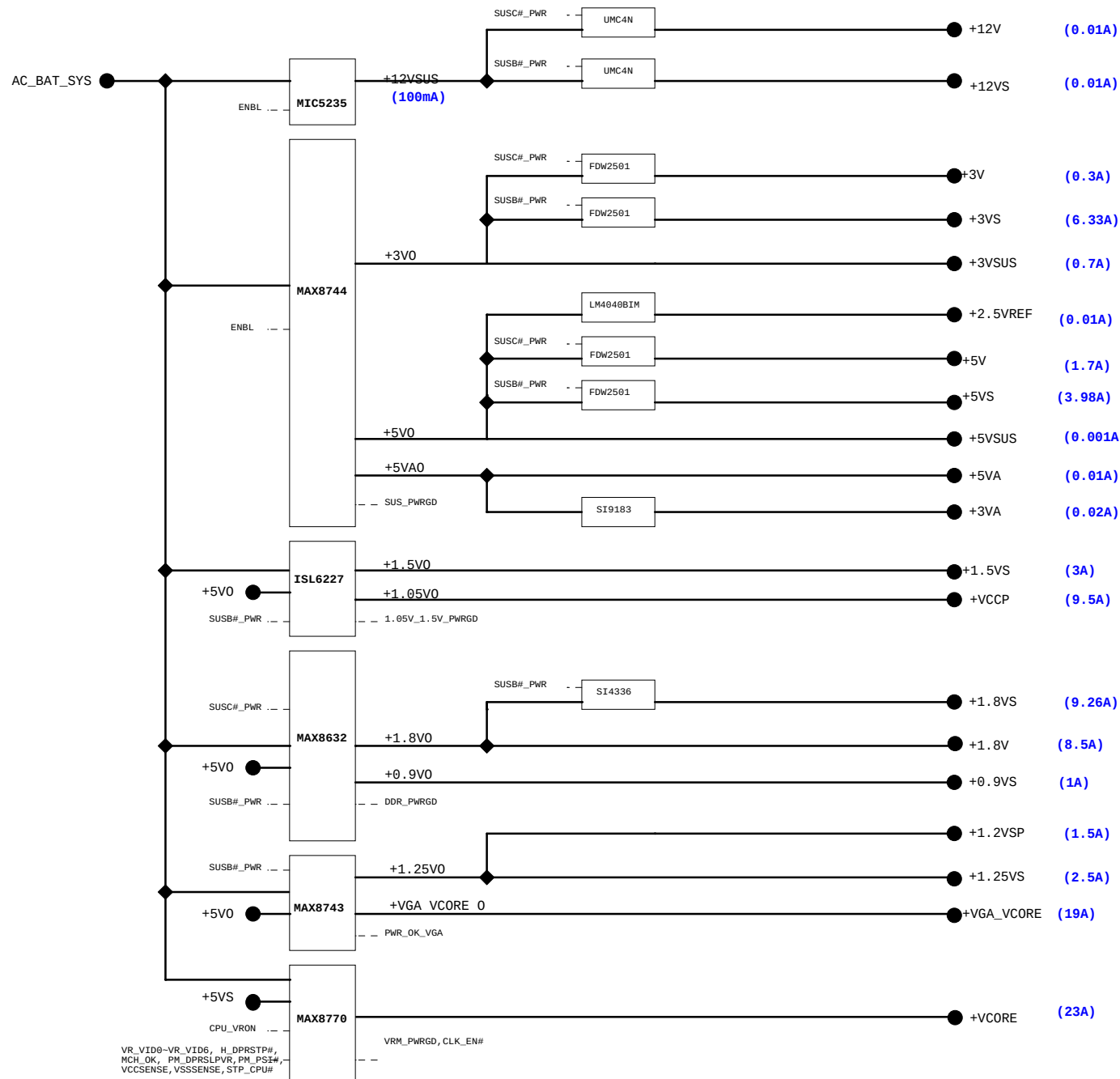
Enable Signal

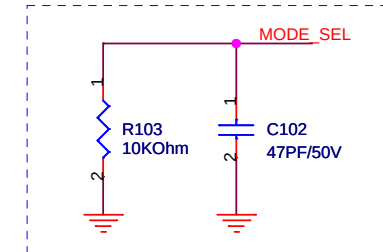
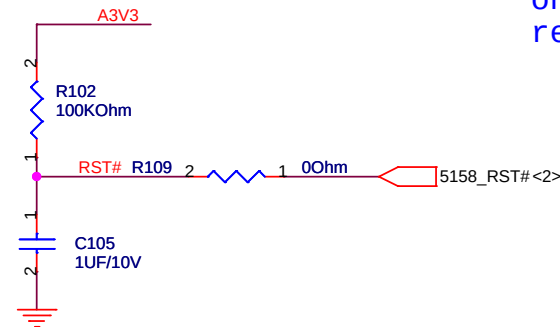
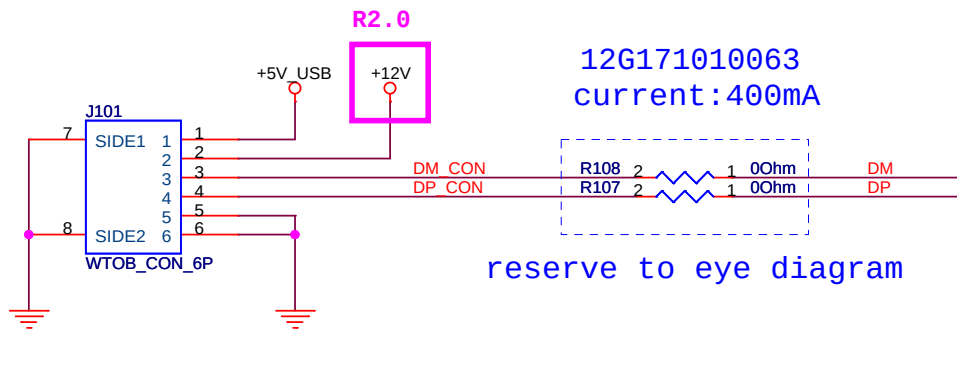


POWER GOOD DETECTOR

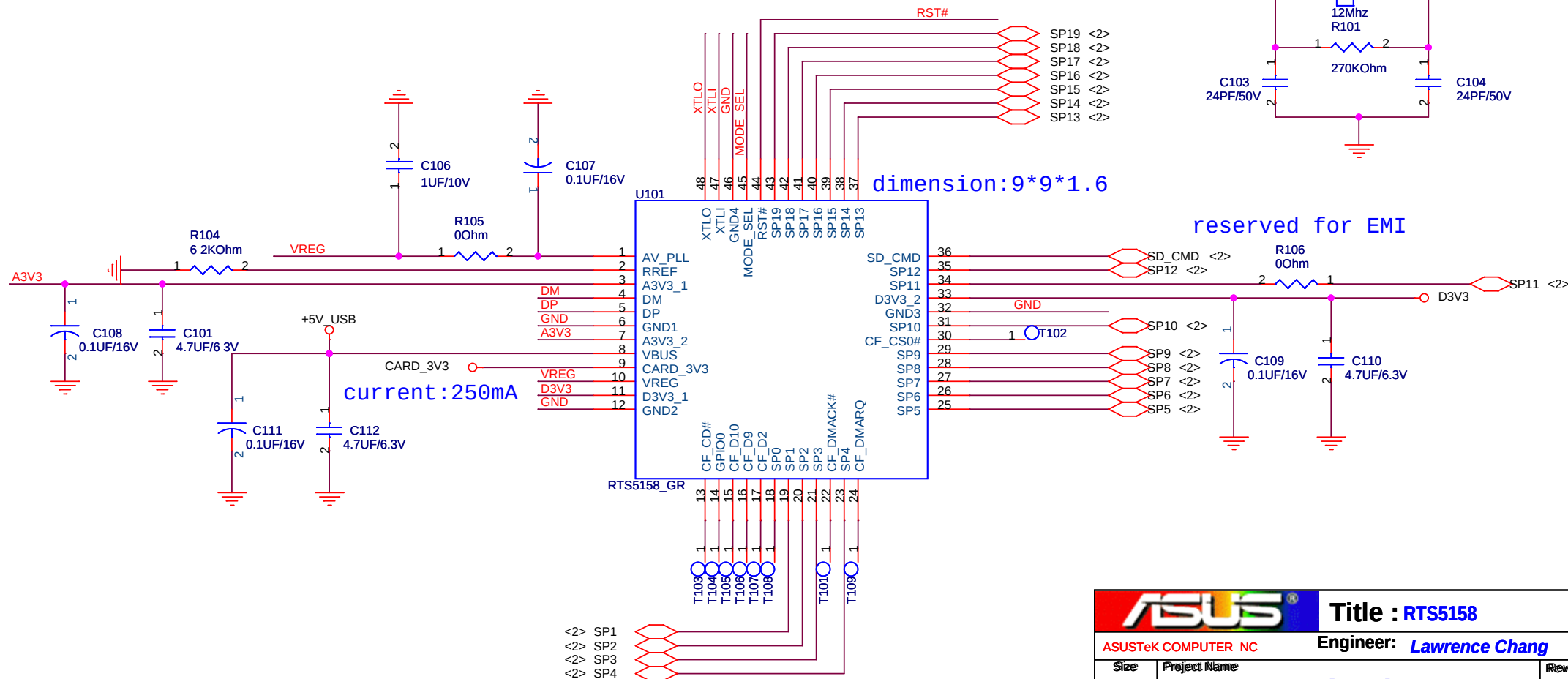
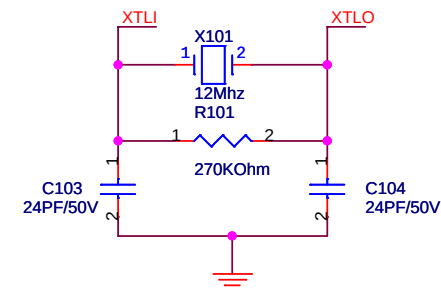




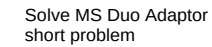
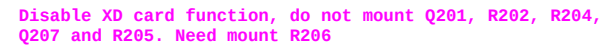
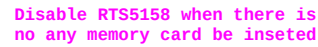




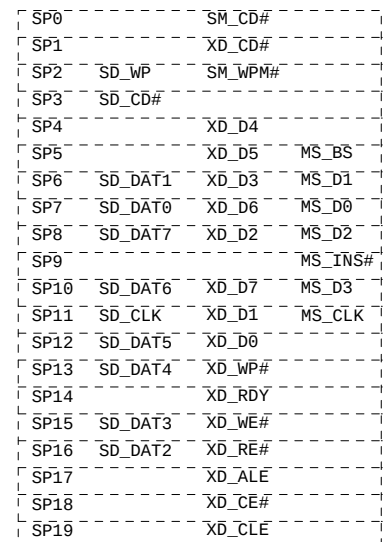
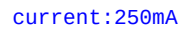
dimension:5*3.2*1.2



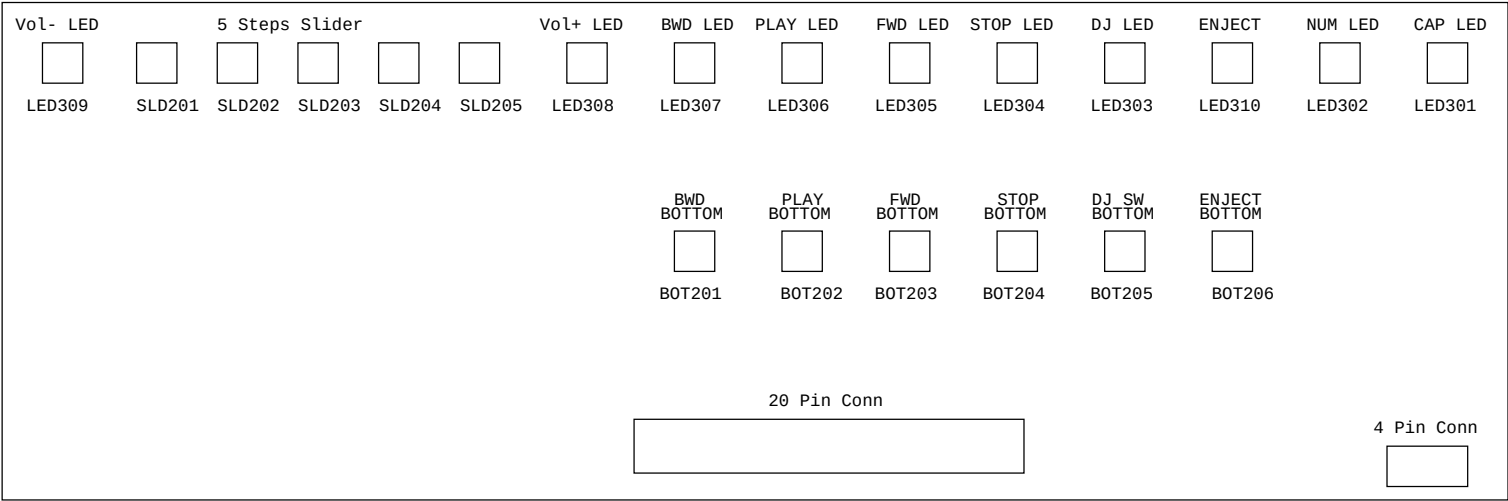
R2.0



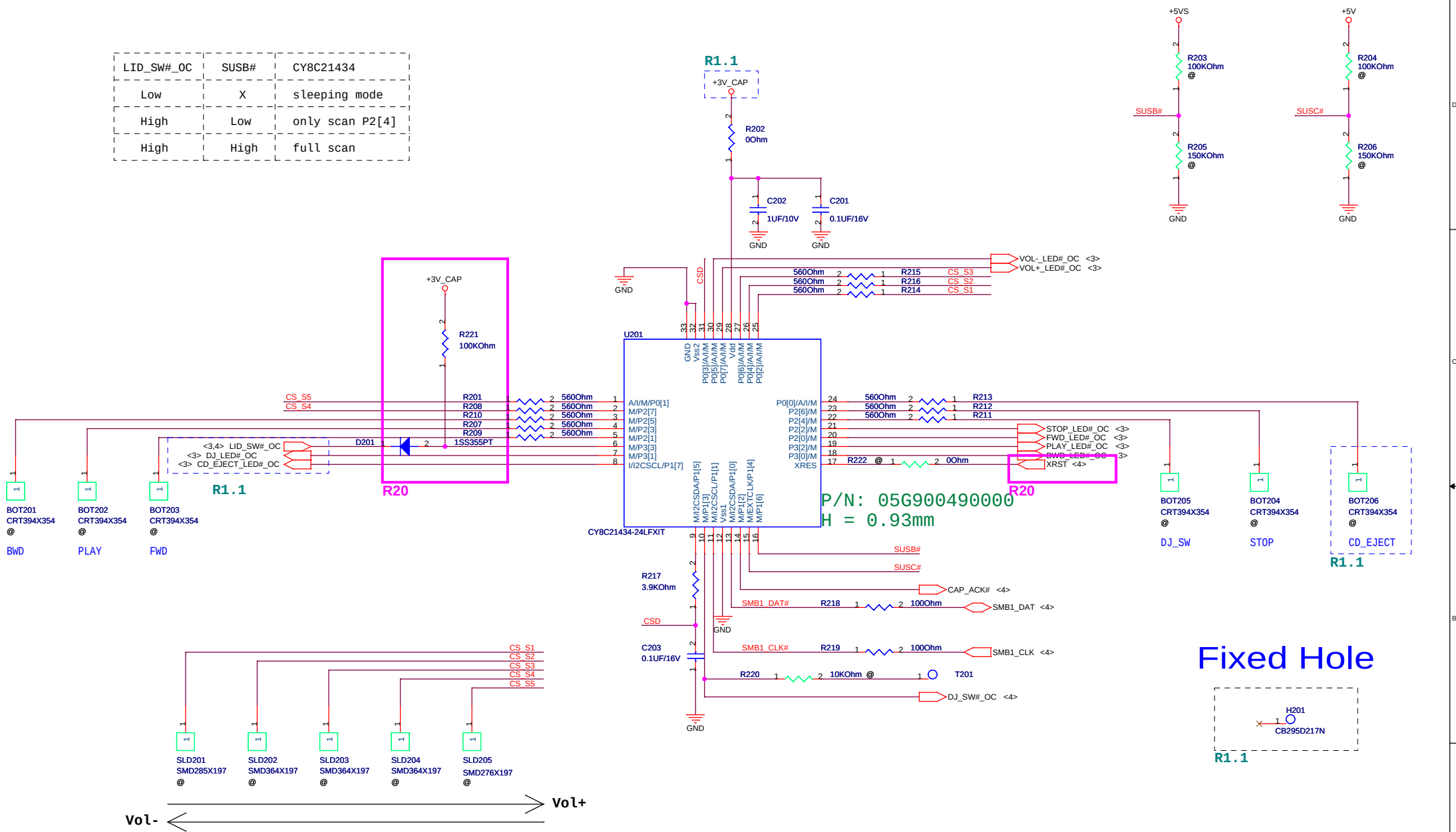
```
current:250mA
```

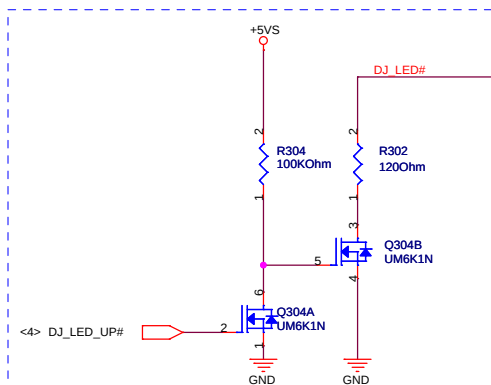
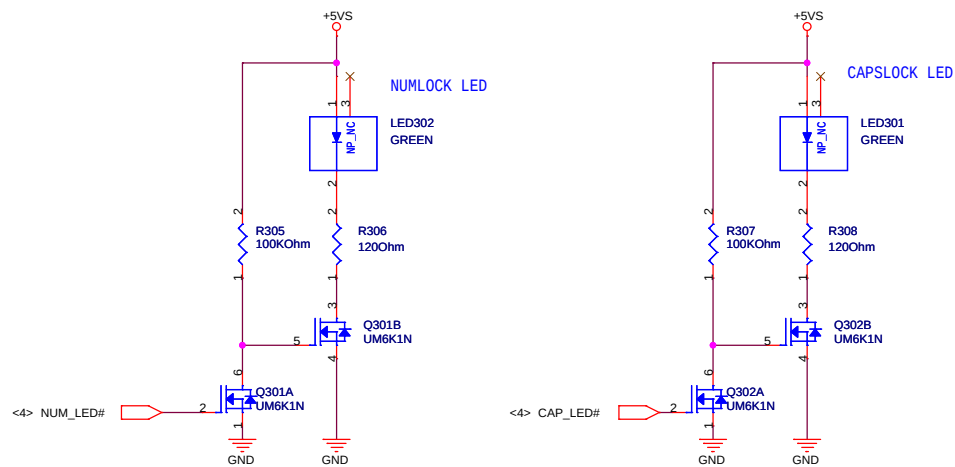


Block Diagram



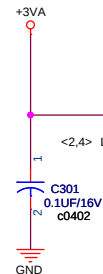
LID_SW#_OC	SUSB#	CY8C21434
Low	X	sleeping mode
High	Low	only scan P2[4]
High	High	full scan





R1.1

current: 2m

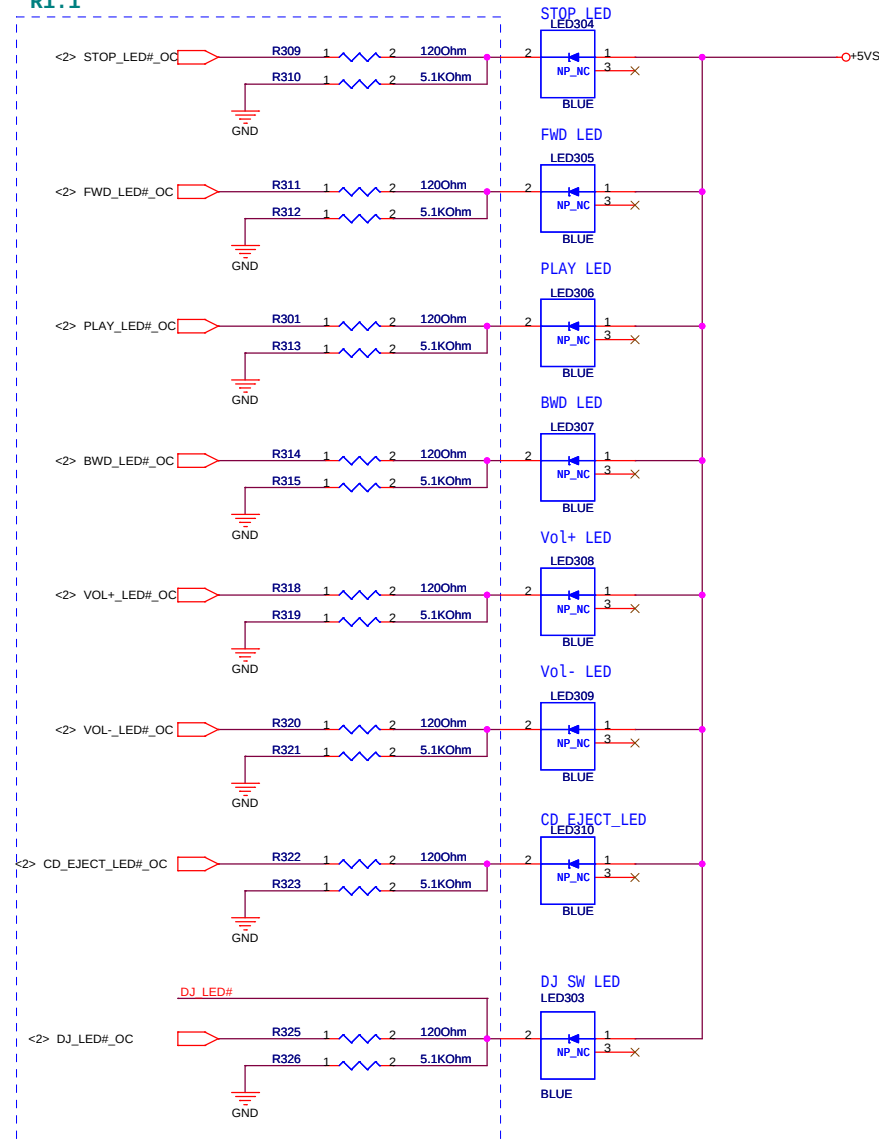


R1.1

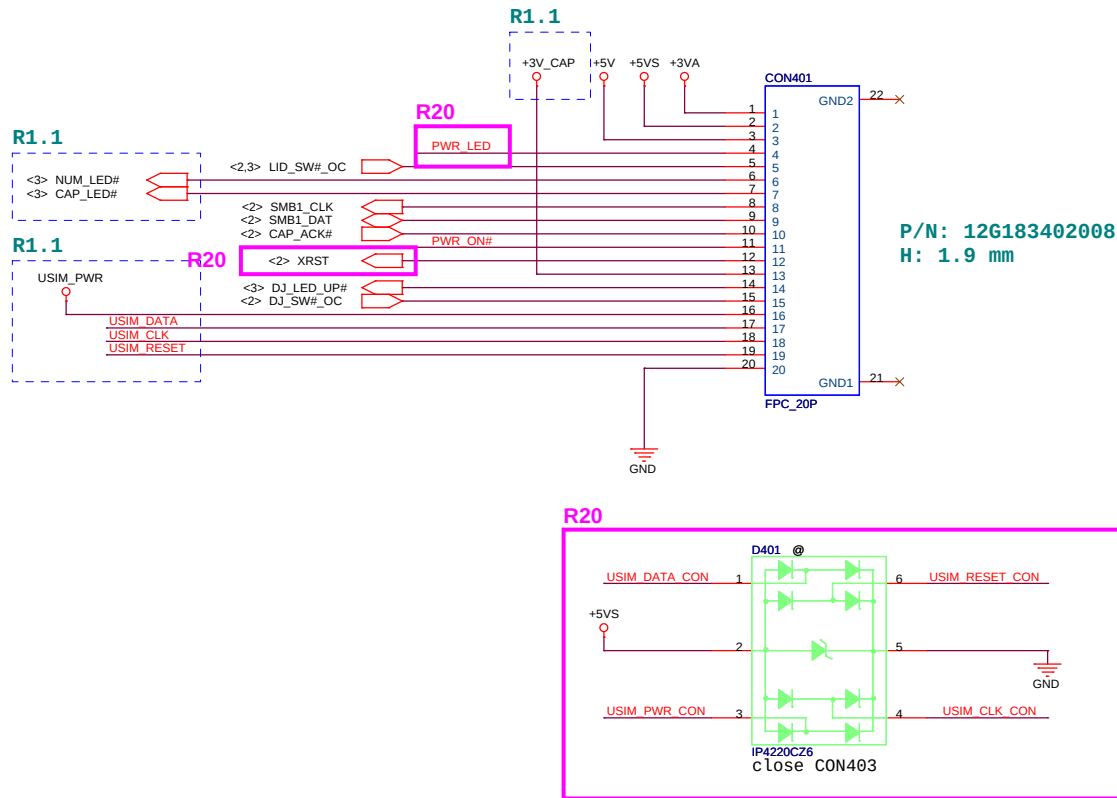
current: 1m

dimension Ø 3*3.1*1.13

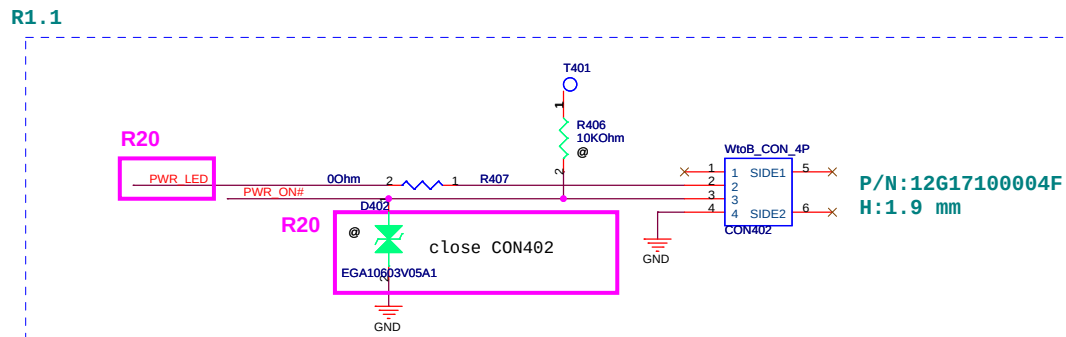
R1.1



Contact to Main Board

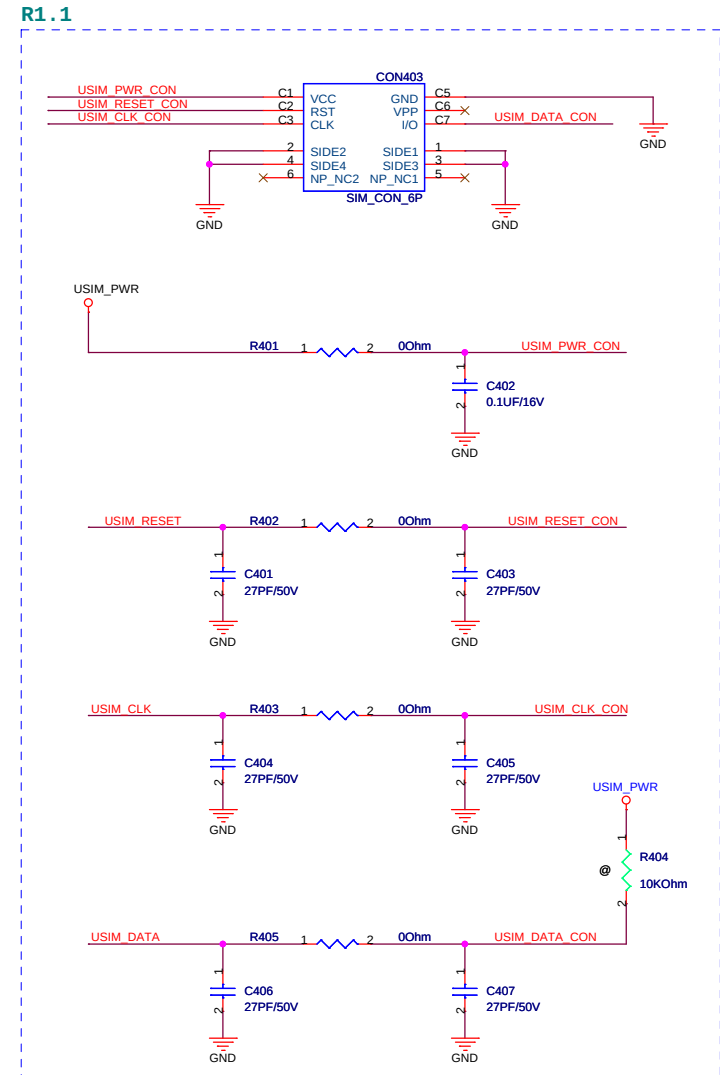


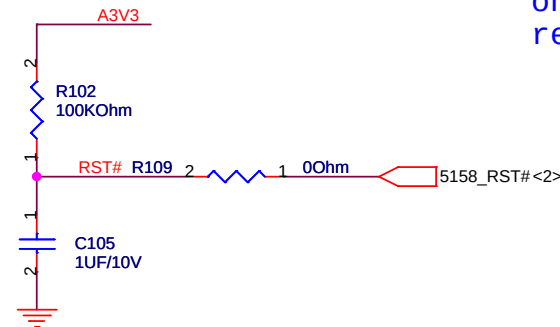
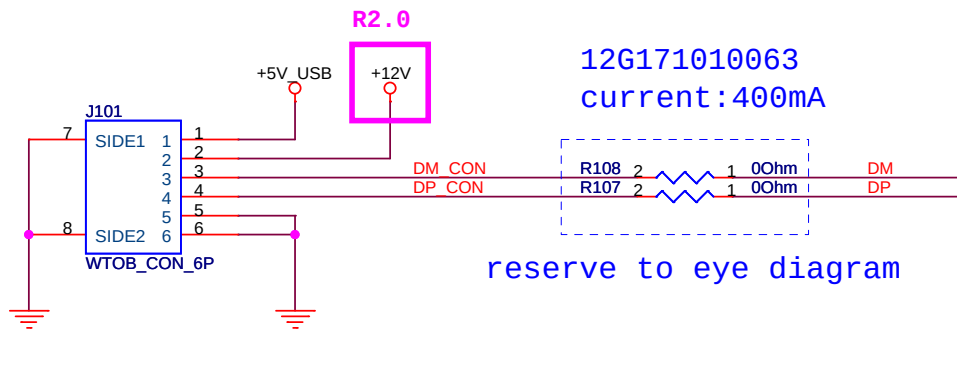
Contact to Power Swith Board



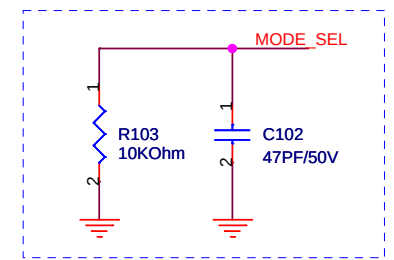
SIM

P/N: 12G2530006P3
H: 1.6 mm

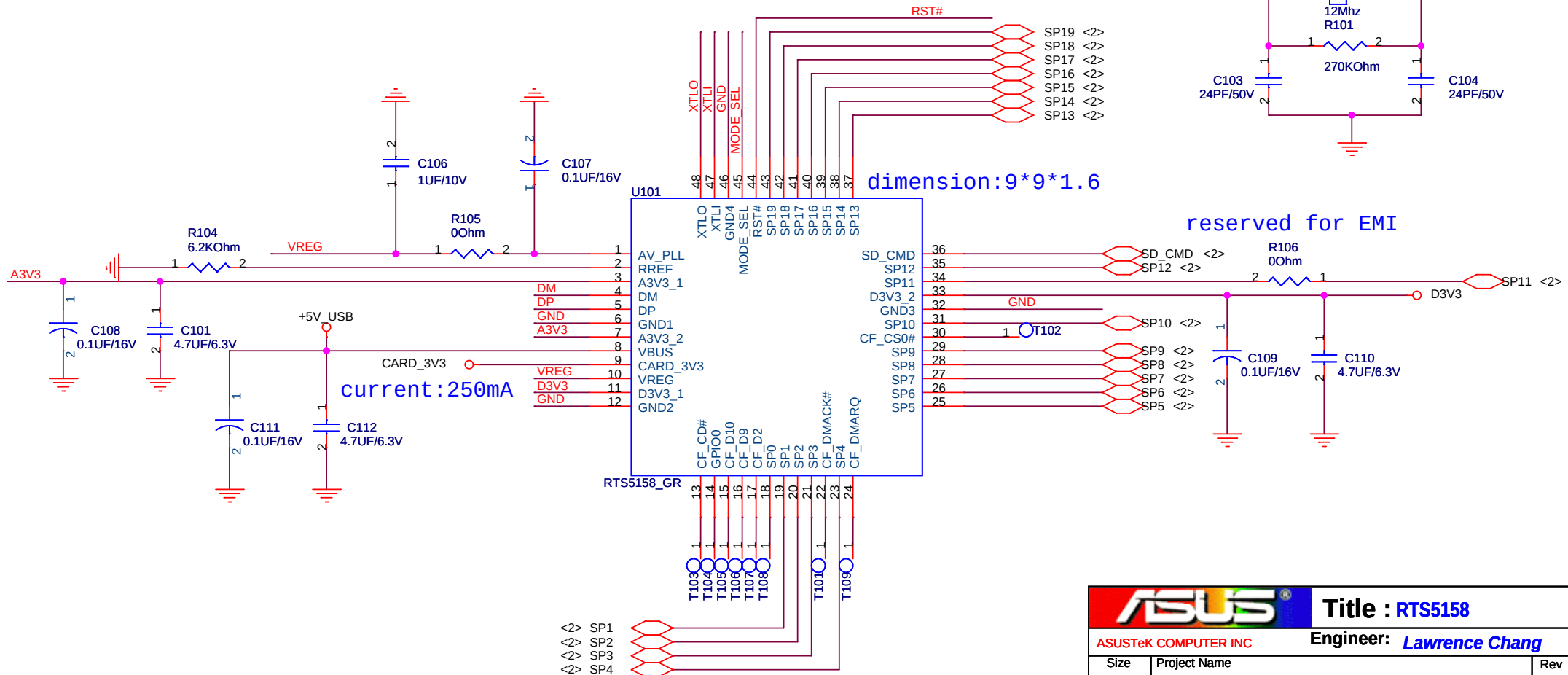
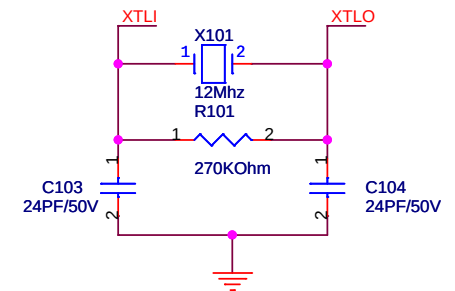




only for all-in-one card reader and power saving

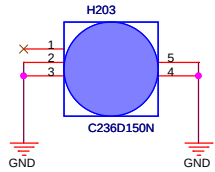
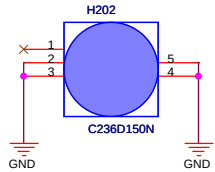


dimension:5*3.2*1.2

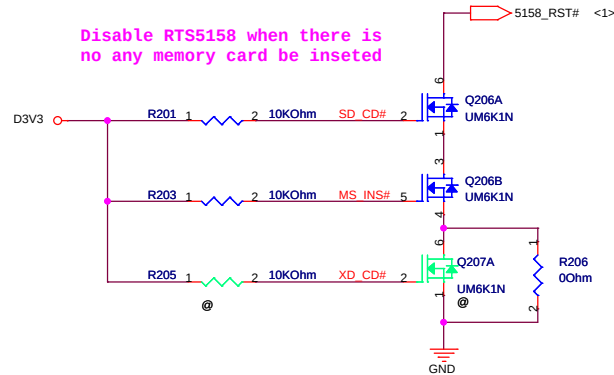


Fixed Hole

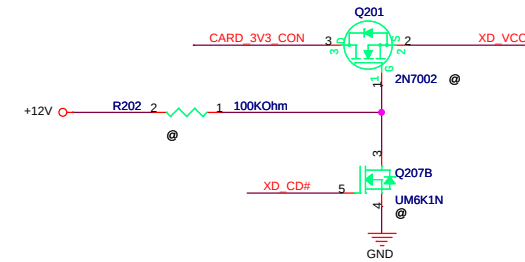
R2.0



Disable RTS5158 when there is no any memory card be inserted



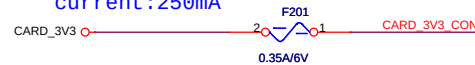
Disable XD card function, do not mount Q201, R202, R204, Q207 and R205. Need mount R206



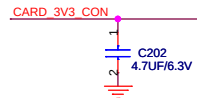
Solve MS Duo Adaptor short problem

R2.0

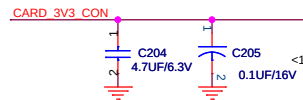
current:250mA



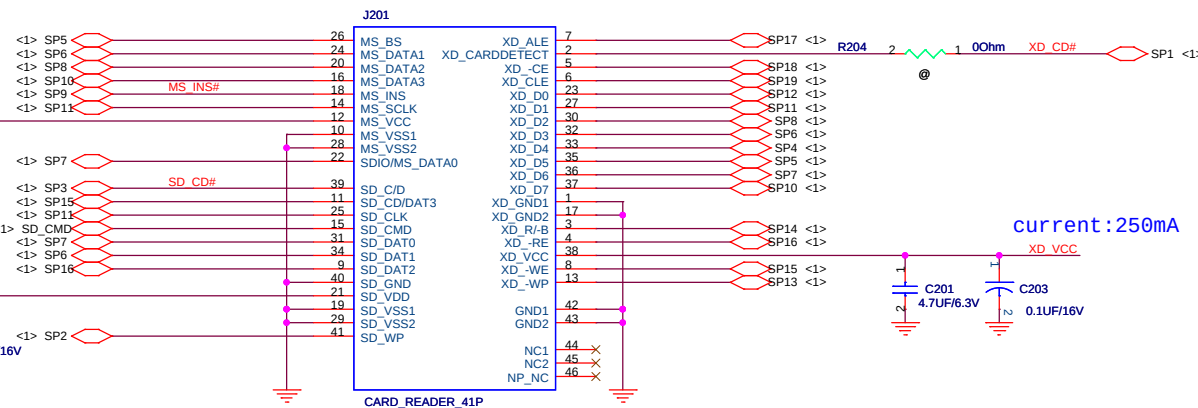
current:250mA



current:250mA



dimension:30*36.5*4.85



SP0	SM_CD#		
SP1	XD_CD#		
SP2	SD_WP	SM_WPM#	
SP3	SD_CD#		
SP4	XD_D4		
SP5	XD_D5	MS_BS	
SP6	SD_DAT1	XD_D3	MS_D1
SP7	SD_DAT0	XD_D6	MS_D0
SP8	SD_DAT7	XD_D2	MS_D2
SP9			MS_INS#
SP10	SD_DAT6	XD_D7	MS_D3
SP11	SD_CLK	XD_D1	MS_CLK
SP12	SD_DAT5	XD_D0	
SP13	SD_DAT4	XD_WP#	
SP14		XD_RDY	
SP15	SD_DAT3	XD_WE#	
SP16	SD_DAT2	XD_RE#	
SP17		XD_ALE	
SP18		XD_CE#	
SP19		XD_CLE	